
ISSCC 2021 Tutorials

Designing Amplifiers for Stability

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Introduction – Viola Schäffer

- 1998 – present: Precision Amplifiers Design Manager, Distinguished Member Technical Staff, Texas Instruments Germany/Tucson
 - Precision signal conditioning including instrumentation and programmable gain amplifiers, power amplifiers, industrial drivers as well as magnetic-based current sensors and precision magnetic sensors.
 - Led multiple technology – circuit co-developments and designed key enabling IP on these new process nodes.
 - Expertise on bipolar / CMOS analog front-end design, dynamic offset cancellation, integrated sensor developments, precision bipolar technology.
 - Holds 18 patents related to her work in analog circuit design.
- 1999: M. Sc Degree in Electrical Engineering, University of Arizona
- Primary author/presenter on 16 and co-author on 30 presentations at TI internal and IEEE conferences; Analog program selection committee at IEEE European Solid State Circuits Conference and International Solid-State Circuits Conference.



Motivation

- ❑ Operational amplifiers are a universal building block and most analog designers will design one
- ❑ An oscillating amplifier is a designer's nightmare and leads to revisions. Gain peaking or settling tails can be detrimental to system performance
- ❑ From practice, more time is spent on assuring the amplifier's stability under all conditions than on other features
- ❑ Return of Investment (ROI): time saved > time spent

ACKNOWLEDGEMENT: Big thanks to [Steve Brantley](#) and [Vadim Ivanov](#) for teaching, mentoring and the foundations for this presentation material.

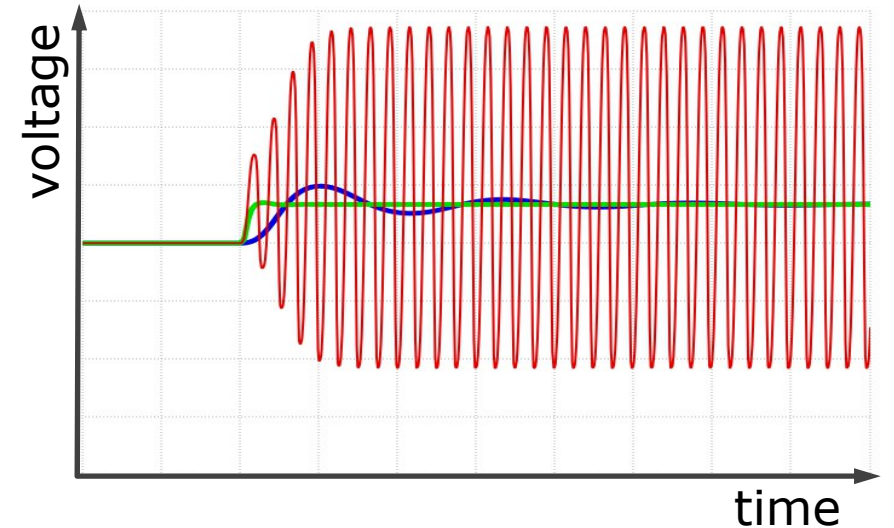
Stability

Definition of *stability*

1 : the quality, state or degree of being stable:

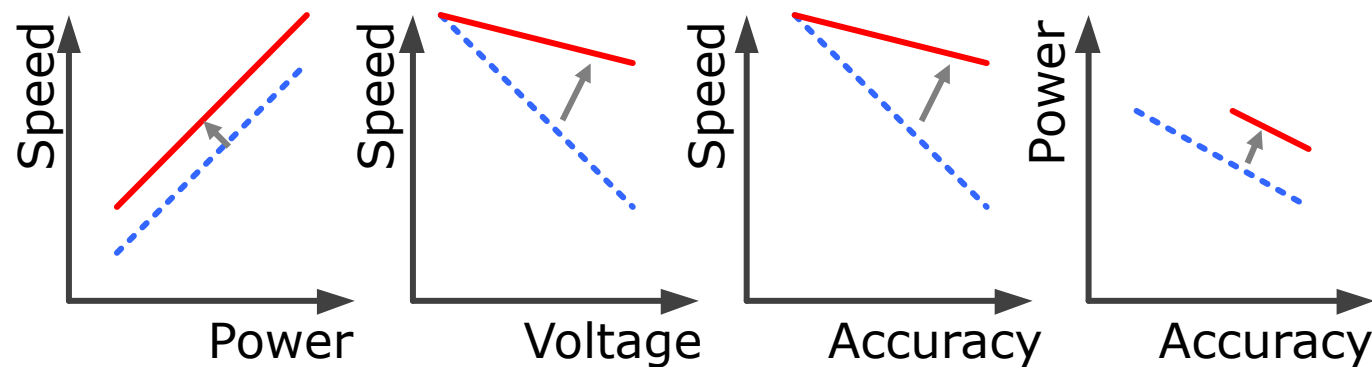
- a) the strength to stand or endure
- b) the property of a body that causes it when disturbed from a condition of equilibrium or steady motion to develop forces or moments that restore the original condition
- c) resistance to chemical change or physical disintegration

□ We will focus primarily on small signal stability, touch upon large signal (conditional) stability. We will not cover stability over temperature or time.



Benefits and Challenges

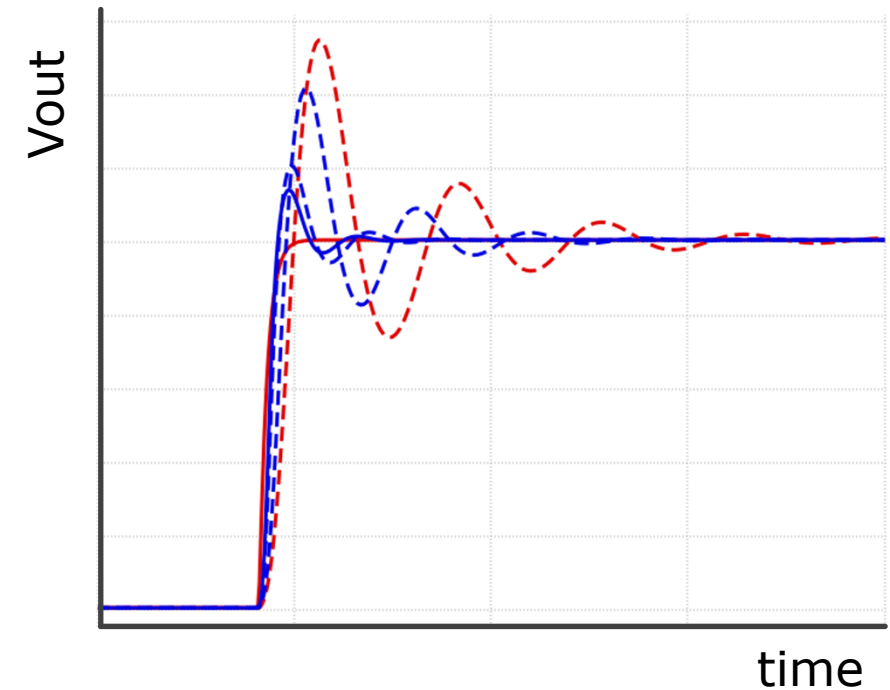
- ❑ There are over 15000 commercially available operational amplifiers and much more integrated inside mixed-signal ICs
- ❑ Variety is due to the many different optimization objectives
- ❑ They must be robust in variety of operating conditions and feedback variations
- ❑ For optimum performance one must trade-off conflicting requirements
- ❑ Poor compensation leads in worse case to system failure and in best case to wasted power and area
- ❑ Concepts are applicable to other circuits (i.e. regulators, references, etc.)



Stable (enough?)

- Stability requires power, die area, complexity and time investments
- Depends on operating conditions, load and system configuration
- Impact key amplifier parameters such as speed, power, settling time but difficult to benchmark

$$FOM: \frac{GBW}{I_Q} \neq \frac{GBW}{I_Q \cdot V_{CC}} \neq \frac{GBW \cdot C_{LD}}{I_Q \cdot V_{CC}} \neq \frac{GBW \cdot C_{LD} \cdot \phi_M}{I_Q \cdot V_{CC}}$$



Outline

- Introduction and background
- Refresher on feedback theory, bode plots and gain and phase margin
- Amplifier frequency response
- Basic compensation techniques
- Alternative (advanced) compensation schemes
- Practical tips and suggestions

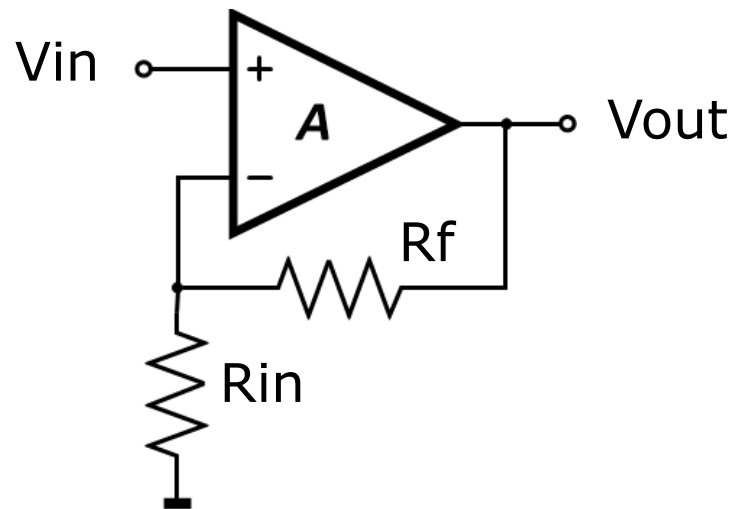
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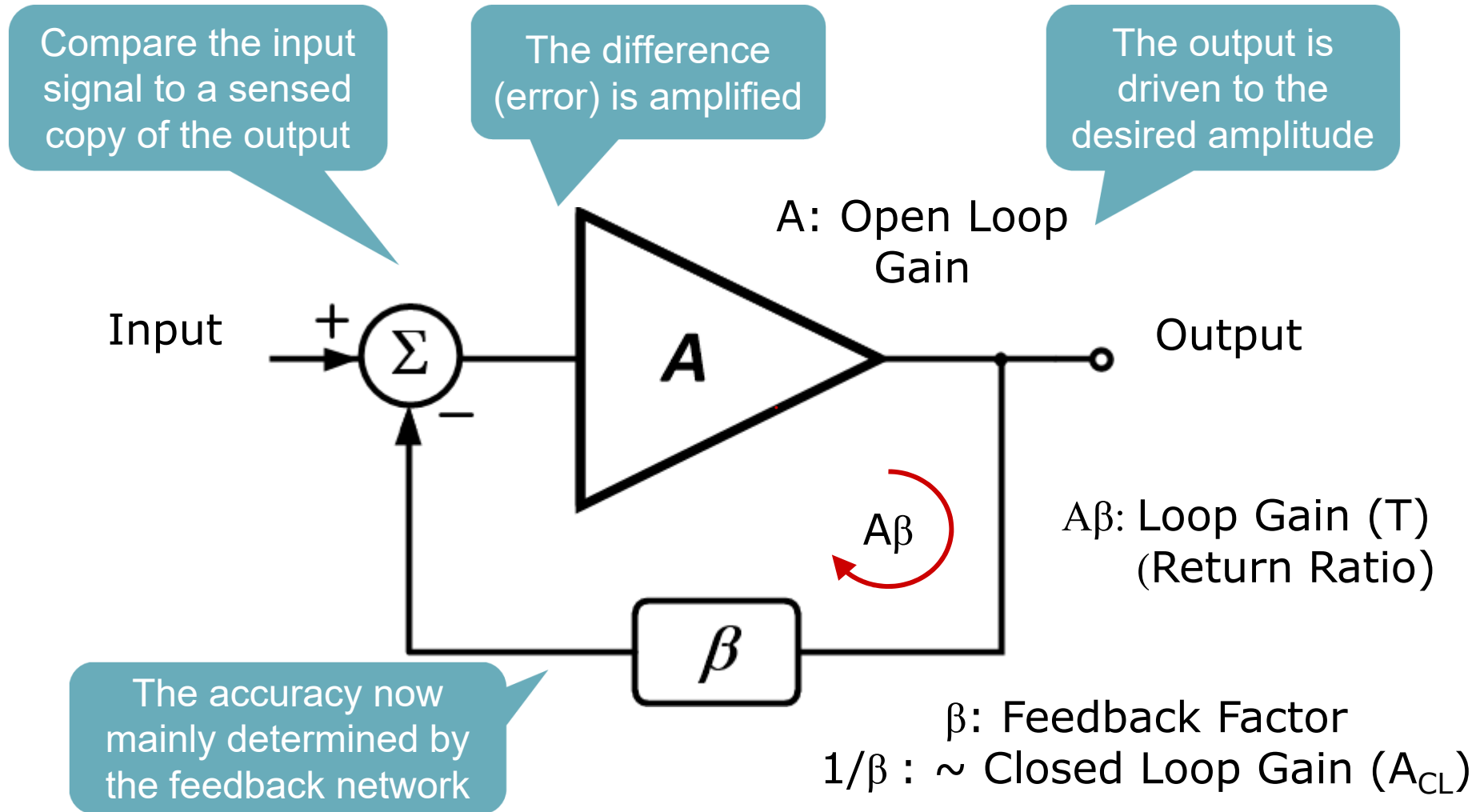
Feedback theory
Poles, zeros
Bode plots
Stability criterion

Ideal Amplifier

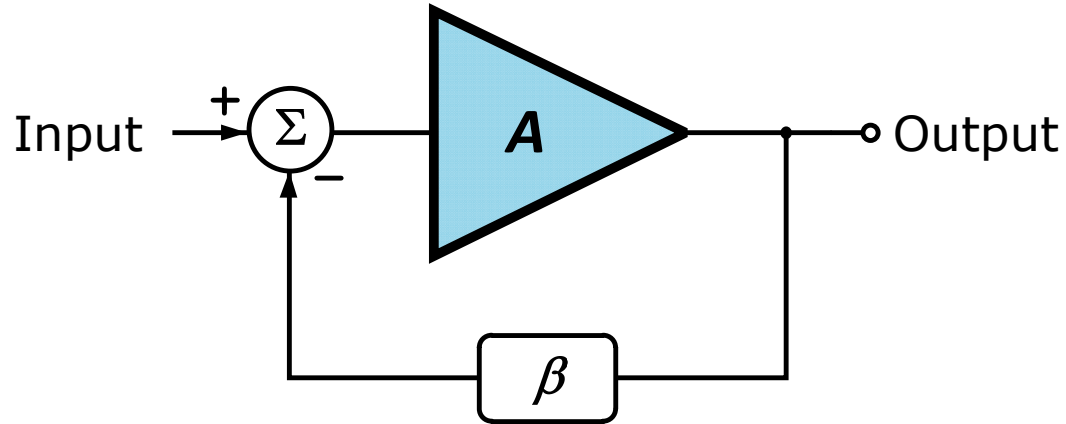
- ❑ Integrated amplifiers can have large gain (80-140dB), but it is very process, temperature, supply dependent
- ❑ Need feedback to eliminate these dependencies. Passive feedback elements such as resistors offer better stability and linearity
- ❑ Many amplifier parameters will be controlled by local feedback loops and therefore amplifiers are interacting multi-loop systems



Feedback Concept



Feedback Concept



$$\text{Gain} = \frac{A}{(1 + A\beta)} = \frac{1/\beta}{(1 + 1/A\beta)}$$

For voltage (series-shunt) feedback:

$$R_{inFB} = R_{in}(1 + A\beta) \uparrow \quad R_{outFB} = \frac{R_{out}}{(1 + A\beta)} \downarrow$$

❑ Negative Feedback

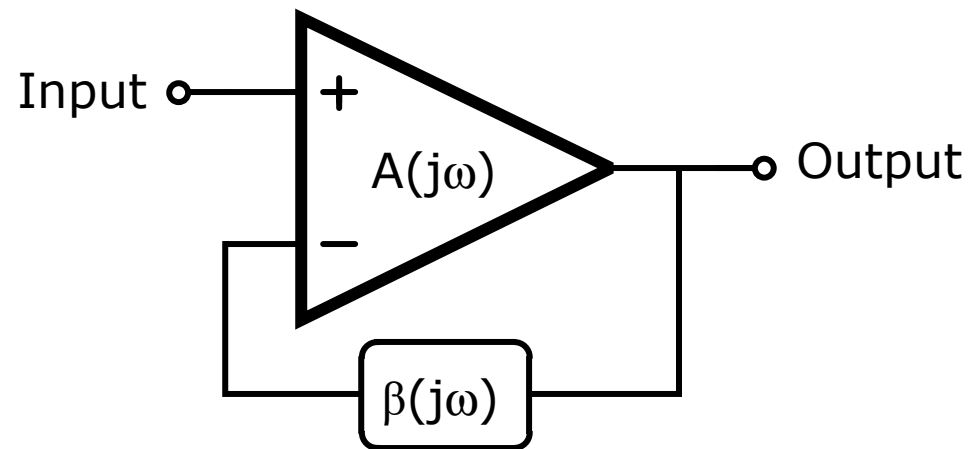
- Increases accuracy (with accurate β)
- Reduces sensitivity to gain variations
- Linearizes system
- Improves input/output impedance
- Provides more usable bandwidth

❑ However

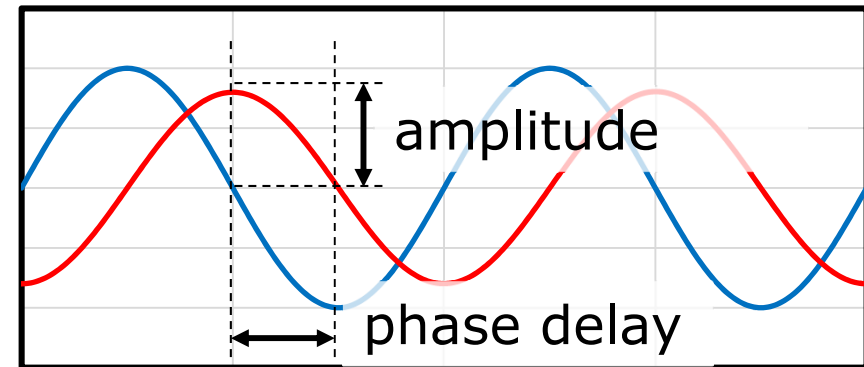
- Lowers achievable gain
- Increases area and complexity
- Can lead to instability

Frequency Dependence

- The open loop gain (A) and feedback factor (β) do not stay constant over frequency
- Feedback loop has to be analyzed over frequency



$$\omega[\text{rad/s}] = 2\pi f[\text{Hz}]$$

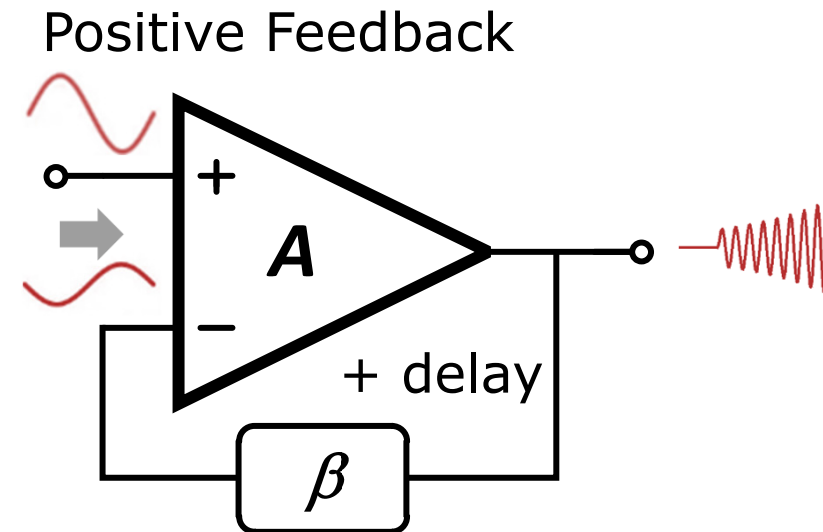
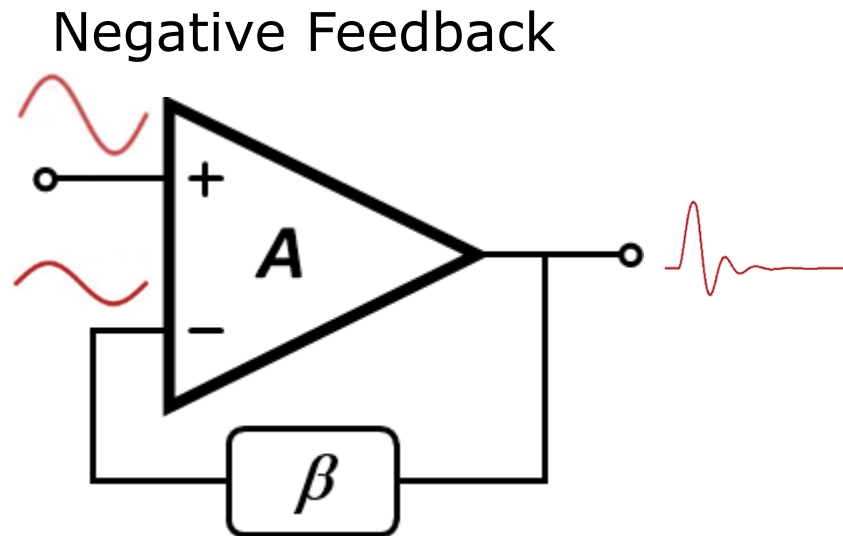


$$CL \text{ Gain} = \frac{V_{\text{output}}}{V_{\text{input}}} = \frac{\frac{1}{\beta(j\omega)}}{\left(1 + \frac{1}{A(j\omega)\beta(j\omega)}\right)}$$

$$T(j\omega) = A(j\omega)\beta(j\omega)$$

Negative and Positive Feedback

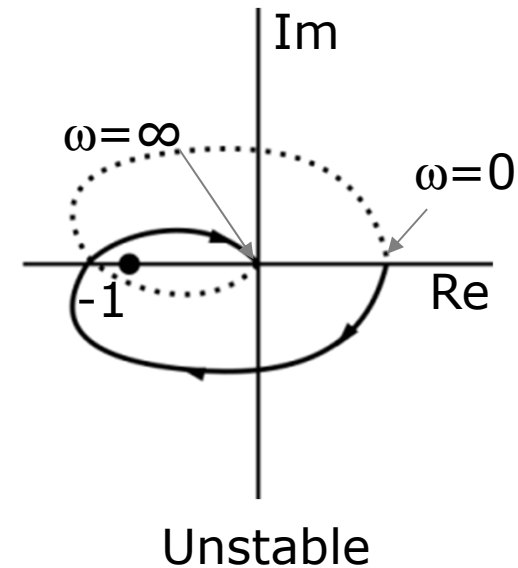
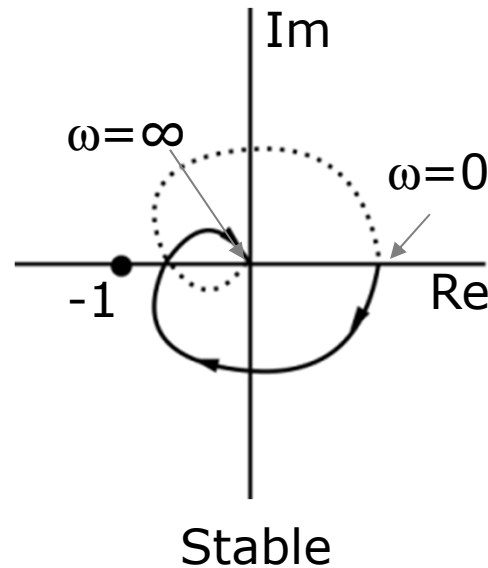
With too much delay in the signal fed back (phase shift) in the system the negative feedback can become positive leading to increasing output and therefore instability.



Feedback Stability: Nyquist Stability Criterion

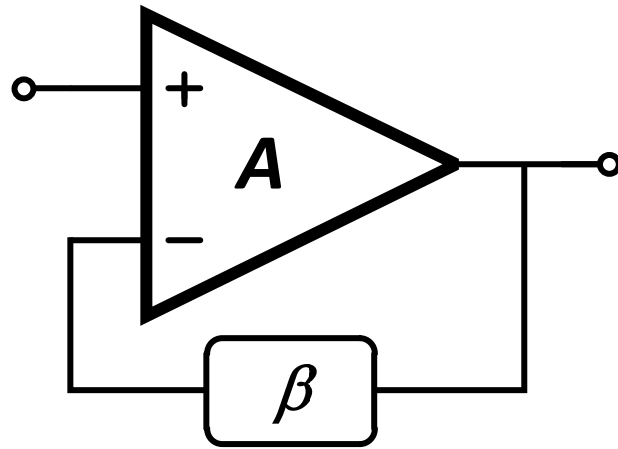
- The Nyquist plot is obtained by tracing in the complex plane the magnitude and phase of the loopgain $T(j\omega)$ from $\omega=0$ to $\omega=j\infty$
- Based on Nyquist Stability Criteria (1932) for a feedback amplifier (without right half plane poles) a necessary and sufficient condition for stability is that the Nyquist plot in the $T(j\omega)$ plane of the loopgain $T(j\omega)$ does not encircle the $-1+j0$ (180°) [1,17]

$$T(j\omega) = A_0 \frac{\left(1 + \frac{j\omega}{\omega_{z1}}\right) \left(1 + \frac{j\omega}{\omega_{z2}}\right) \dots}{\left(1 + \frac{j\omega}{\omega_{p1}}\right) \left(1 + \frac{j\omega}{\omega_{p2}}\right) \dots}$$



A practical simplification

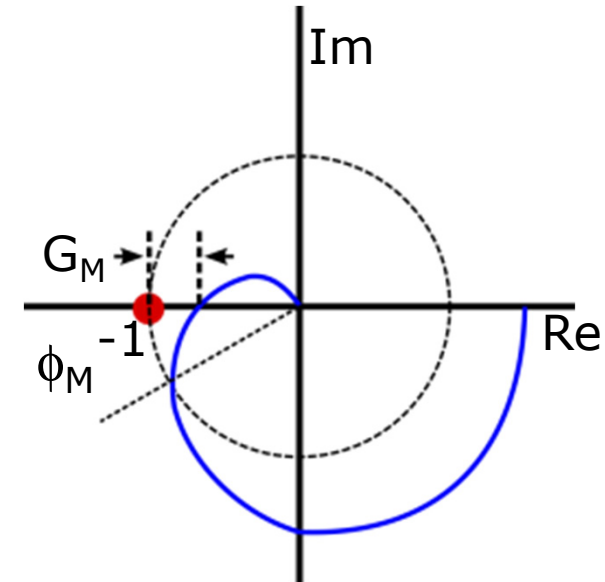
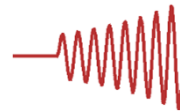
For most practical feedback amplifier cases we aim to assure that the frequency where the phase is inverted (-180°) the magnitude of the loopgain is attenuated (below $1=0\text{dB}$)



$|T|$ at $f_{180} < 1$



$|T|$ at $f_{180} > 1$



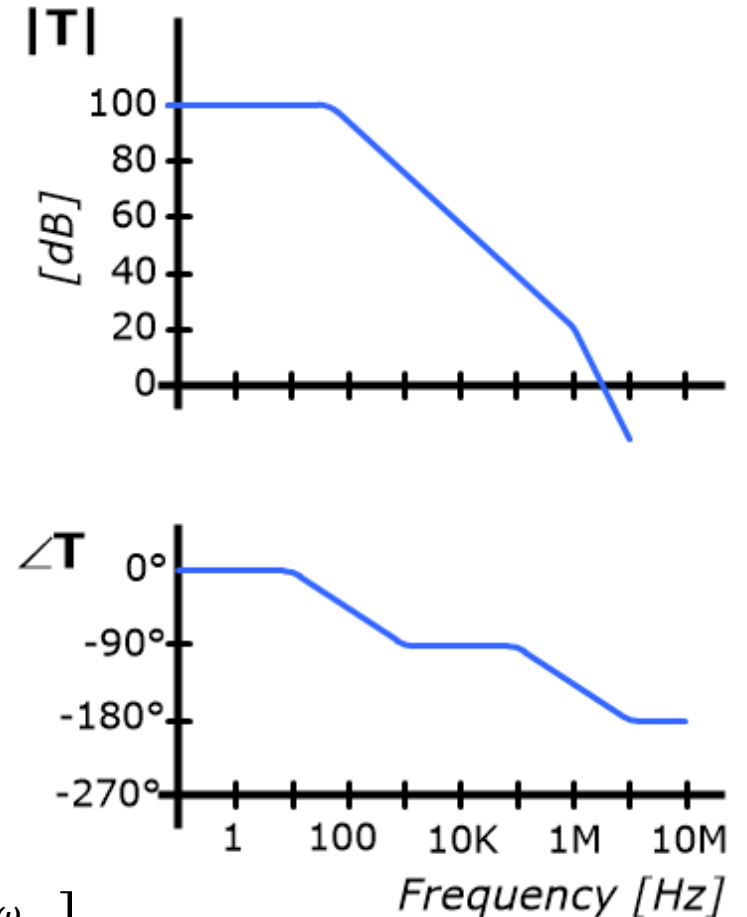
Bode Plots: Refresher

- A graphical method to represent the small signal frequency response of a system **at a particular operating point**
- Useful to understand trends for stability by looking at gain or loopgain magnitude and phase
- Always follow up by transient simulations and disturbances!

$$T(j\omega) = A_0 \frac{\left(1 + \frac{j\omega}{\omega_{z1}}\right) \left(1 + \frac{j\omega}{\omega_{z2}}\right) \dots}{\left(1 + \frac{j\omega}{\omega_{p1}}\right) \left(1 + \frac{j\omega}{\omega_{p2}}\right) \dots}$$

$$|T| = 20 \log_{10} |T(j\omega)|$$

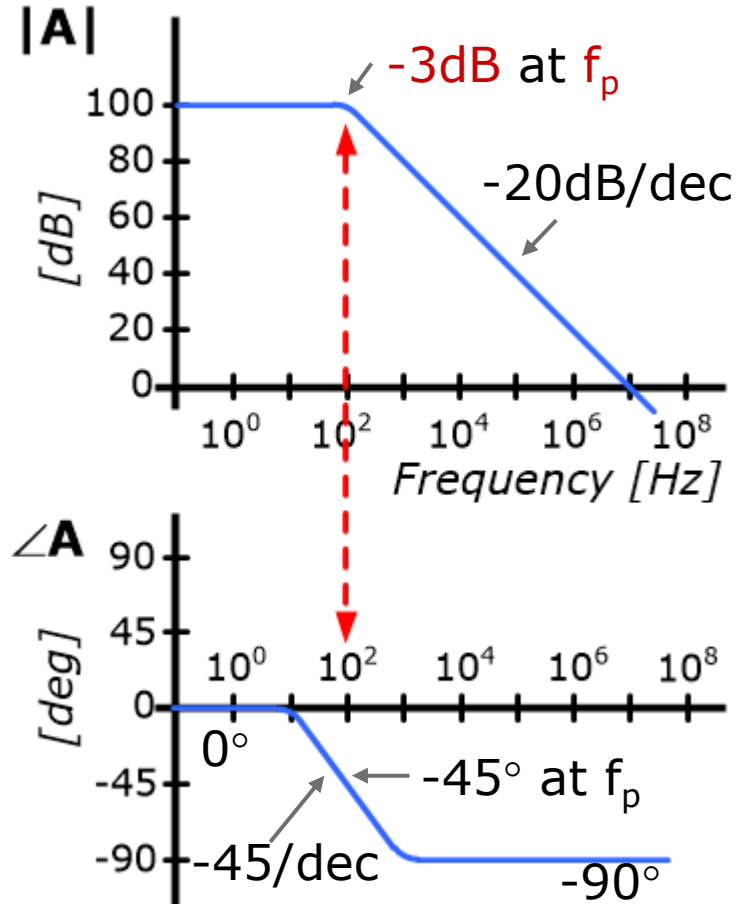
$$\angle T = \sum \tan^{-1} \left[\frac{\omega}{\omega_{z1,2..}} \right] - \sum \tan^{-1} \left[\frac{\omega}{\omega_{p1,2..}} \right]$$



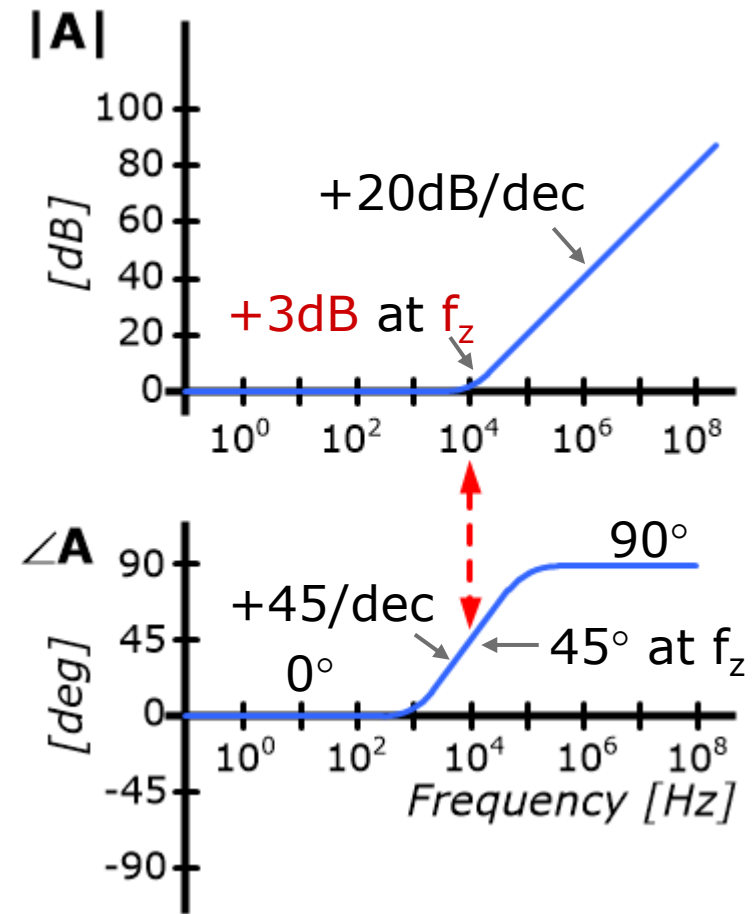
$$\omega[\text{rad/s}] = 2\pi f[\text{Hz}]$$

Bode Plots: LHP Pole and Zero

Bode Plot: LHP Pole

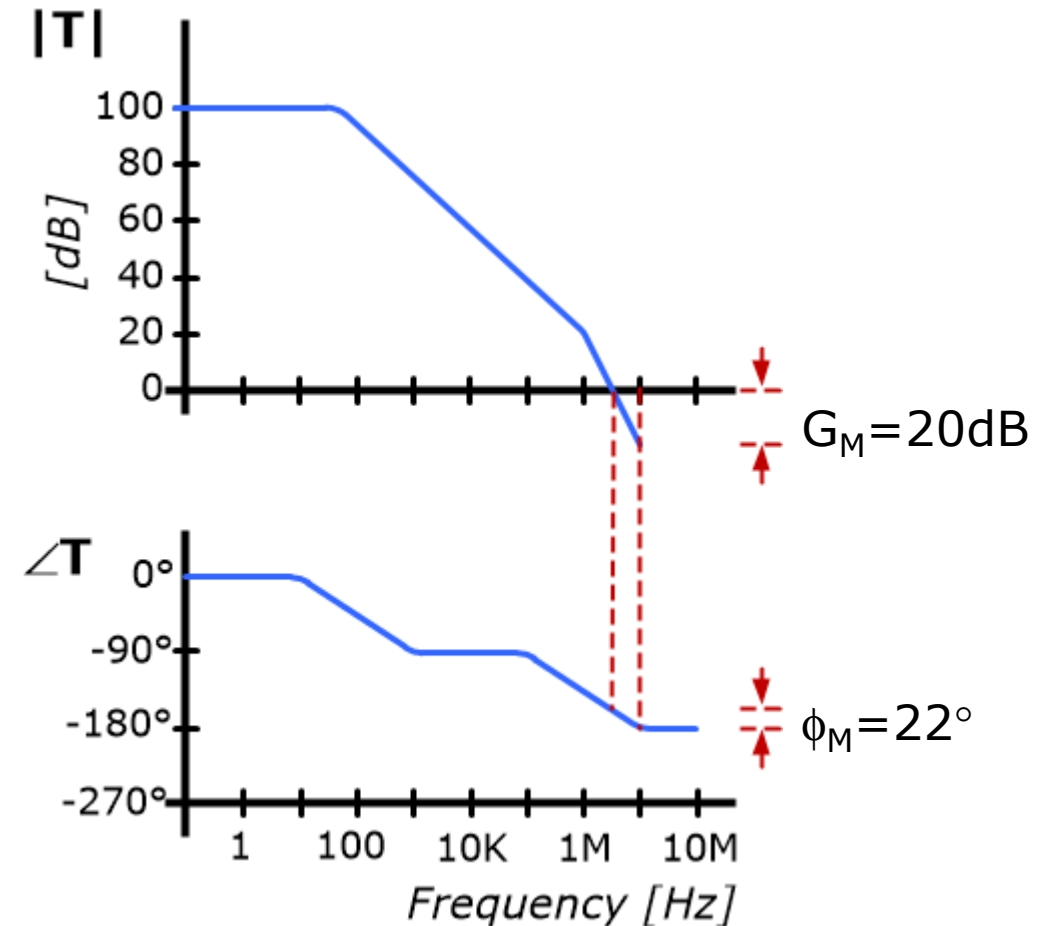


Bode Plot: LHP Zero

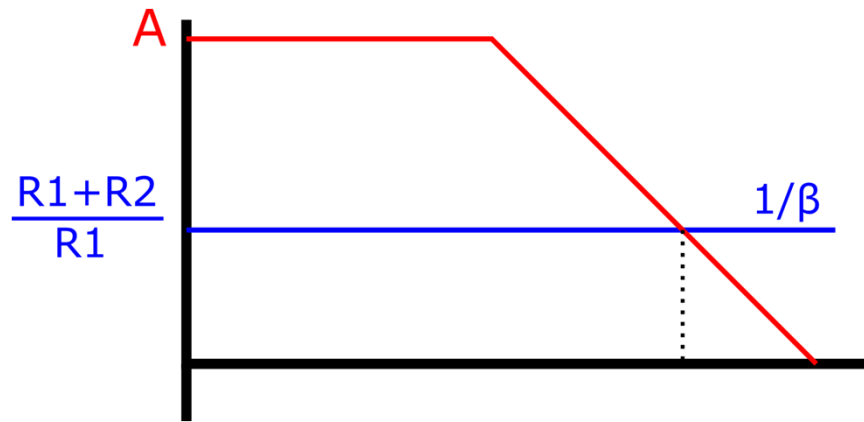
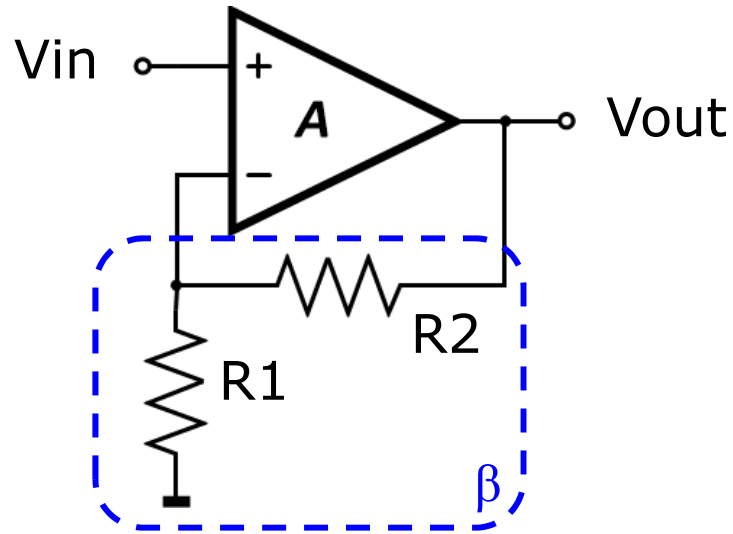


Gain and Phase Margin

- From Nyquist Criterion the degree of a system's stability can be quantified by:
 - Phase Margin (ϕ_M): The phase at the frequency where the loopgain (T) is 0dB (referenced to -180°)
 - Gain Margin (G_M): The attenuation where the phase of the loopgain (T) has reached 180°



Op amp Feedback Stability: Graphical Method

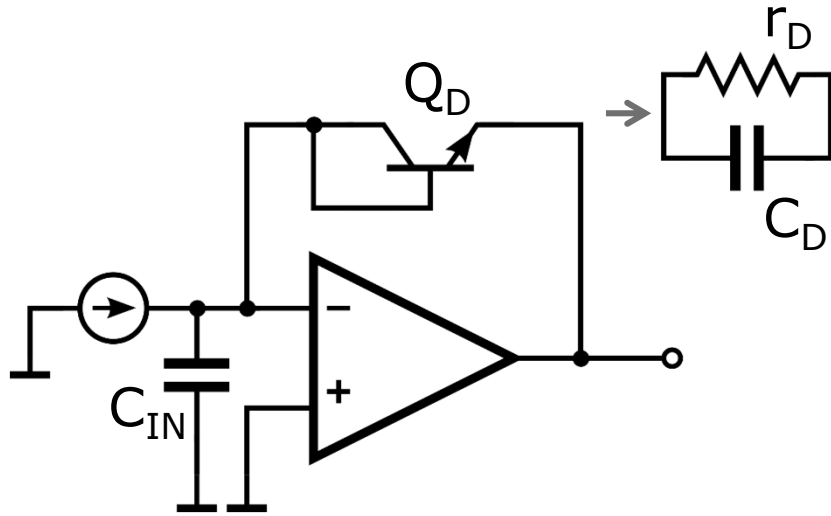


- Loopgain $T = A\beta$
- $\text{Log}(T) = \text{Log}(A) + \text{Log}(\beta)$
- **$\text{Log}(T) = \text{Log}(A) - \text{Log}(1/\beta)$**

Use above equation in a plot:

- Separately plot $20\text{Log}(A)$ and $20\text{Log}(1/\beta)$
- Loopgain is difference of curves, unity gain is where lines cross
- For stability loopgain needs to cross unity with 20dB /decade roll-off

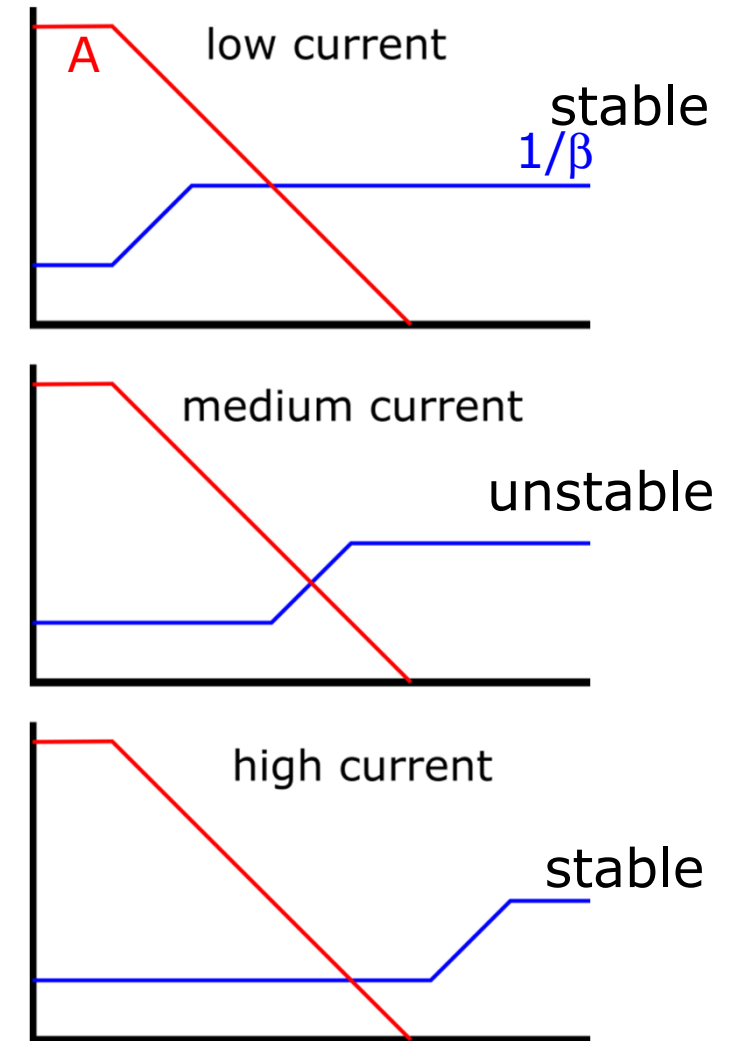
Op amp Feedback Stability: Graphical Method



$r_D = 1/g_m$ of Q_D (current dependent)

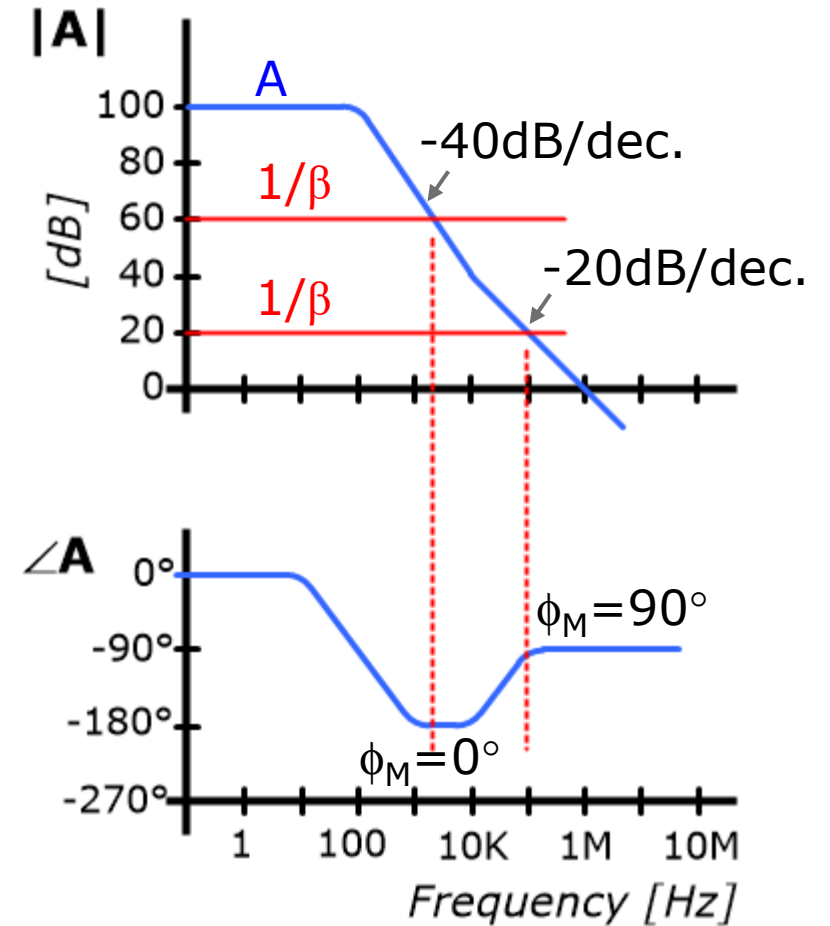
$C_D =$ parasitic diode capacitance

$$f_z = \frac{1}{2\pi r_D (C_D + C_{IN})} \quad f_p = \frac{1}{2\pi r_D C_D}$$



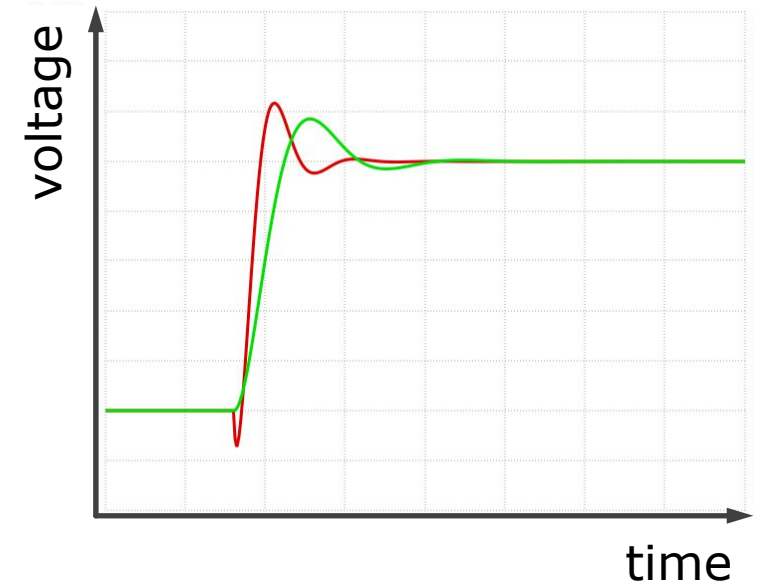
Op amp Feedback: Conditional Stability

- Stable if feedback network is restricted to certain values
 - Unstable with CL gain > 20 (26dB)
 - Stable with CL gain < 20 (26dB)
- Much improved power efficiency
- Careful in any condition that can lead to loss of open loop gain!



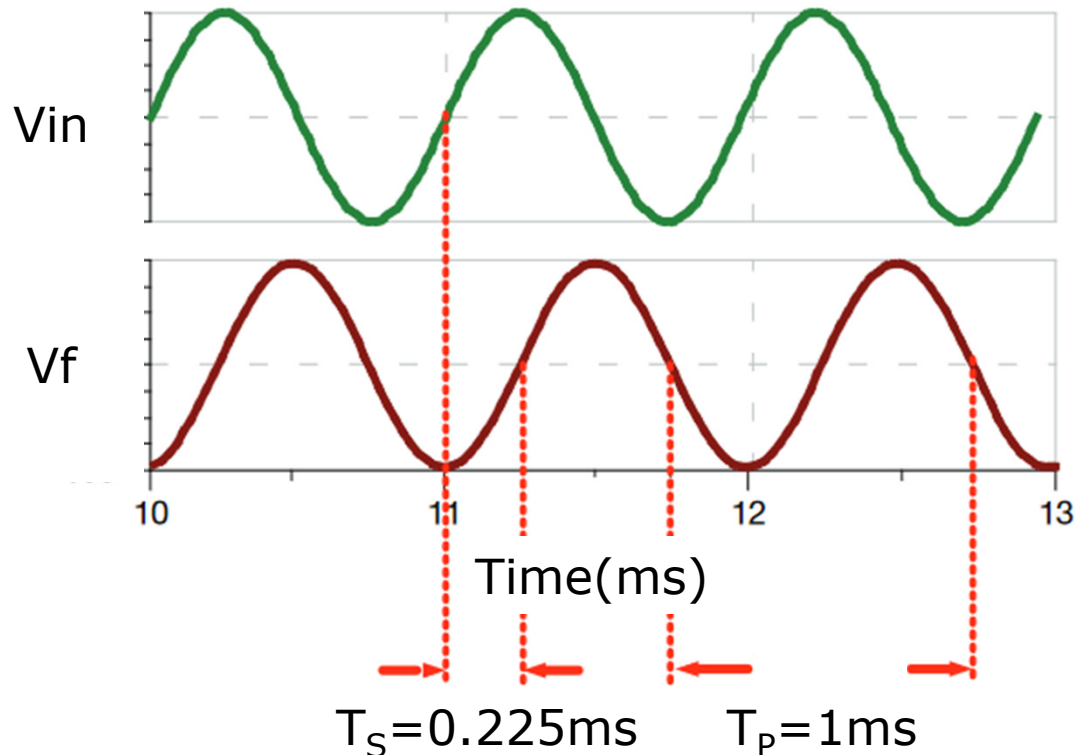
Bode Plots: LHP vs RHP Pole and Zero

- Pole magnitude decreases with frequency, zero increases with frequency
- LHP pole or RHP zero phase decreases with frequency
 - RHP zero reduces phase margin!
 - Commonly found in circuits (i.e. Miller capacitance feed-forward)
- RHP pole or LHP zero phase increases with frequency
 - RHPP lead to system instability
 - Not known [?] in amplifiers



Relating to Transient Response [19]

Time to phase shift



T_S : time shift from input to output signal

T_P : period of signal

θ : phase shift of the signal from input to output

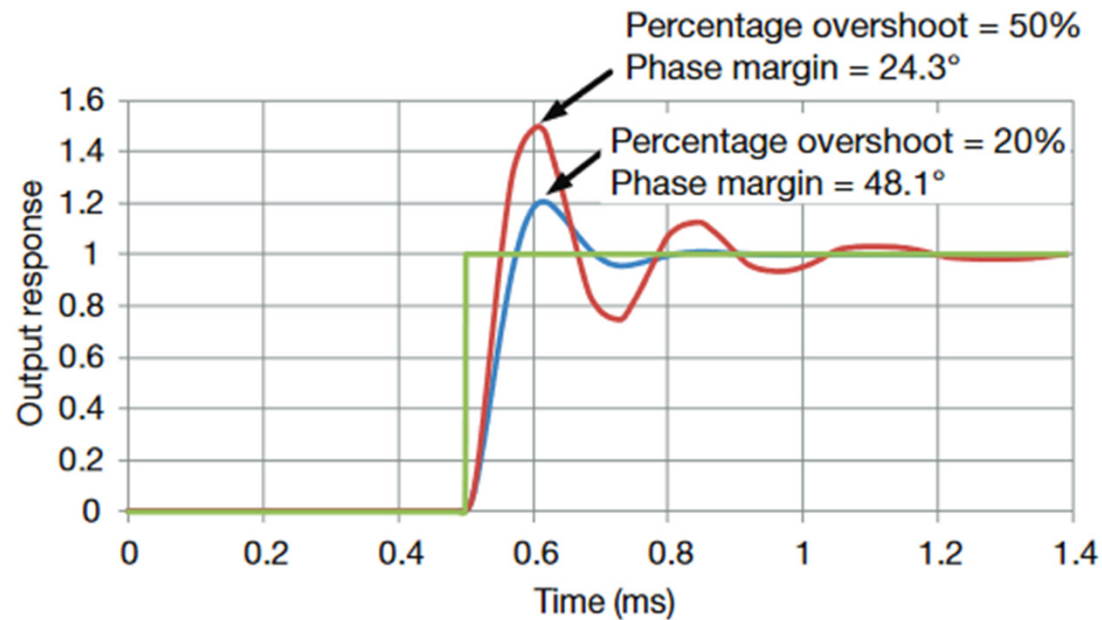
To calculate phase shift in degrees:

$$\theta = \frac{T_S}{T_P} \cdot 360^\circ = \left(\frac{0.225\text{ms}}{1\text{ms}} \right) \cdot 360^\circ = 81^\circ$$

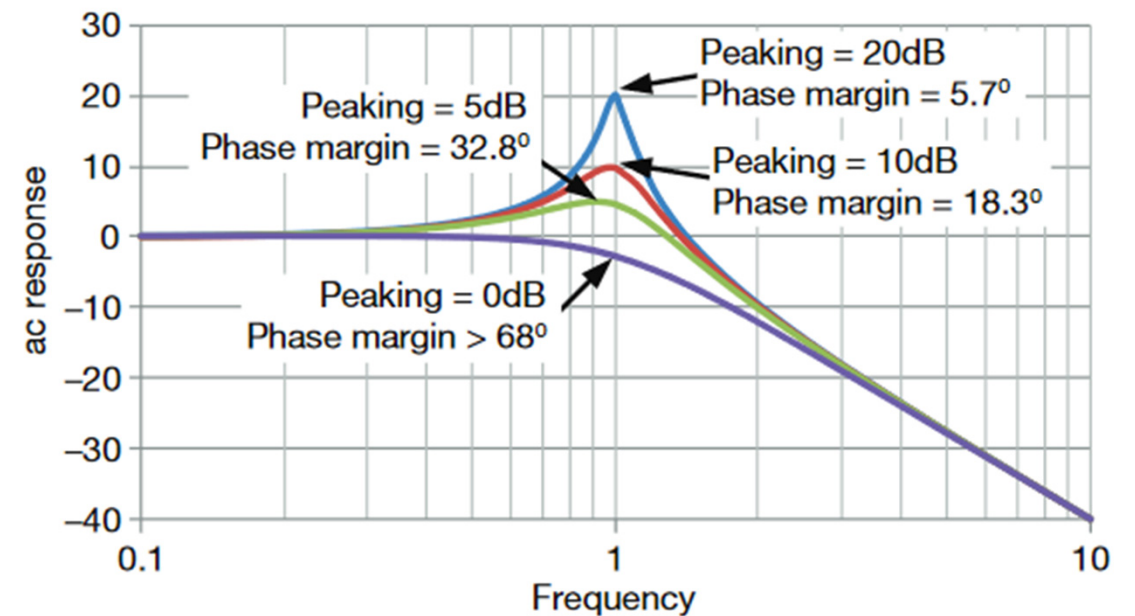
Relating to Transient Response [19]

Dominant (2 pole) system responses

Small-step overshoot



AC Response vs. Frequency

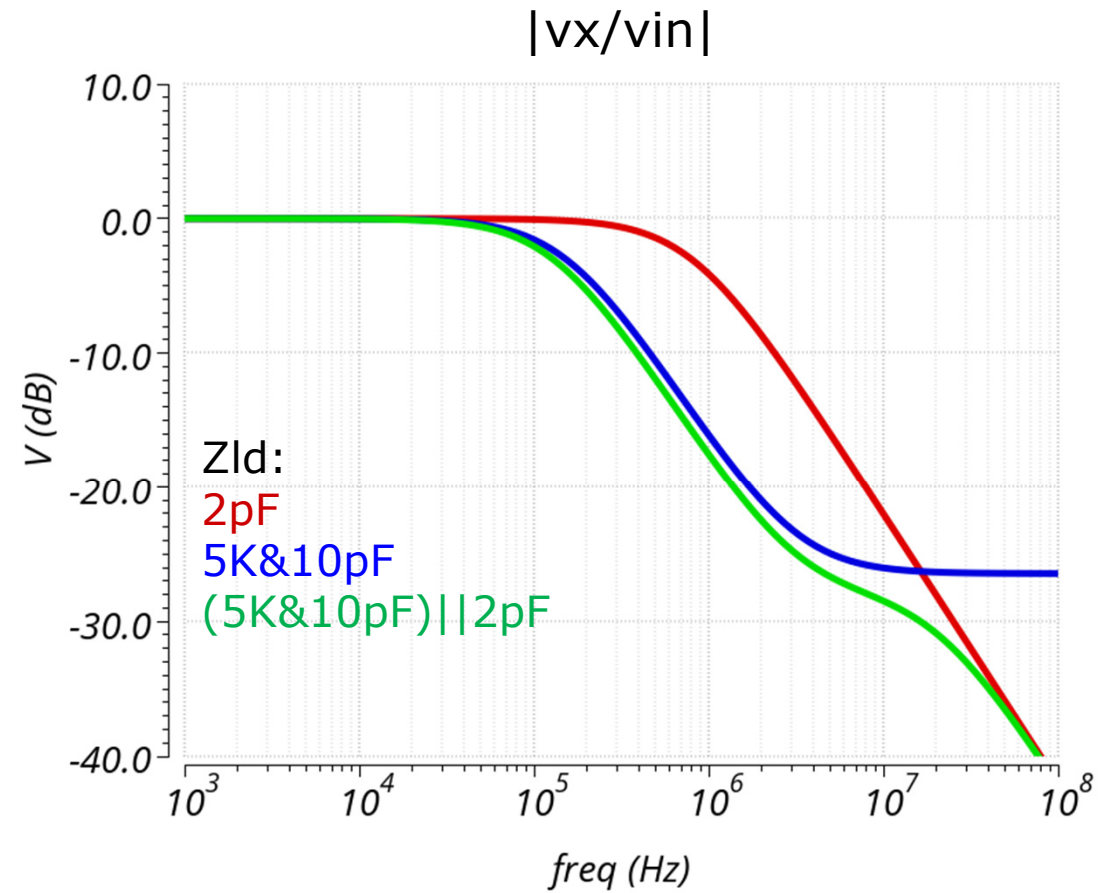
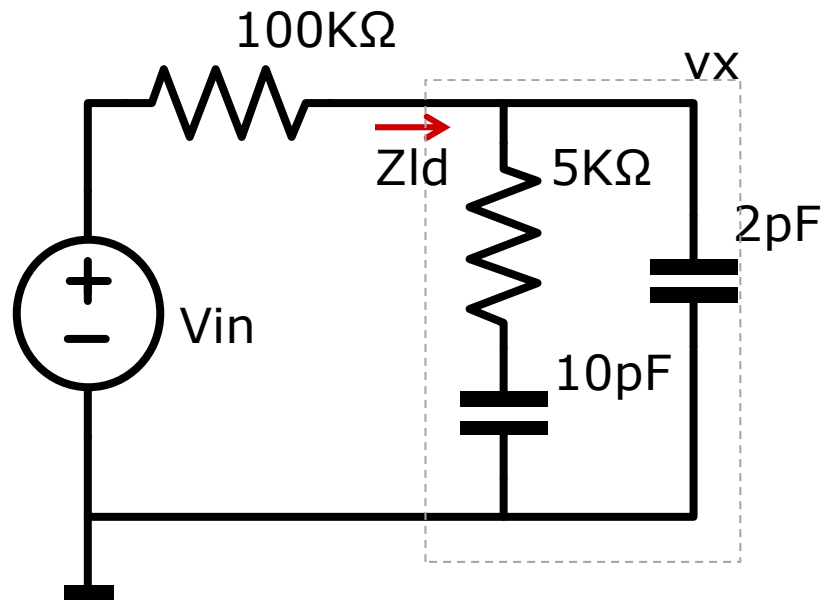


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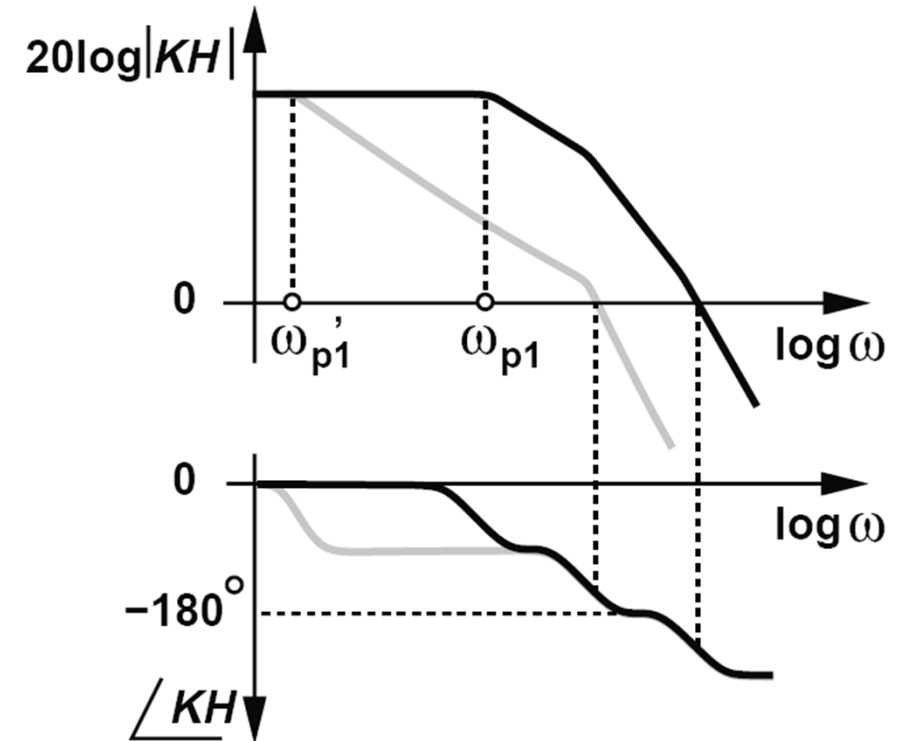
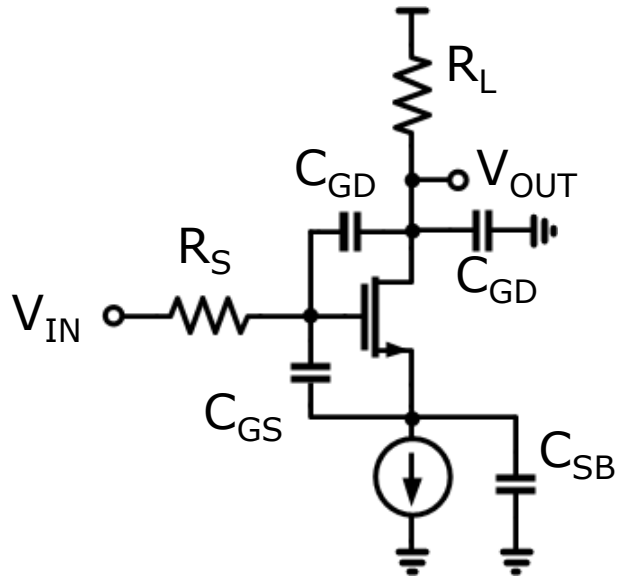
Causes of poles and zeros
Number of gain stages

Trends to Remember



Frequency response of an amplifier^[20]

- Parasitic capacitances as well as multiple gain paths introduce poles and zeros in the frequency response.



Frequency compensation

- Introduces elements in the feedforward or feedback path to ensure needed phase and gain margin for stability
- Reduces dependency on parasitic elements and their variation over process/temperature/supply in the circuit
- Always comes with reduction of power to bandwidth ratio

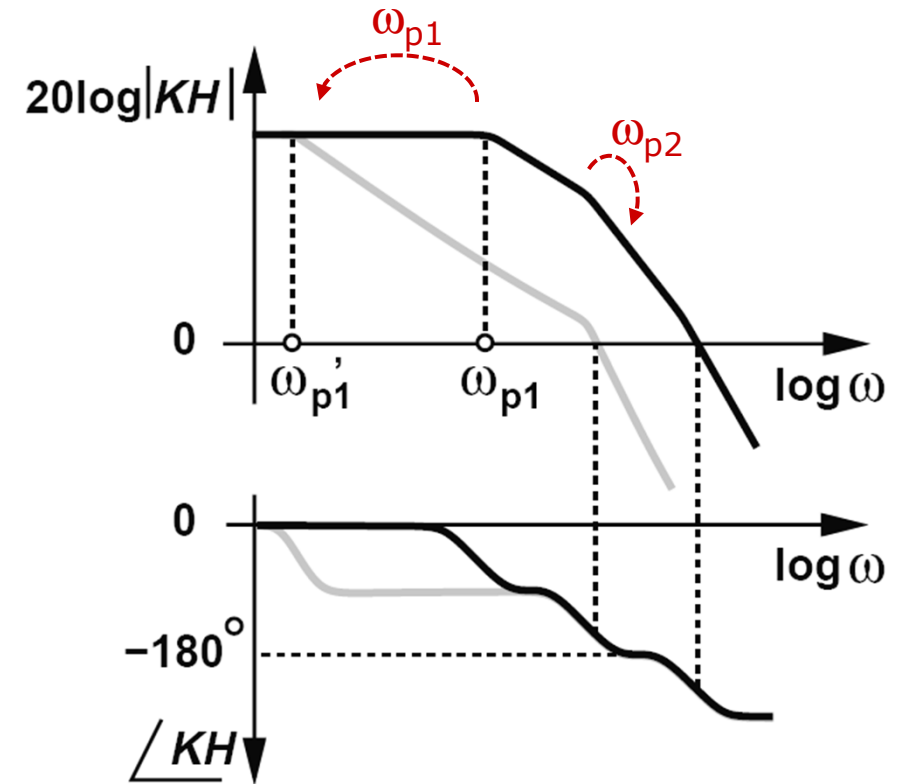
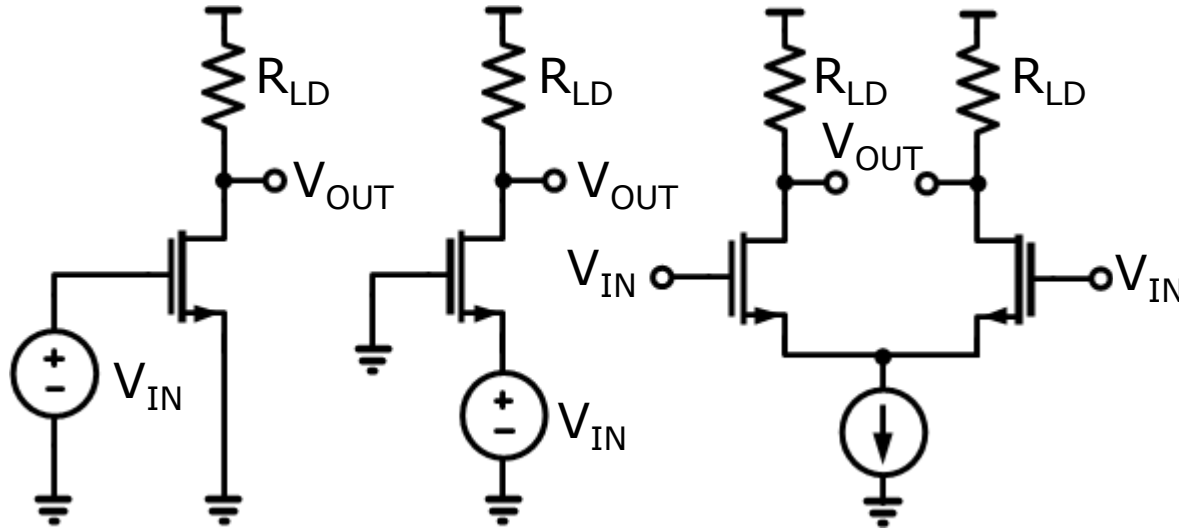


Figure [20]

Amplifiers: Number of stages

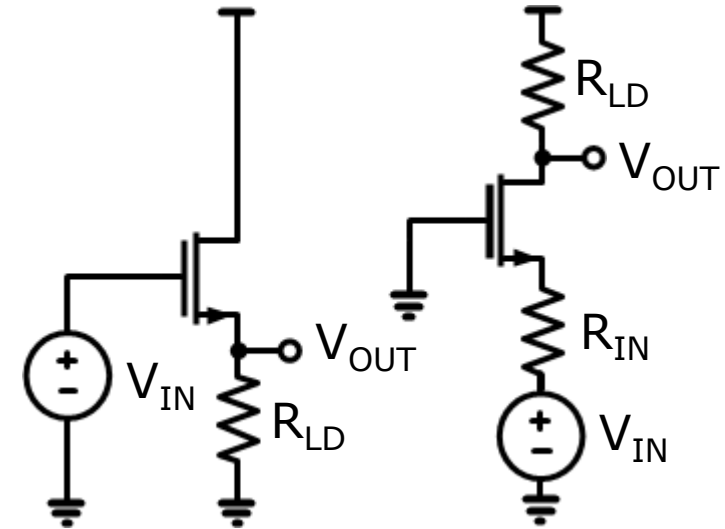
- 1 Stage Amplifiers
 - Simplicity, stability and power efficiency (++)
 - Low gain, high output impedance, limited output range, load sensitivity, limited current drive capability (--)
- 2 Stage Amplifiers
 - Lower output impedance, output range, load insensitivity, input and output ranges decoupled, current drive capability (++)
 - Simplicity, cap load drive, settling time, power efficiency (+)
 - Limited gain, higher offset and noise, distortion, current drive capability (bipolar amplifiers) (-)
- 3 Stage Amplifiers
 - High gain, lower offset and noise, less distortion, current drive capability (bipolar amplifiers) (++)
 - Complexity, power efficiency, large and small signal stability concerns (--)

What is a gain stage in the signal path?



□ Gain stages:

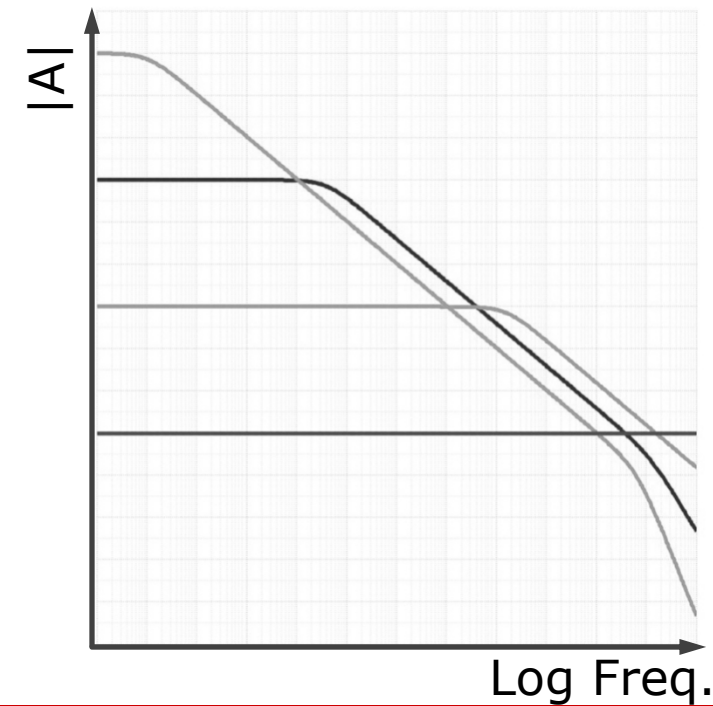
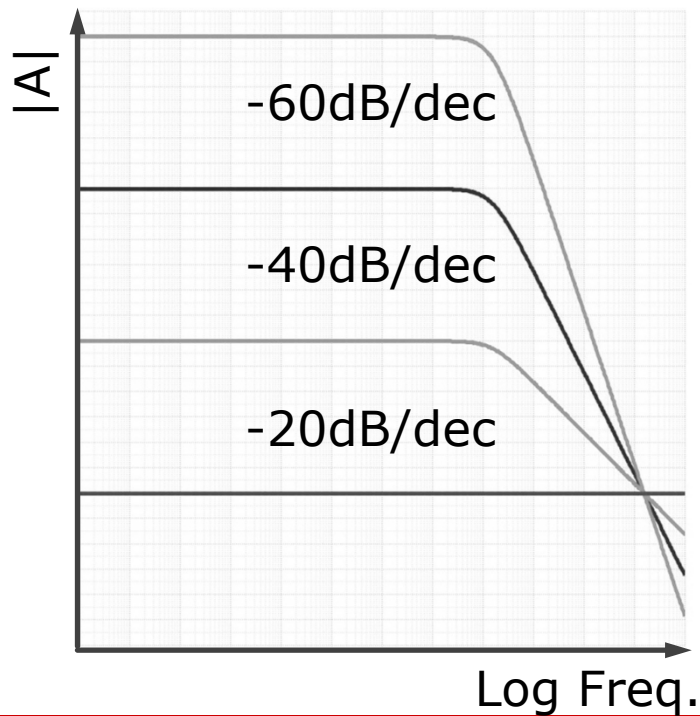
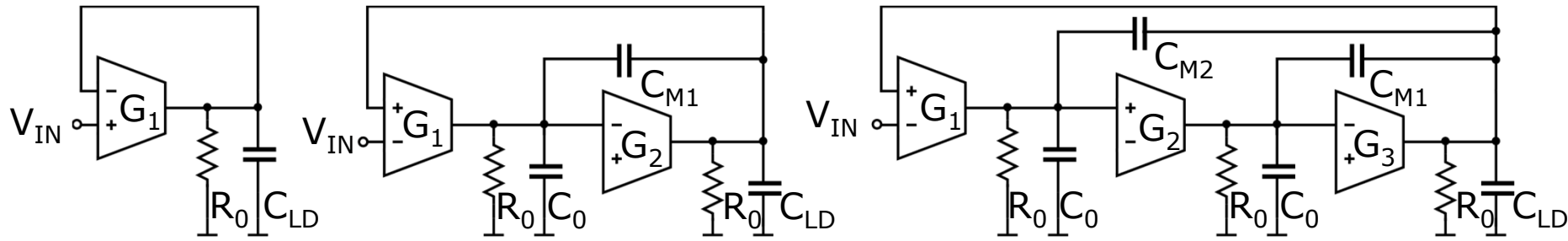
- common source
- common gate if $Z_{in} \ll Z_L$
- differential stage



□ Not gain stages:

- common drain (follower)
- common gate if $Z_{in} > Z_L$

More stages $\rightarrow$ lower achievable bandwidth



Outline

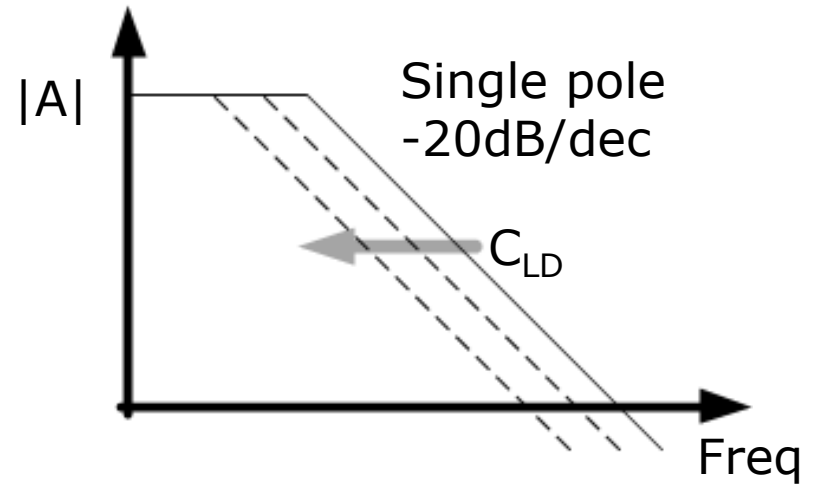
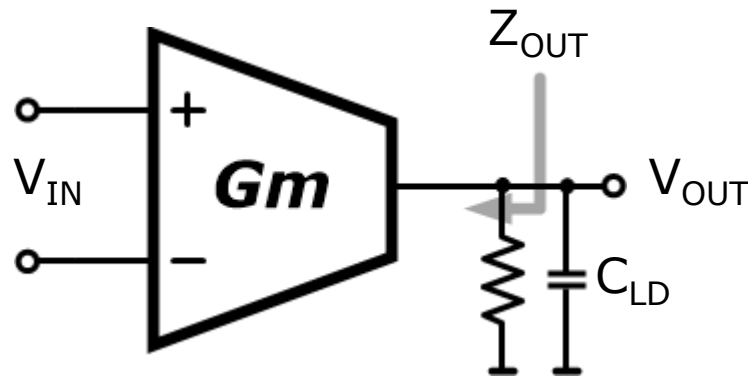
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compensation techniques
2 stage amplifier example

Types of compensation

- ❑ Load Compensation – the load capacitor provides compensation
- ❑ Shunt Compensation – a (large) capacitor to kill gain of a stage
- ❑ Parallel (Lead) Compensation – additional RC in gain path to cancel out pole introduced by load
- ❑ Miller Compensation – using a capacitor across a high gain stage to provide the dominant pole in the frequency response. This has many variants
- ❑ Feed-forward Compensation – using a faster (lower gain) element or stage to bypass high gain stages particular frequencies
- ❑ Countless combination of the above

Load compensation (single stage amplifier)

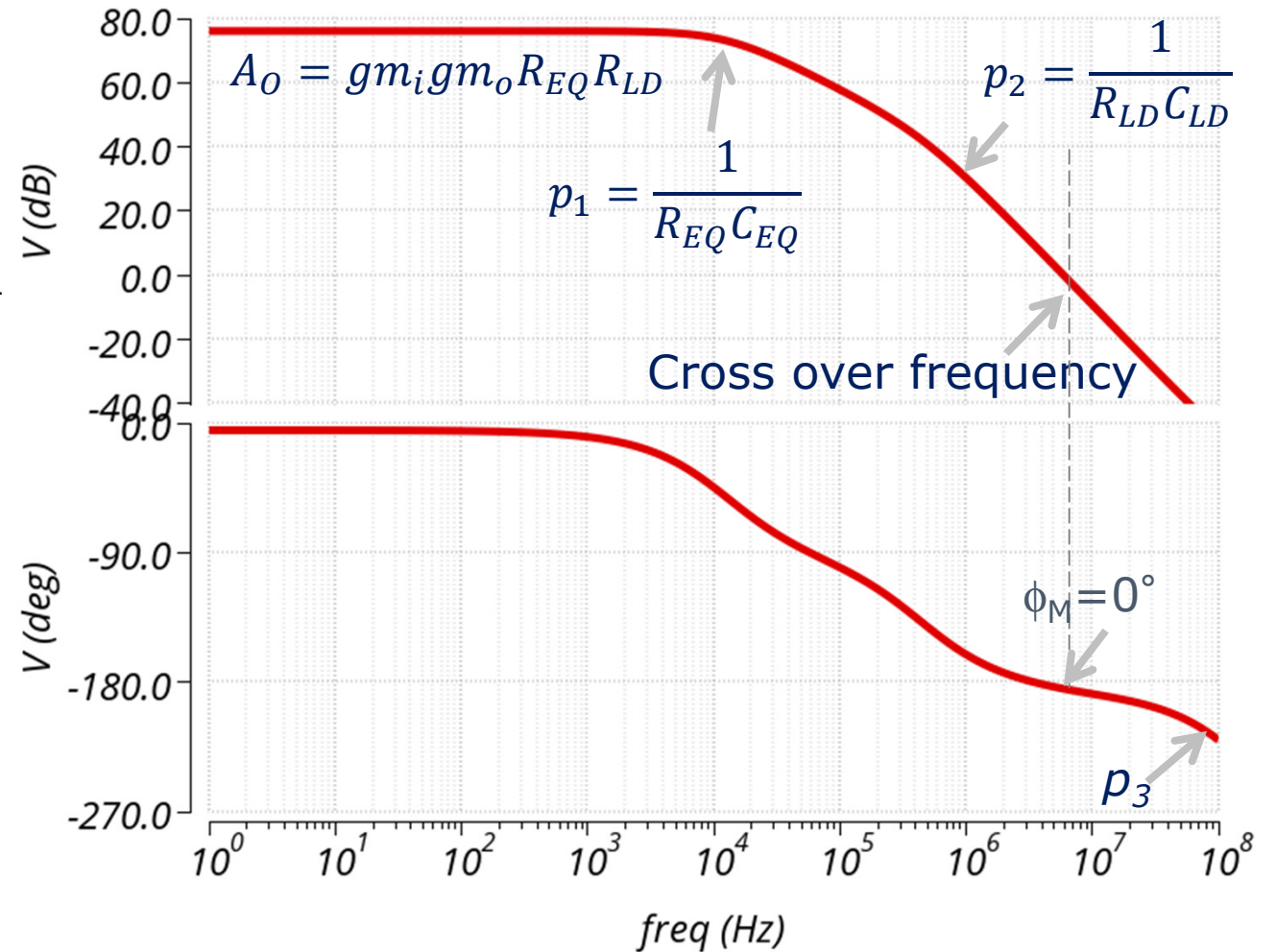
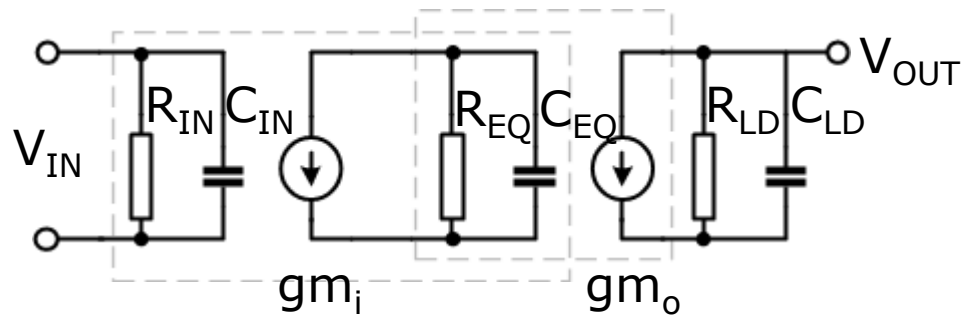
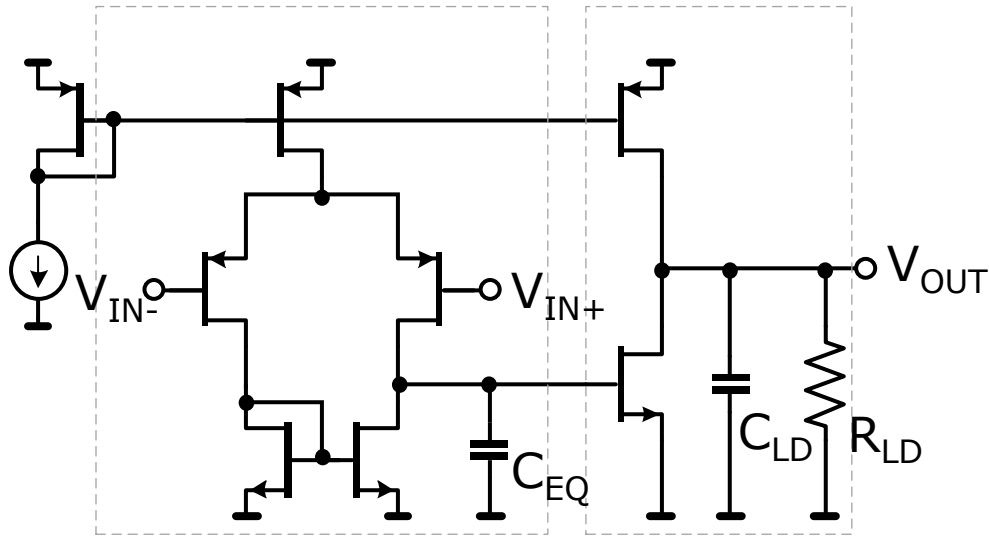


$$A_v = G_m Z_{out} \cong \frac{G_m R_{ld}}{1 + j\omega/\omega_c}$$

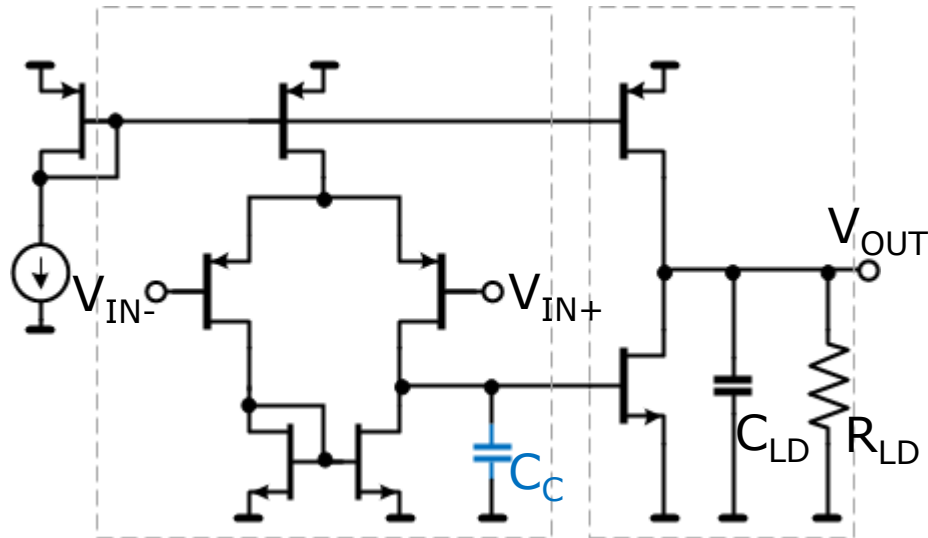
$$\omega_c = \frac{G_m}{C_{LD}}$$

Large load capacitors simply reduce Gain-Bandwidth but the Phase Margin still 90°
Stable but limited gain and output impedance is high.

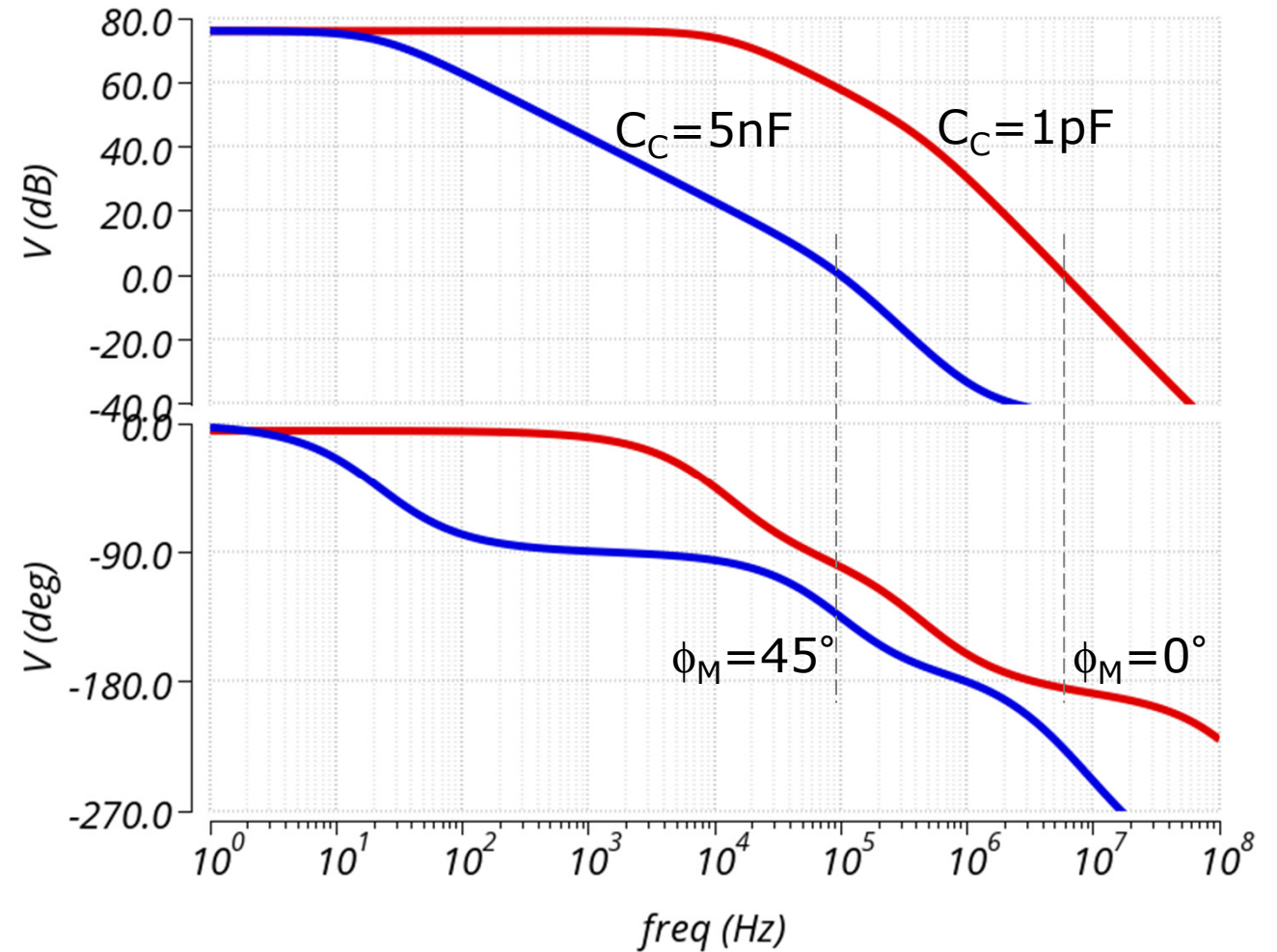
Two stage amplifier: Frequency response



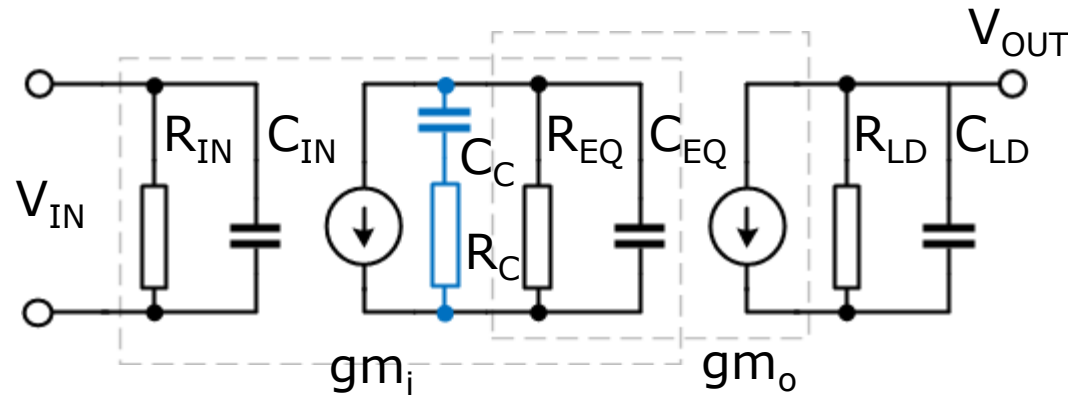
Two stage amplifier: Shunt compensation



Not practical for ICs.



Parallel (lead) compensation [3]



Original poles (w/o comp):

$$p_1 = \frac{1}{R_{EQ}C_{EQ}} \quad p_2 = \frac{1}{R_{LD}C_{LD}}$$

Compensated poles/zeros:

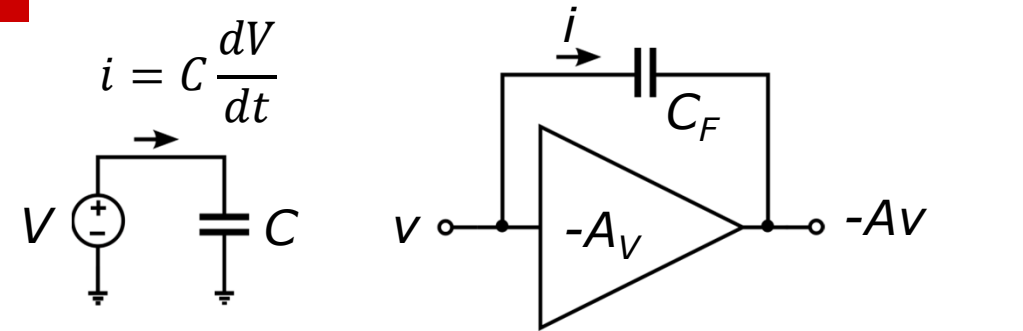
$$p_1 = \frac{1}{R_{EQ}C_C} \quad p_2 = \frac{1}{R_C C_{EQ}} \quad z_1 = \frac{1}{R_{EQ}C_{EQ}} \quad z_2 = \frac{1}{R_C C_C}$$

$$\text{Select } R_C \cdot C_C = R_{LD} \cdot C_{LD}$$

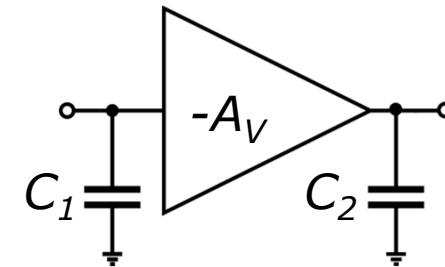
- ❑ Introduces zero that cancels out the pole generated by load
- ❑ Best for power efficiency
- ❑ C_C required can be large
- ❑ Load conditions are not always controllable
- ❑ Error in cancellation introduces pole/zero doublet which can lead to settling delays

Miller Effect

- Identified in 1920 [2]
- Negative feedback with a capacitor as the feedback element
- Impedance seen at input is multiplied by gain
 - Gain is also function of frequency
 - A capacitor is effectively a short at high frequencies $\Rightarrow$ phase is inverted (negative to positive)

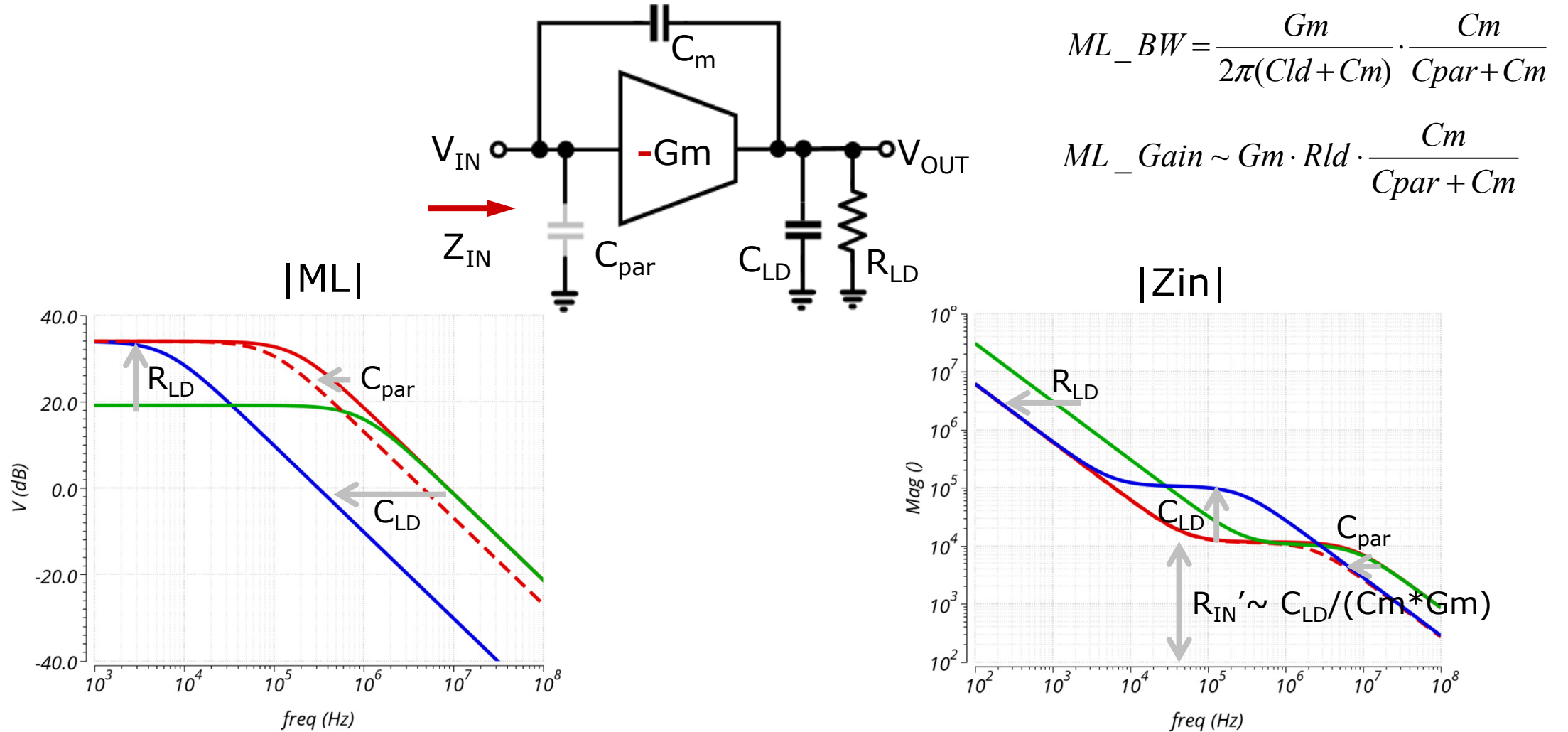


$$i = C \frac{d[v - (-A_V v)]}{dt} = C \frac{dv[1 + A_V]}{dt} = C_M \frac{dv}{dt}$$

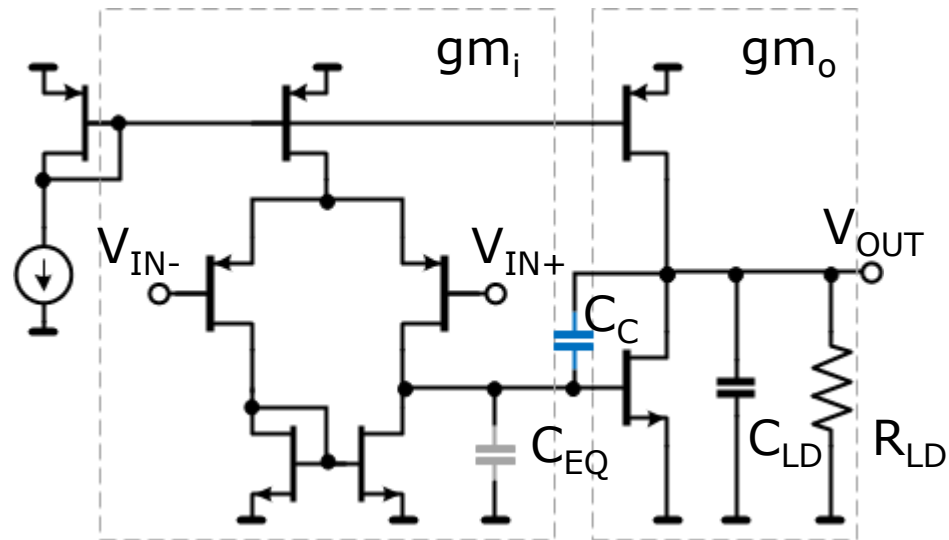


$$C_1 = C_F(1 + A) \quad C_2 = C_F \frac{(1 + A)}{A}$$

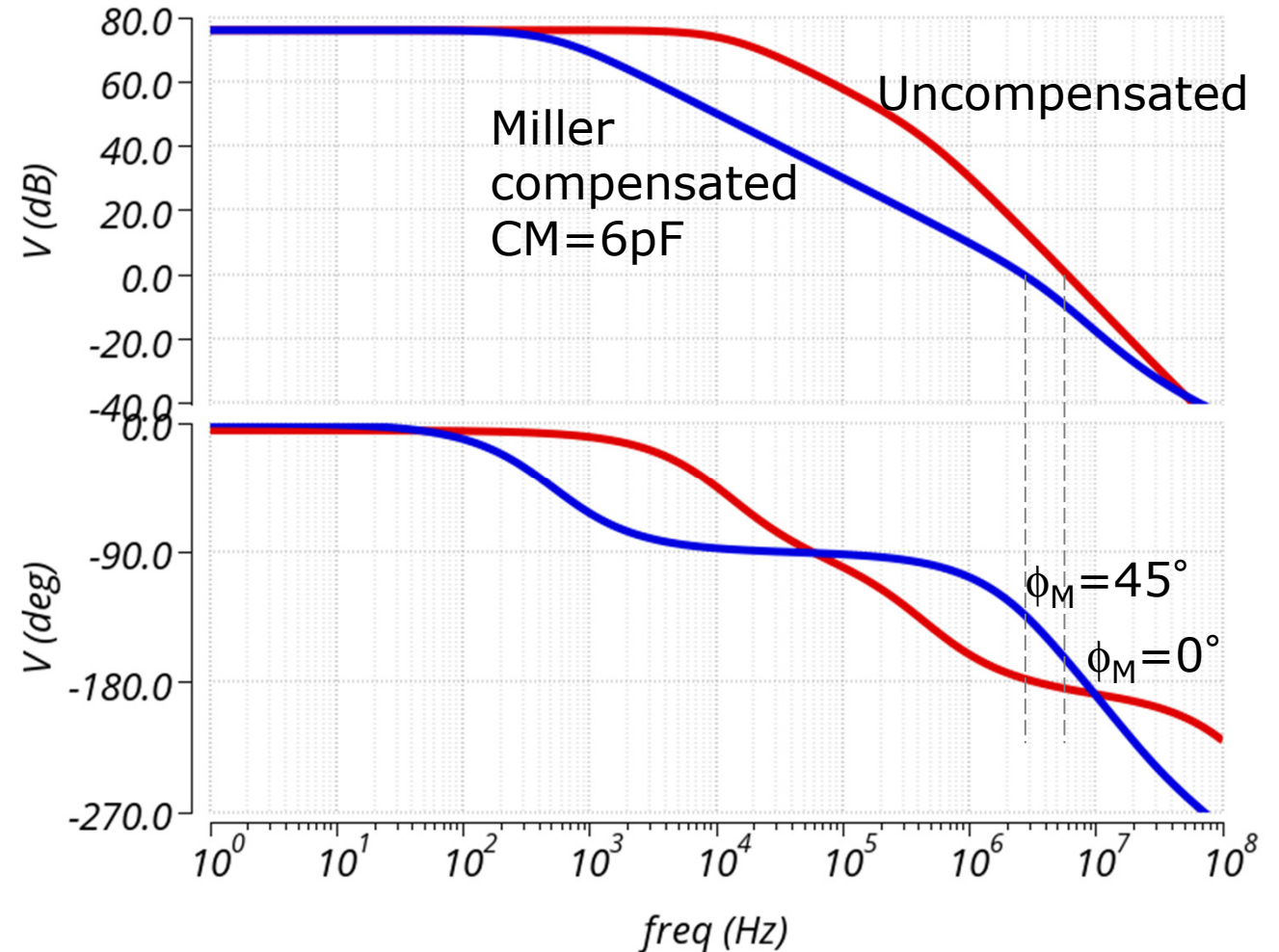
Miller Capacitor



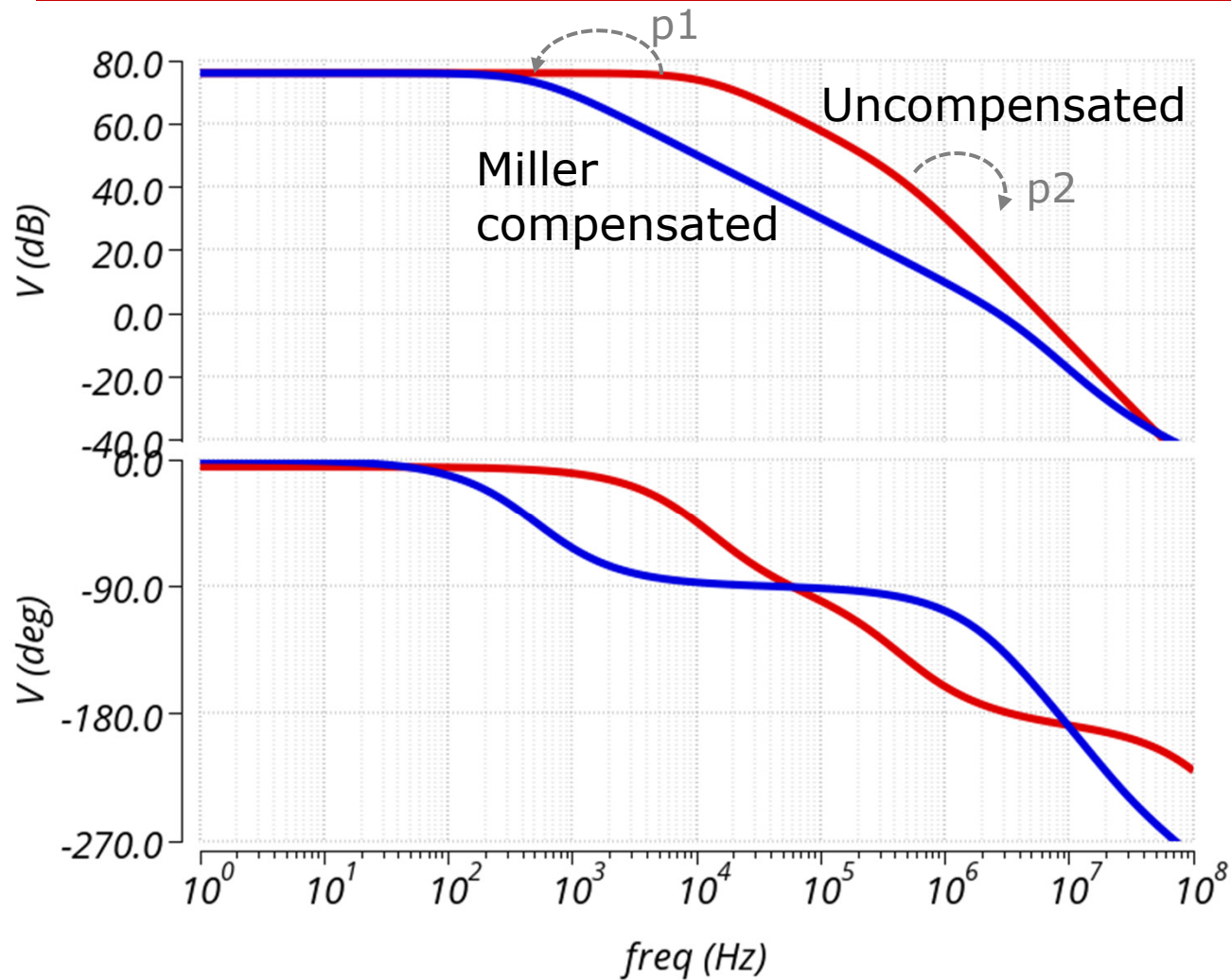
Two stage amplifier: Miller compensation



Very effective. Most commonly used compensation technique.



Two stage amplifier: Miller compensation



Poles moved apart due to Miller Compensation

Dominant pole:
$$p_1 = \frac{1}{gm_o R_{EQ} R_{LD} C_C}$$

Output pole:
$$p_2 = \frac{gm_o}{C_{LD}}$$

But introduced right half plane zero that increases magnitude and reduces phase:

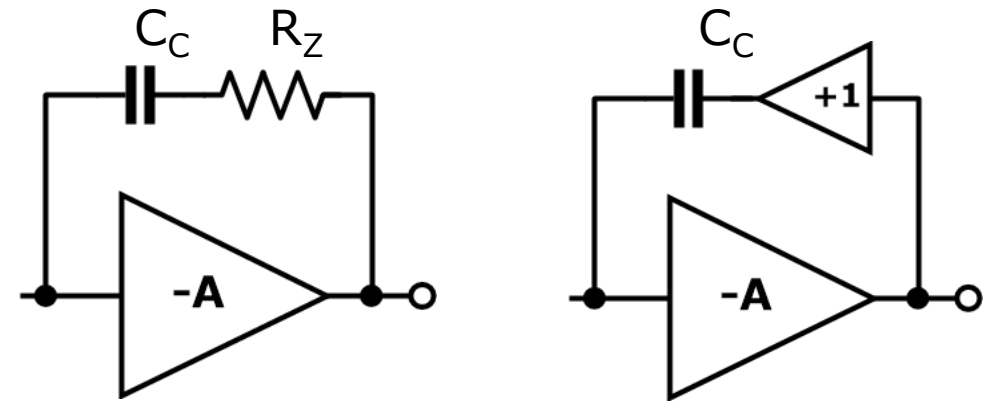
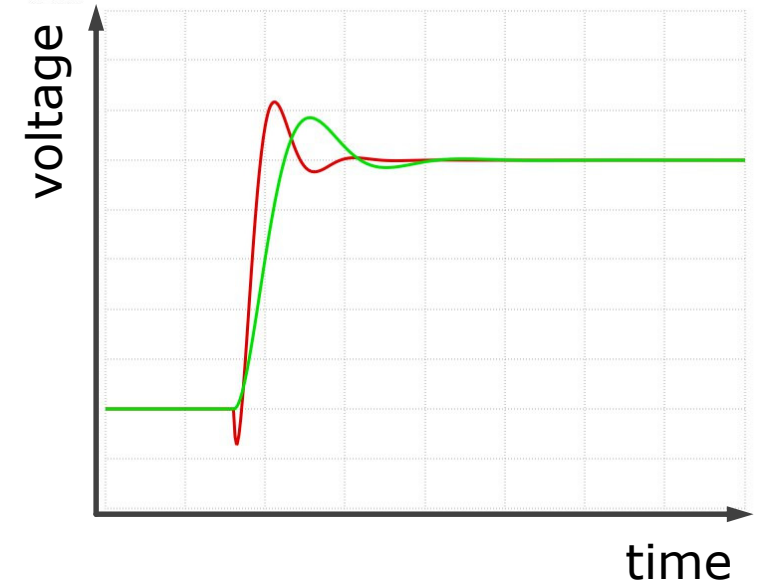
$$z_1 = \frac{gm_o}{C_C}$$

Unity gain bandwidth:
$$\omega_c = \frac{gm_i}{C_C}$$

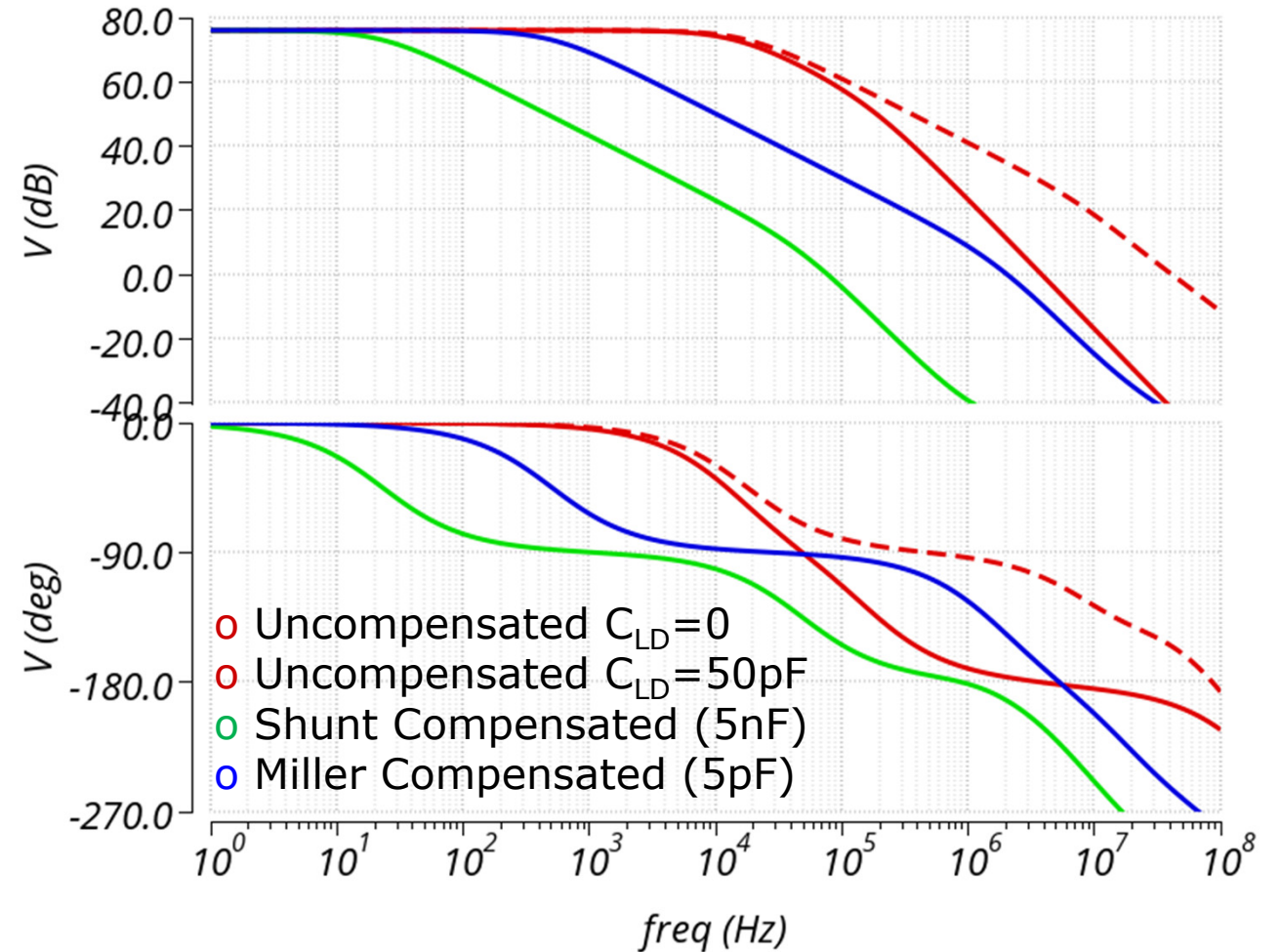
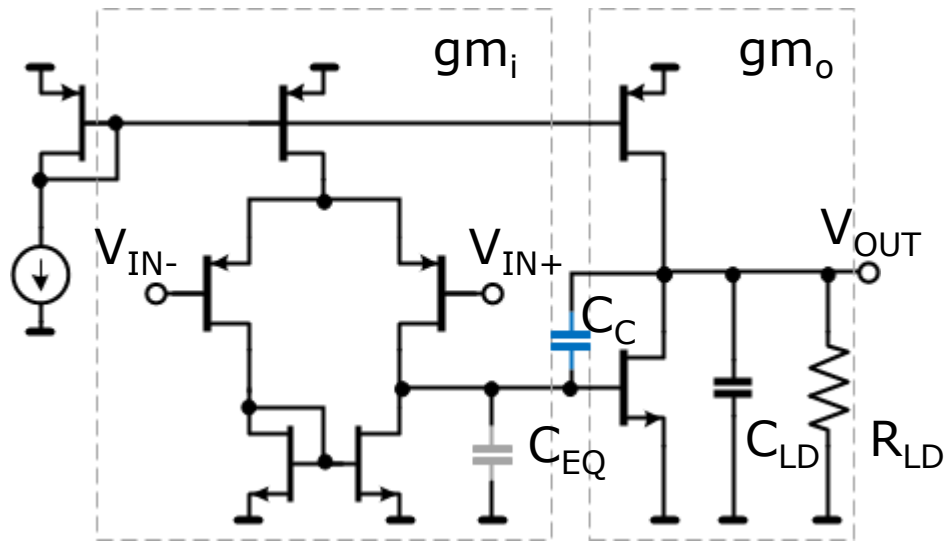
Methods to eliminate the RHP zero

- There are two parallel paths two output $\Rightarrow$ RHP zero in frequency response. RHPZ provides gain but phase loss $\Rightarrow$ strong degradation of stability
 - Can use a buffer to eliminate feed-forward action. Output impedance of buffer is not negligible.
 - Can add a resistor to cancel out the zero. Typically select R_z to be equal to output transistor transconductance.

$$z_1 = \frac{1}{C_c \left(\frac{1}{g_{m_o}} - R_z \right)}$$



Two Stage Amplifier: Compensation Comparison

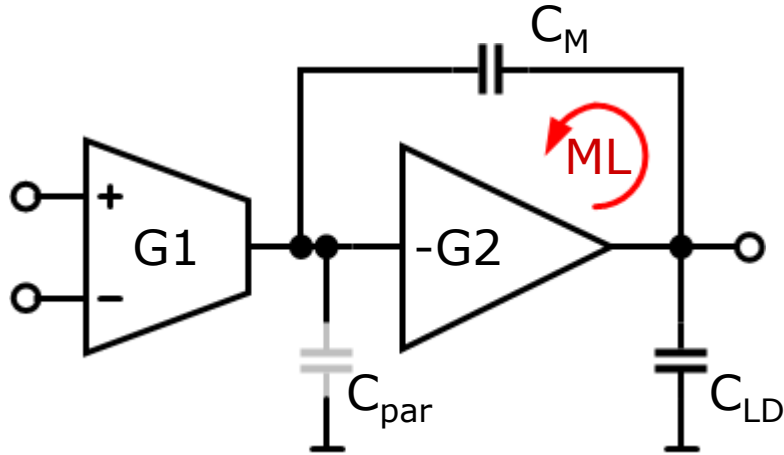


Outline

- Introduction and background
- Refresher on feedback theory, bode plots
- Types of frequency compensation
- A practical design example of a two stage amplifier
- Alternative (advanced) compensation schemes
- Practical tips and suggestions

Nested Miller
Multipath nested Miller
Advanced Miller topics
Useful combinations

Two Stage Amplifier Miller Comp (revisited)

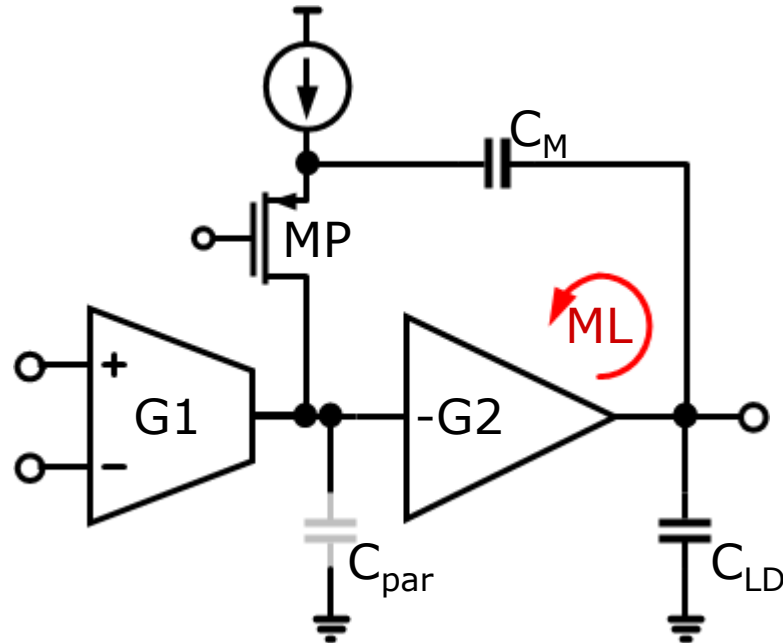


- Single dominant pole response set by C_M as long as Miller loopgain (ML) is above 1
- Second pole, at output node:
 $\omega_2 \sim G2 / (C_{LD}(1 + C_{par}/C_M))$
- Load capacitance and parasitics impact ML bandwidth! Make $C_M > C_{par}$ to maximize.
- Feedforward causes RHP zero at $\omega_z = G2/C_M$
- Load resistance lowers Miller-loop gain! Keep ML_{Gain} above 1

$$ML_{Bandwidth} \sim \frac{G2}{2\pi(C_{LD} + C_M)} \cdot \frac{C_M}{C_{par} + C_M}$$

$$ML_{Gain} \sim G2 \cdot R_{LD} \cdot \frac{C_M}{C_{par} + C_M}$$

Cascoded Miller [8,9]

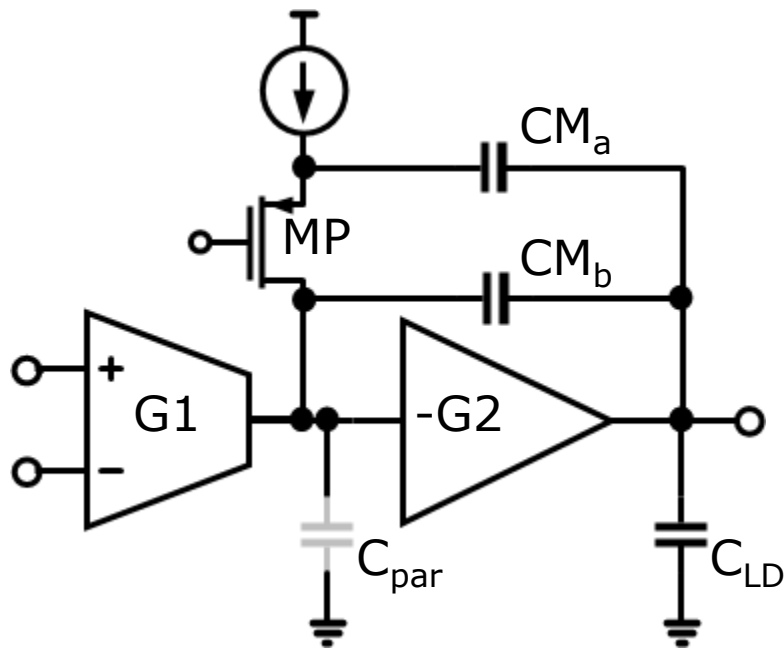


$$ML_{Bandwidth} \sim \frac{G2}{2\pi(CLD + CM)} \cdot \frac{CM}{Cpar + \cancel{X_M}}$$

$$\omega_z = \frac{gm_{MP}}{2\pi \cdot C_M}$$

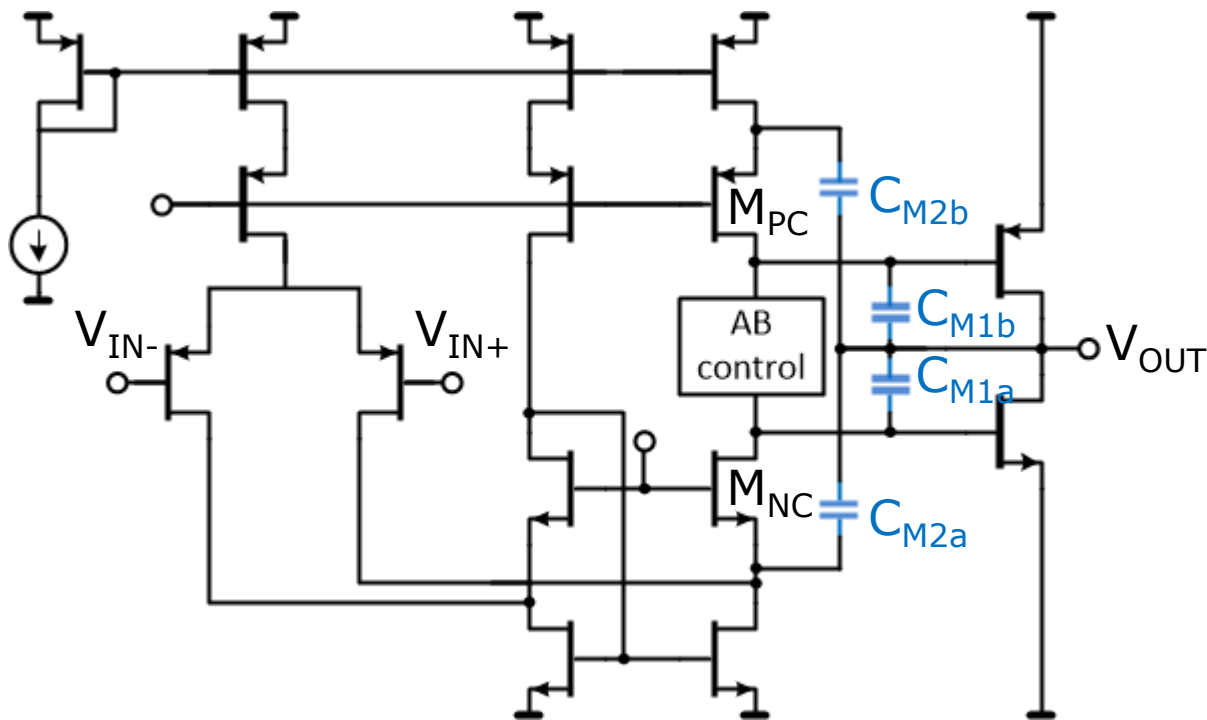
- ❑ Adds gain to Miller Loop $\Rightarrow$ much improved load capacitance drive
- ❑ Blocks the RHP zero
- ❑ Cascode gm (MP) must be higher than $G1$ to be effective
- ❑ Adds LHP zero (improves phase margin)
- ❑ Can have stability issues when $C_{LD} = C_M$ (poor damping in Miller-loop)
- ❑ Watch for stability during slew (avoid operating MP w/o current)

Split Cascoded Miller



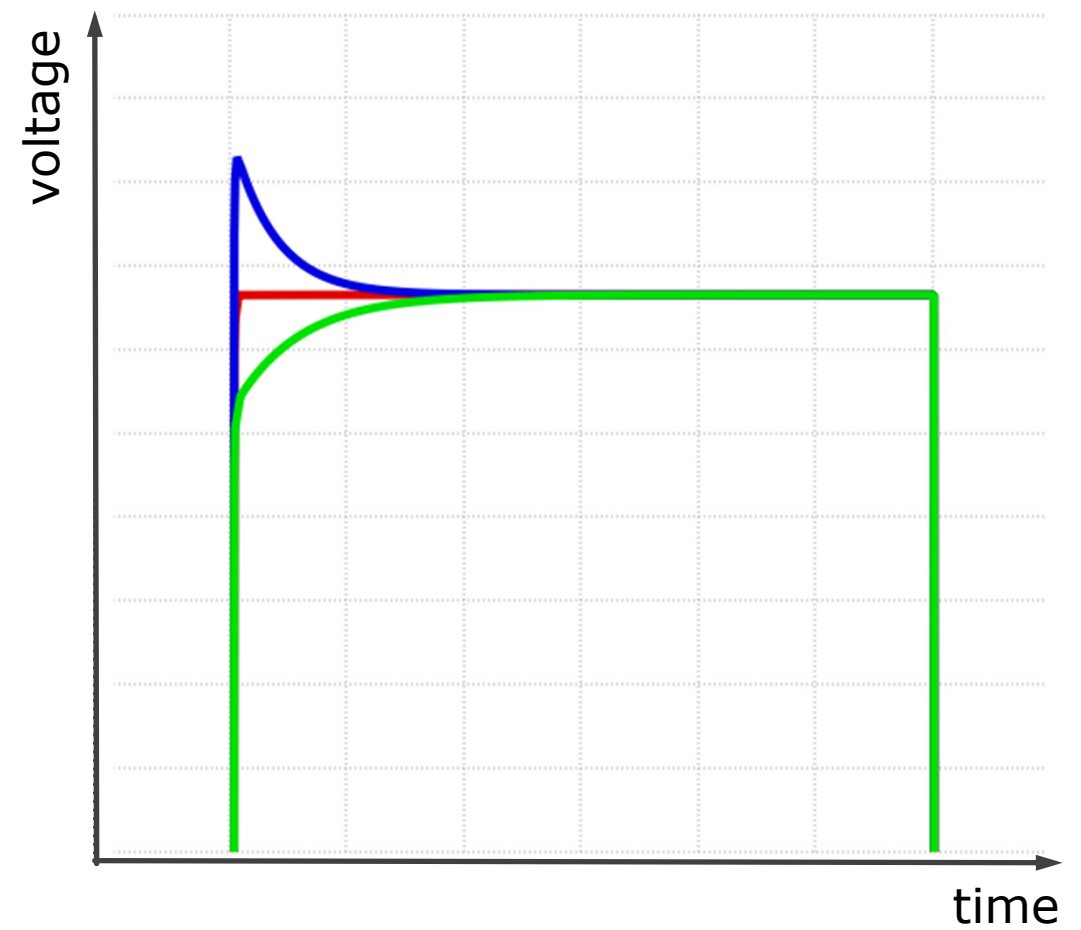
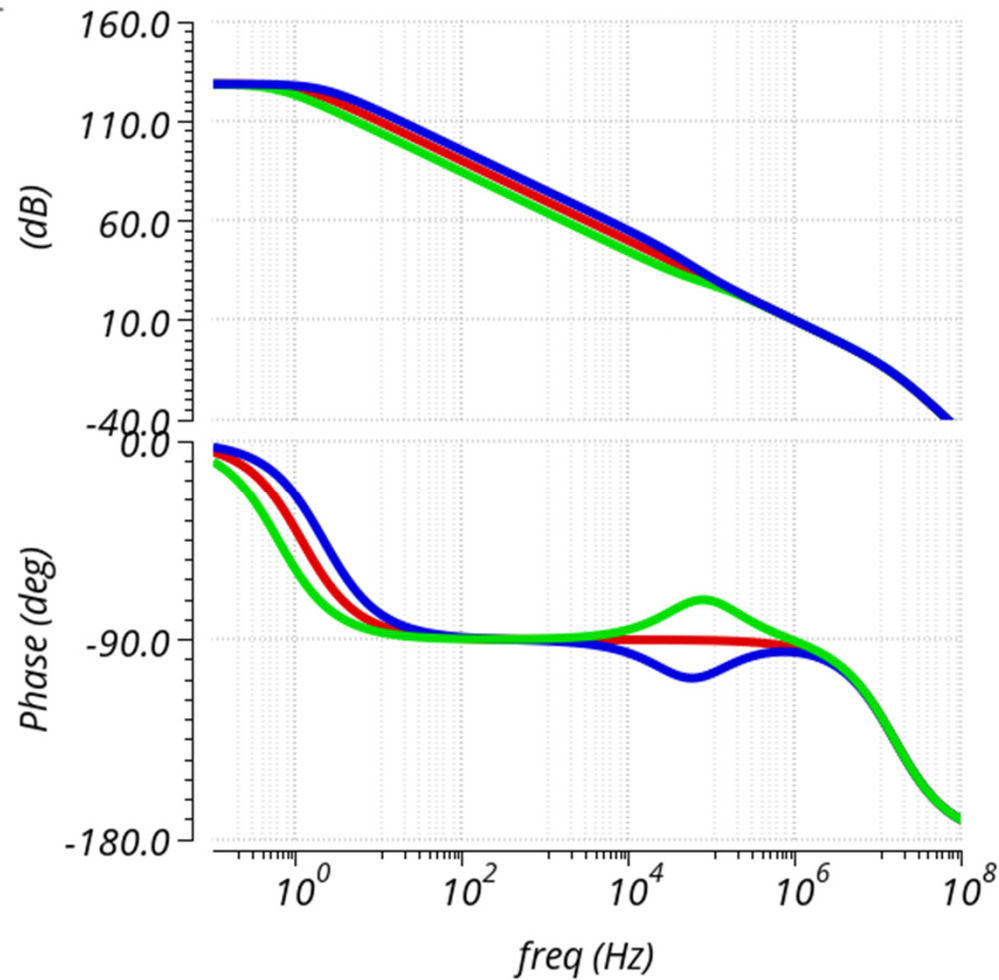
- Improved stability at $C_{ld}=CM$
- $CM=CM_a+CM_b$
- CM_a/CM_b typically around 1 to 6
- CM_b reduces cap load drive but improves Miller loop damping at small C_{LD}
- Feedback around cascode (to increase gm) can eliminate need for CM_b , but requires extremely high BW loop and stability during slewing not addressed

“Hybrid” Cascoded Miller

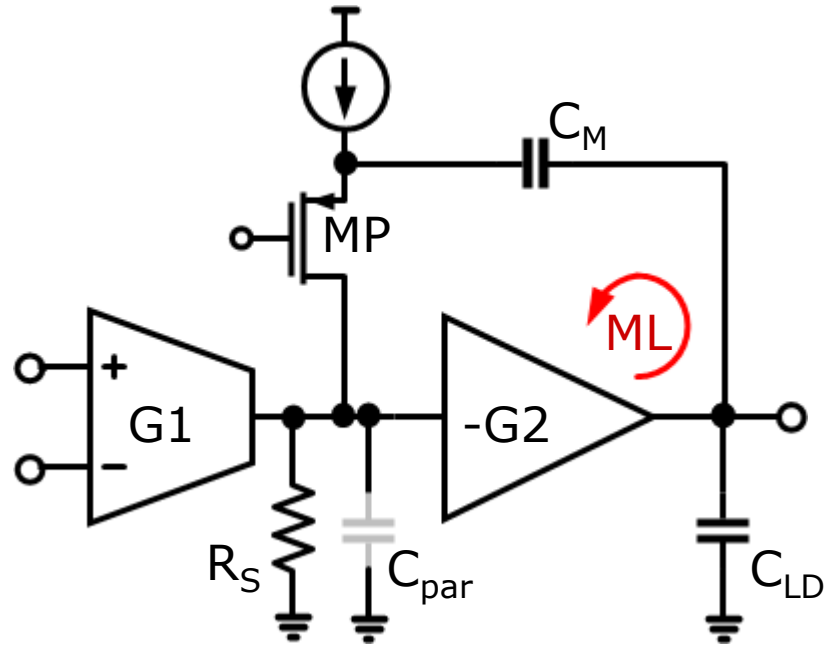


- Increased (2x) effective G_m
- Cascode $G_m > \text{input } G_m$
- $CM2x > CM1x$ for improved Cld drive
- Avoid LHP zero by $CM2a=CM2b$ and $Gm_{MNC}=Gm_{MPC}$
- Pole/zero doublets can exist if not met
- Must be evaluated for different load capacitances and load currents!

Pole-Zero Doublets

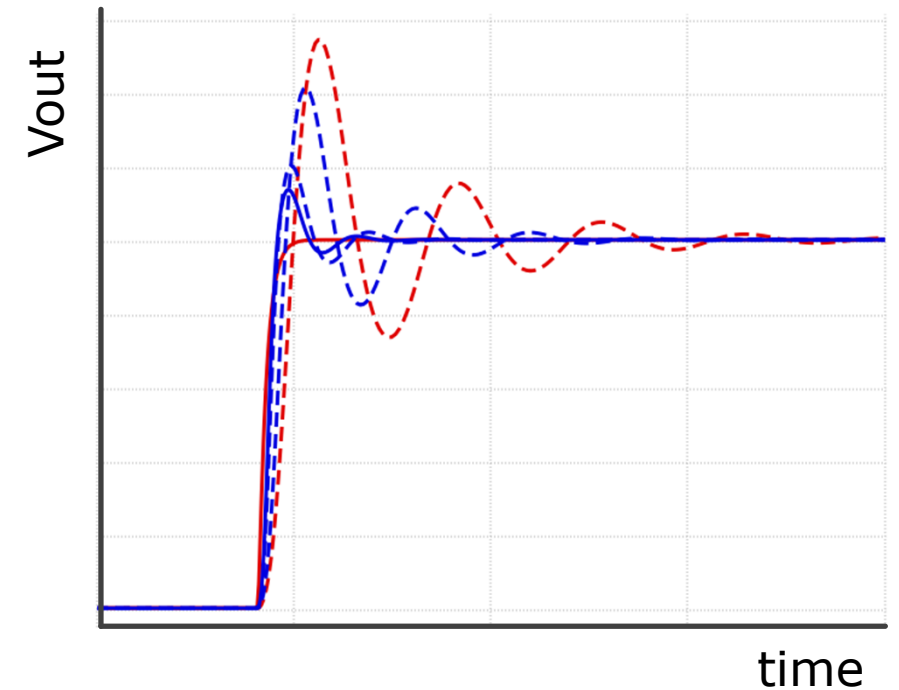
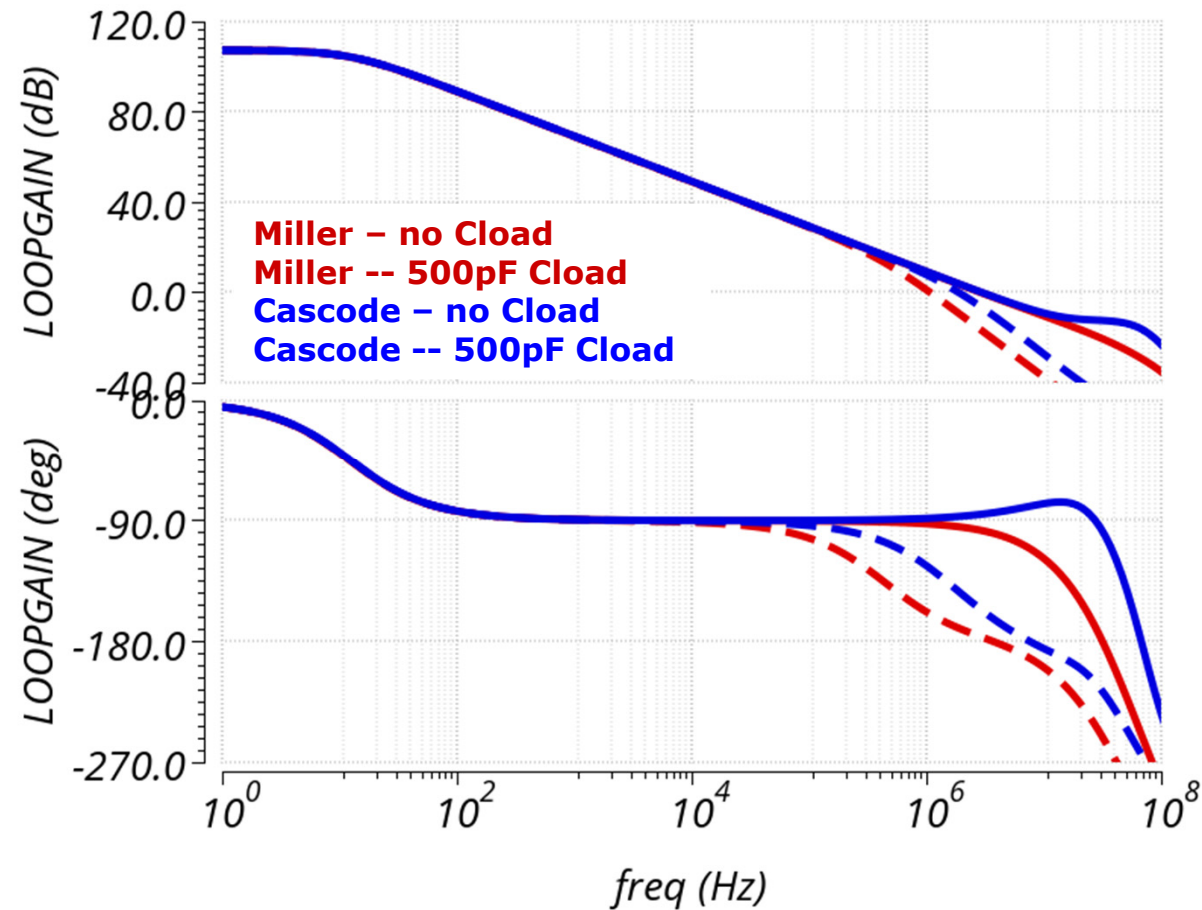


Cascoded Miller: Reduce Gain [10]



- ❑ Reduce 1st stage gain by R_S
- ❑ Cascoded Miller reduces cap loading on first stage and increases Miller loop gain
- ❑ Cld sets dominant pole at high C_{LD} and BW is reduced
- ❑ Can have issue to drive resistive loads (not enough gain)
- ❑ Watch for stability issues at $C_{LD}=C_M$!
- ❑ Many examples in literature to control impedance over frequency of internal high impedance node [10,12,13,14 ..]

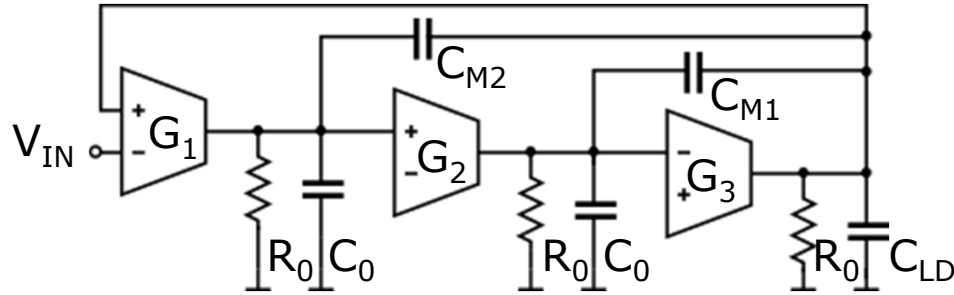
Cascoded Miller Comparison Simulations



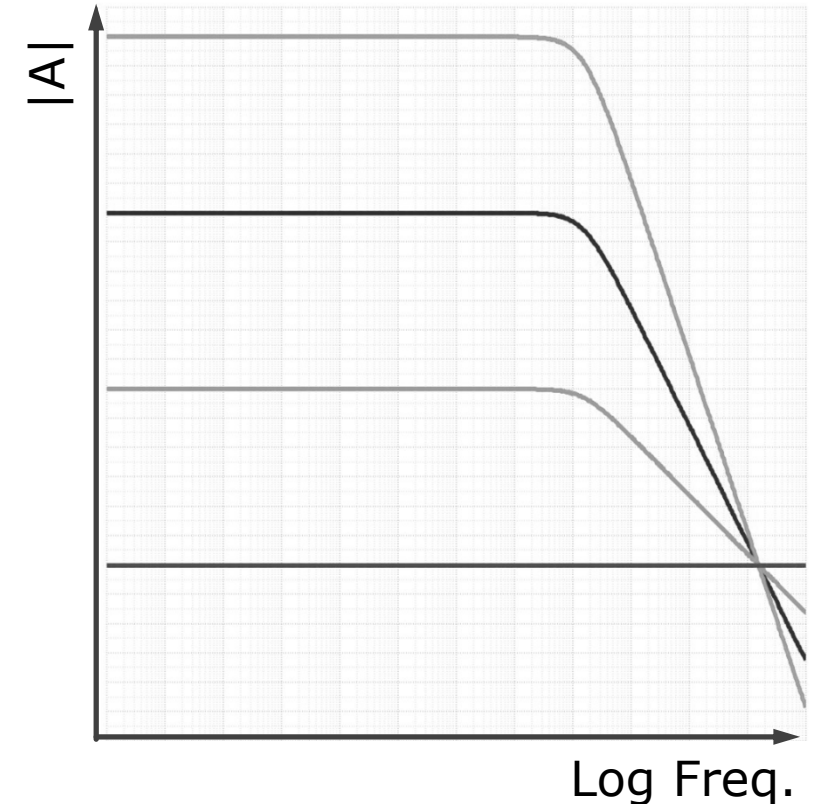
Miller -- no Cloud
Miller -- 500pF Cloud
Cascode -- 500pF Cloud
Hybrid Cascode 0.5:0.5 -- 500pF Cloud
Hybrid Cascode 0.9:0.1 -- 500pF Cloud

Need more gain

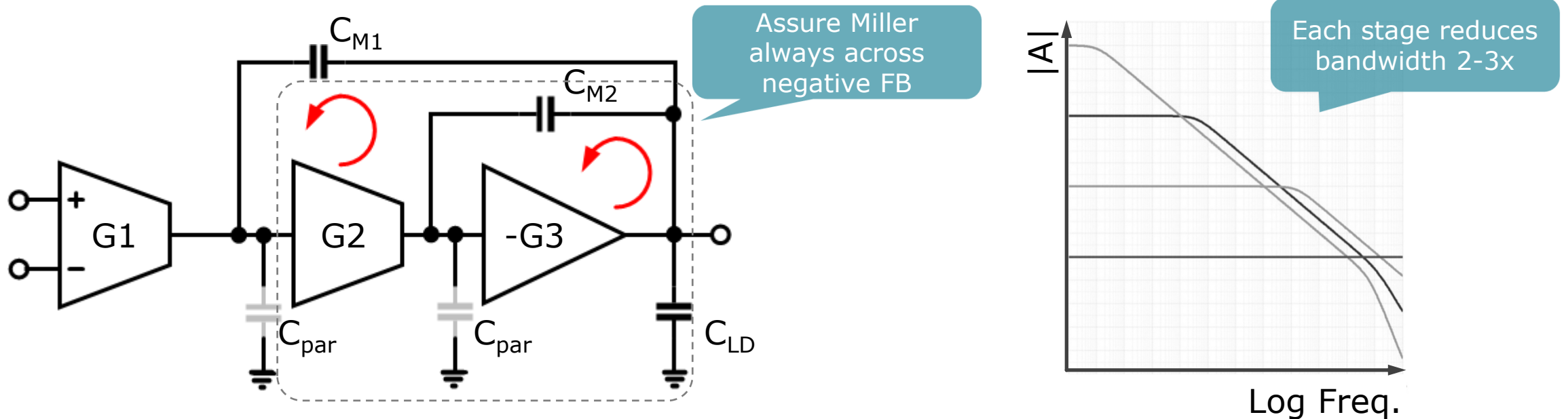
Additional gain stages



- 3 stages provide more gain, lower offset, wider input / output ranges, more current drive capability and lower distortion
- They also lead to 3 high frequency poles that need to be compensated
- For power efficiency always explore adding gain to existing stages (i.e. gain boost) before adding additional stages
- More difficult to achieve in bipolar amplifier especially if need rail-to-rail operation (R_{π})

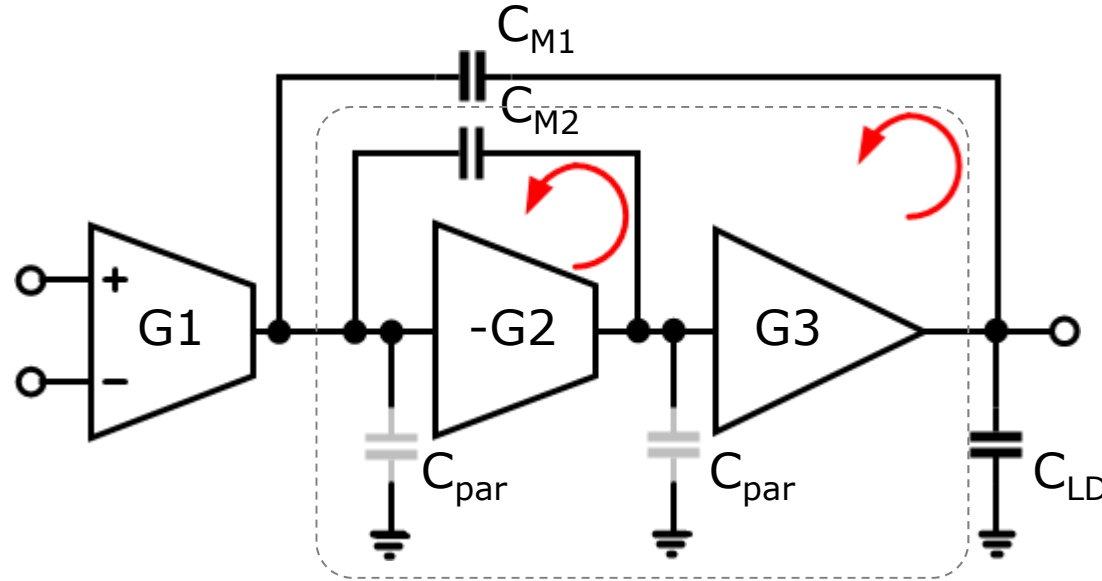


Nested Miller Compensation



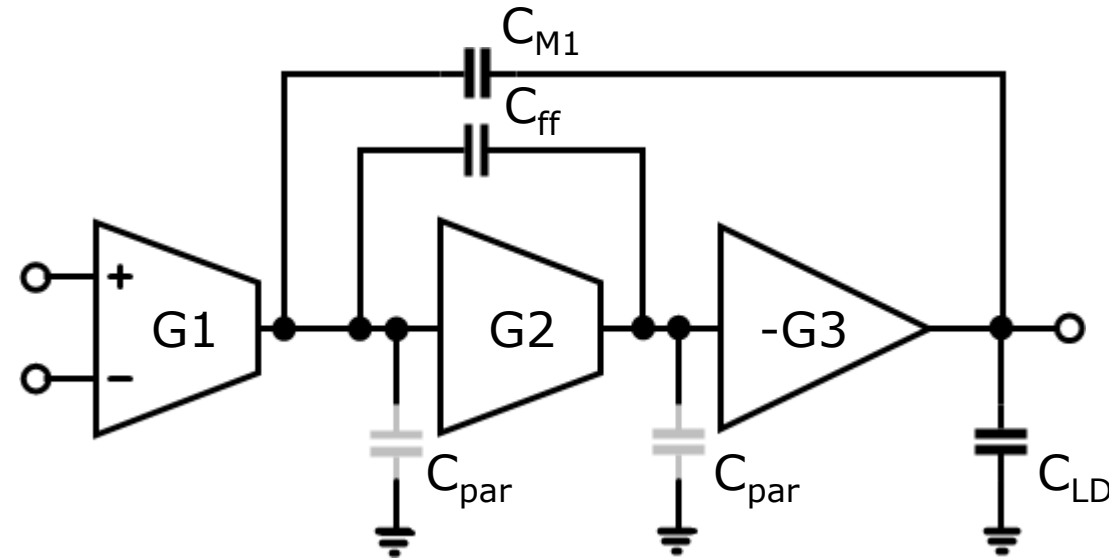
- 3 stages $\Rightarrow$ 3 high frequency poles $\Rightarrow$ "Nested Miller compensation"
 - Set inner loop (G_2 - C_{M2} - G_3) $\sim 2x$ amplifier bandwidth to assure C_{M1} ML gain.
 - Output pole (G_3/C_{LD}) should be yet $\sim 2x$ higher than that
- Concept extends to 4 or more stages, with BW of the inner loops $\sim$ doubling with the addition of each gain stage

Reverse Nested Miller Compensation



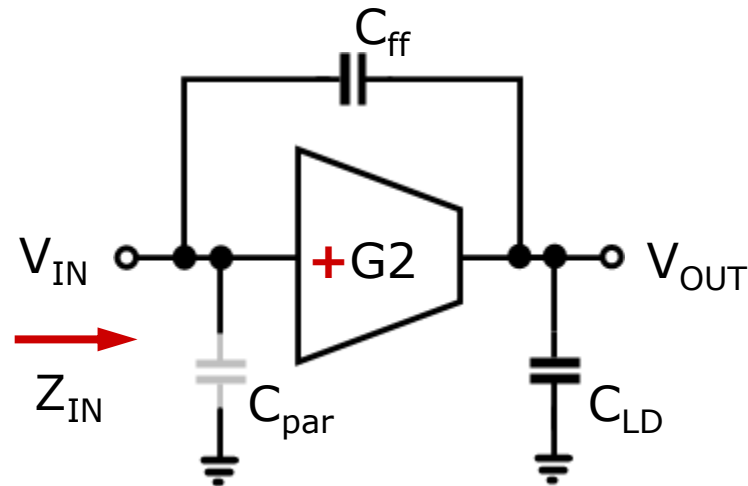
- ❑ Good for power efficiency
- ❑ Strong C_{LD} dependency due to limited gain in G_3
- ❑ Loading effects can be mitigated by isolation resistors
- ❑ Non-inverting output stage!

Feed-forward Compensation

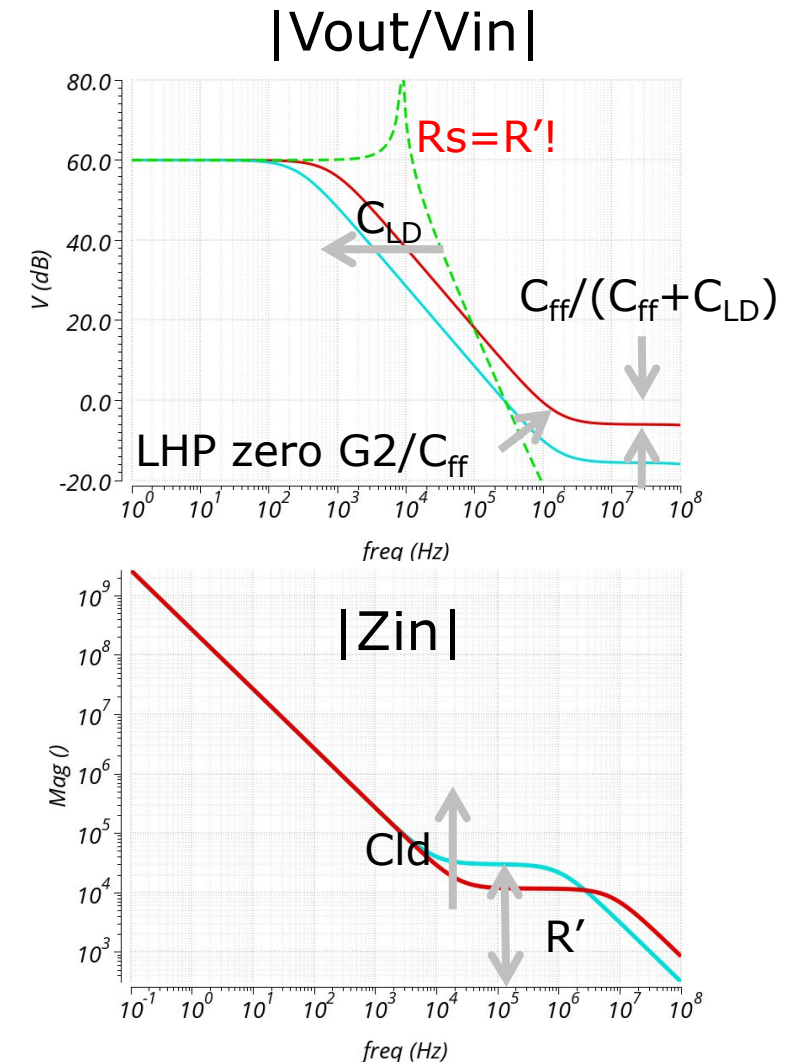


- C_{ff} bypasses G2 at high frequencies
- Impedance looking into second stage is negative!
- Look for distortion effects

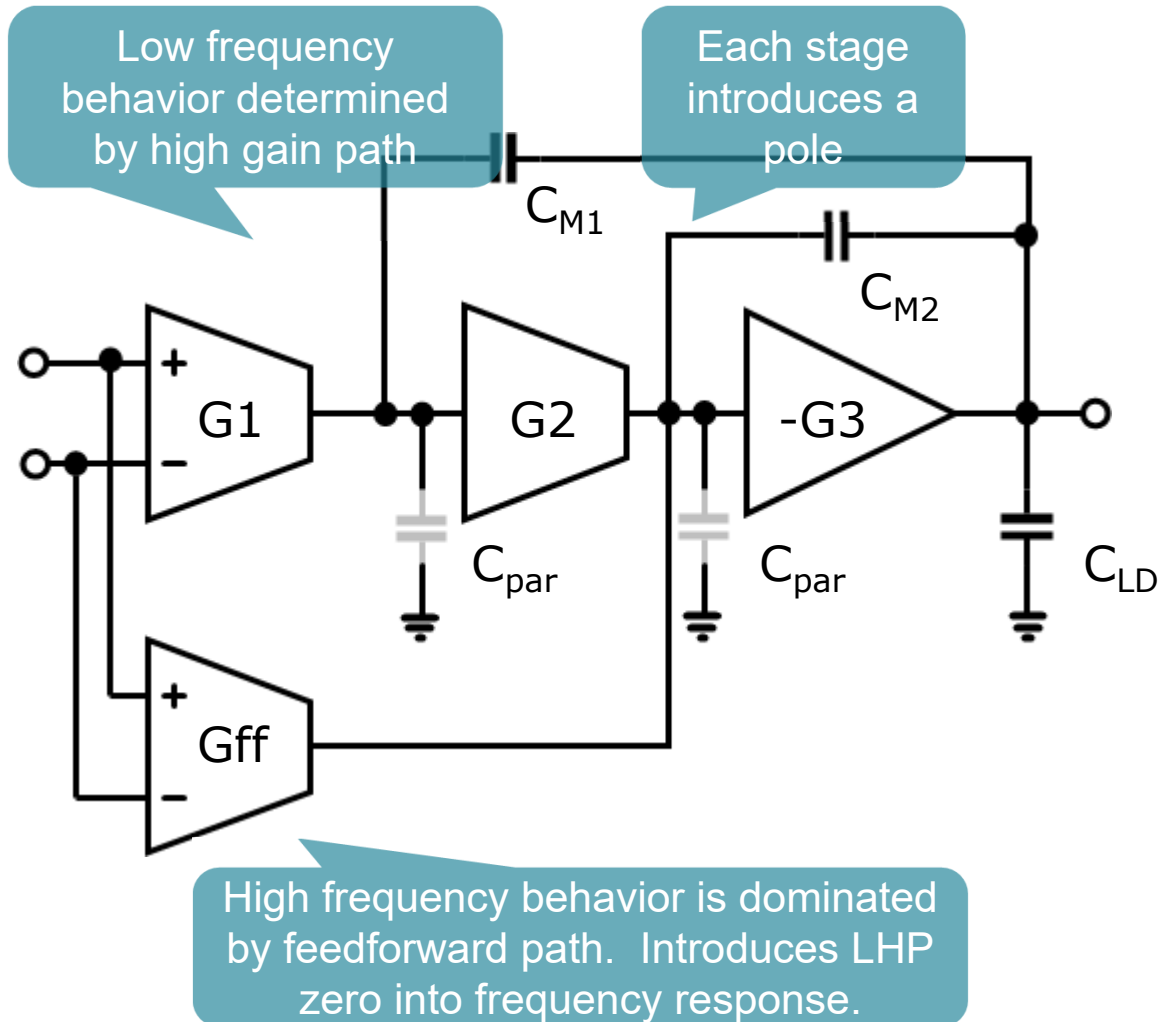
Feed-forward around a gain stage



- Also known as a negative impedance converter
- $R' = -(C_{ff} + C_{LD}) / (C_{ff} * G2)$
- Source impedance must be larger than $|R'|$



Multipath NM (Feedforward) Compensation [3,4]



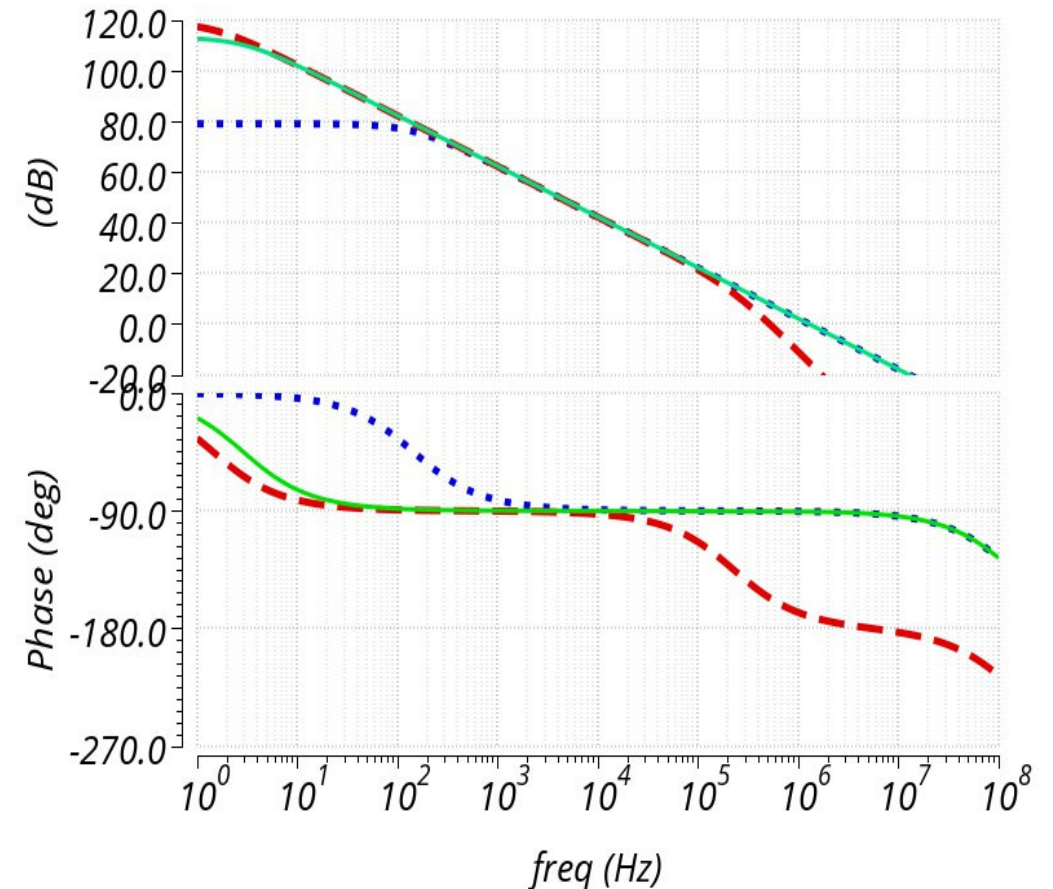
- Combines the dynamic response and speed-to-power ratio advantage of a two-stage structure with the additional gain and better distortion of a three-stage amplifier
- High frequency response dominated by feed-forward stage; offset and offset drift by the high gain path
- Can be extended to multiple stages
- Watch for large signal stability issues!

Multipath Nested Miller Compensation

- ❑ Great for power efficiency
- ❑ Introduces pole-zero doublet that can lead to settling issues. To exactly cancel pole-zero doublet:

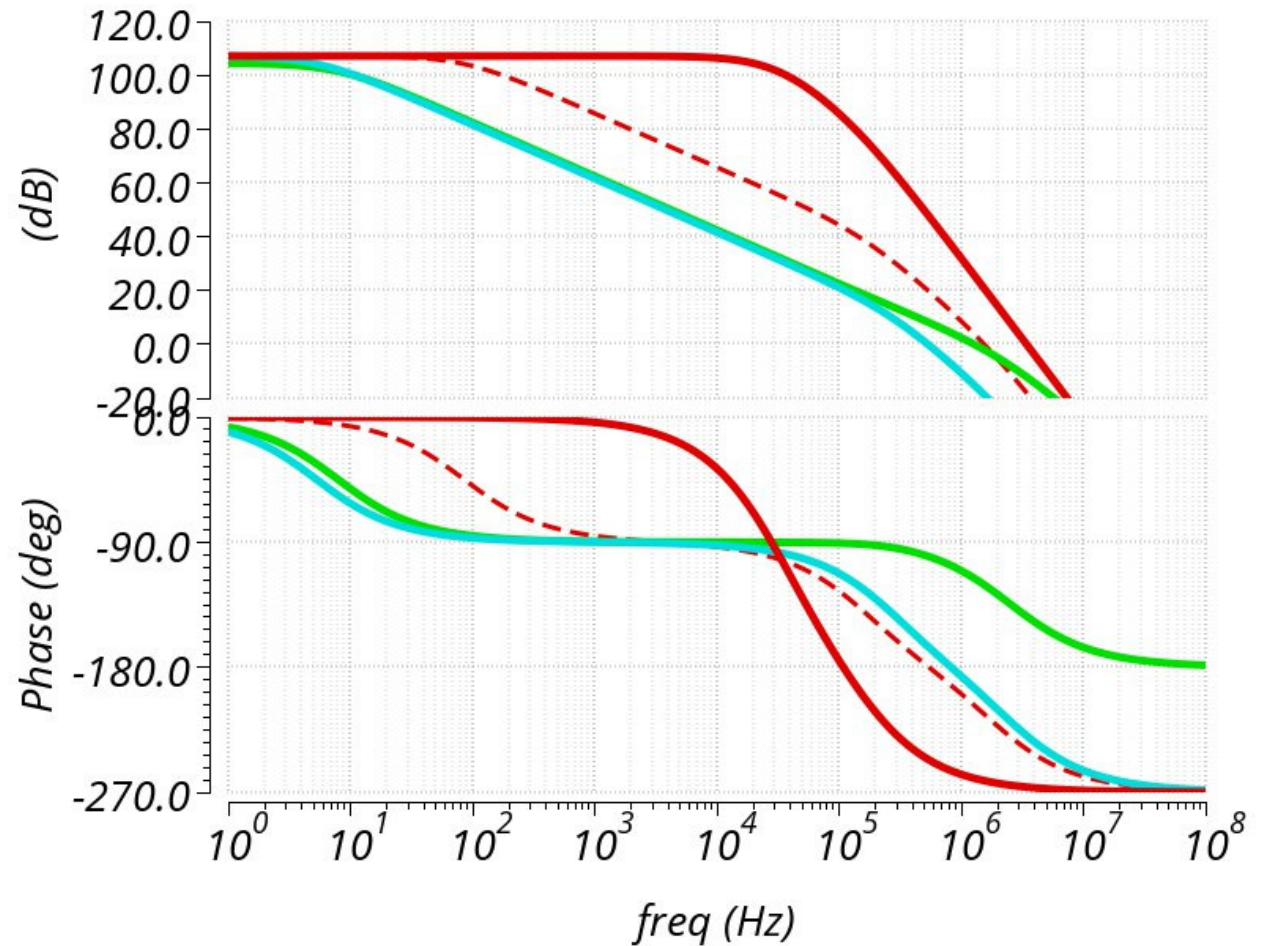
$$\frac{G_1}{G_{ff}} = \frac{C_{M1}}{C_{M2}}$$

- ❑ Keep stich frequency close to bandwidth to mitigate effects of imperfect matching
- ❑ Output impedance is complex! Can have issues with large load caps and driving ADCs.



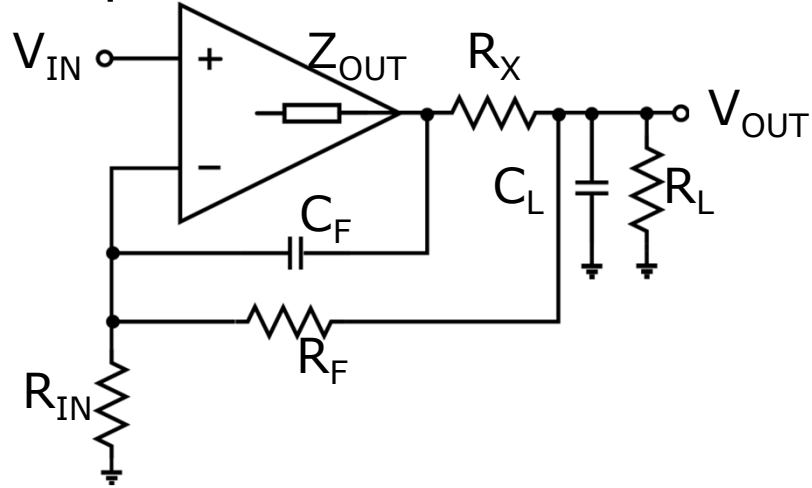
Multipath Nested Miller Compensation

- Compare 3 stage amplifiers
 - Uncompensated
 - Nested Miller compensated
 - Multipath Nested Miller compensated
- In example (50pF C_{LD})
 - 3x better bandwidth
 - 40 degrees more phase margin

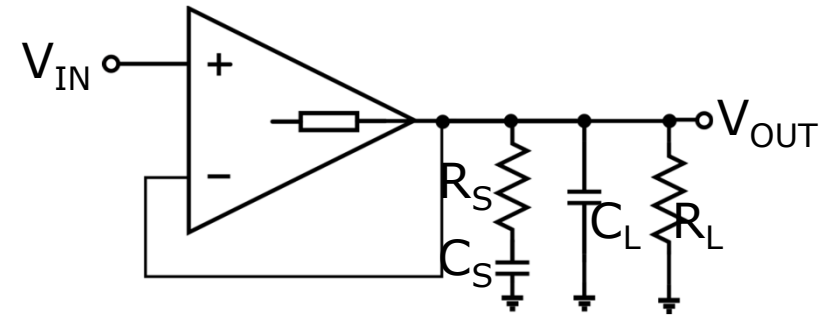


External Techniques to Improve C_L Drive

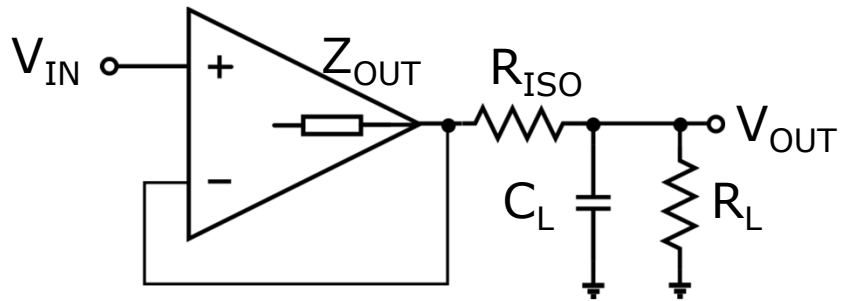
In-loop



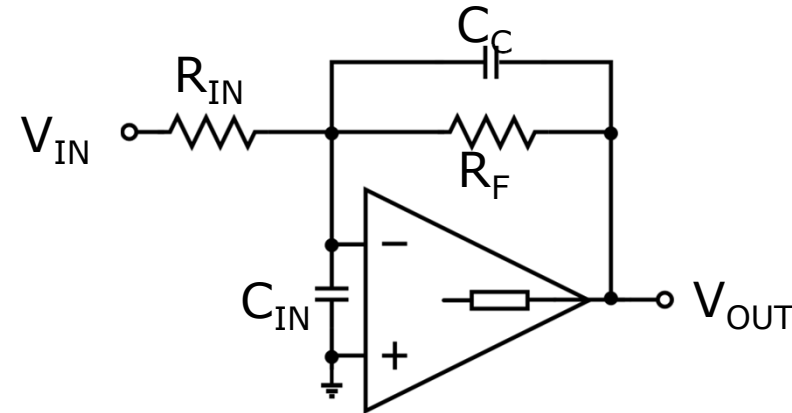
Snubber



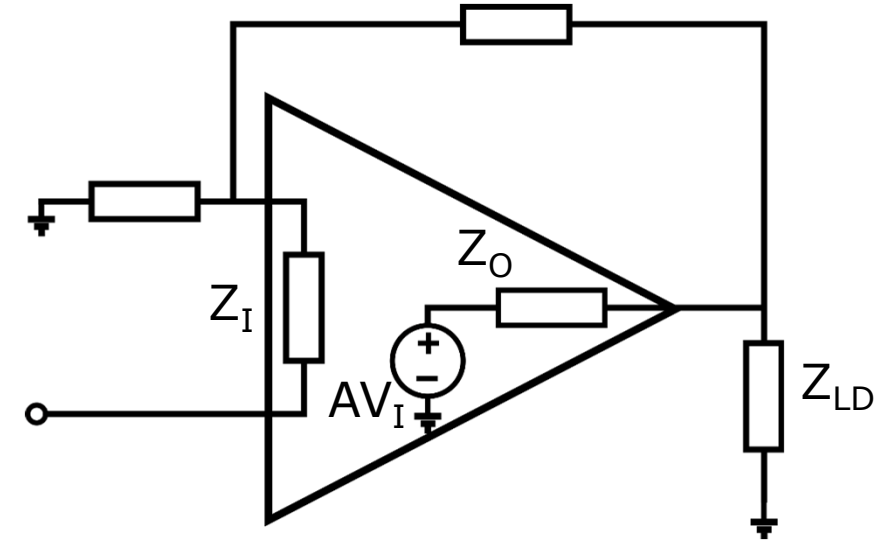
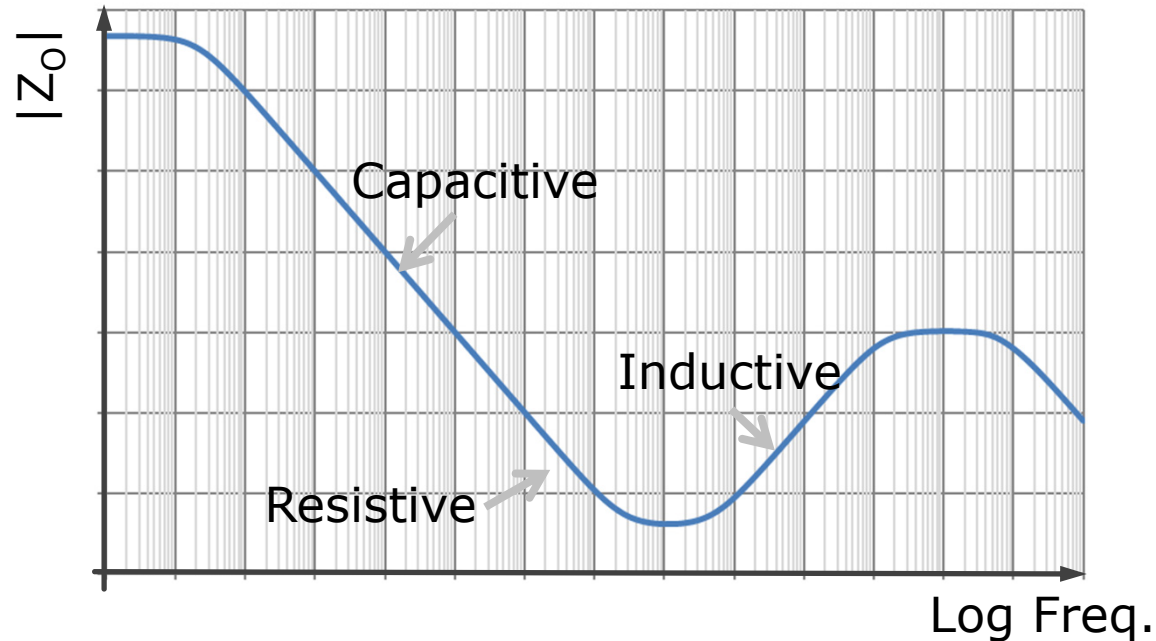
Isolation



Input Cap loading



Output Impedance Frequency Response ^[16]

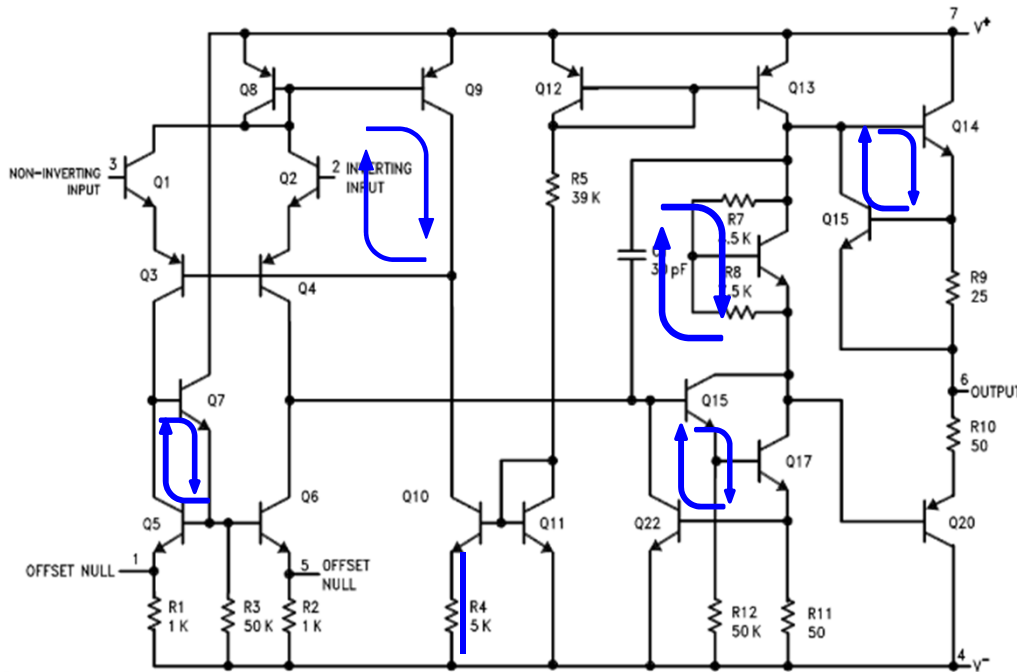


- ❑ Rail-to-rail output stages and multipath structures introduce complex output impedance
- ❑ Region L_O appears inductive and can interact with capacitive loading

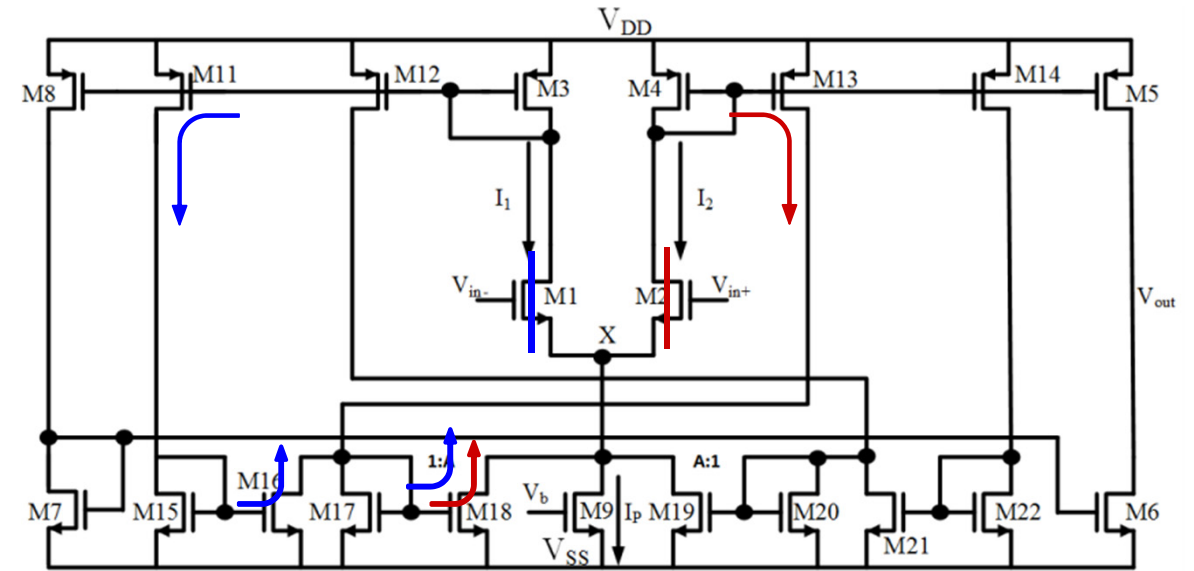
Outline

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- Practical tips and suggestions

Amplifiers have multiple feedback loops



[*] LM741 datasheet

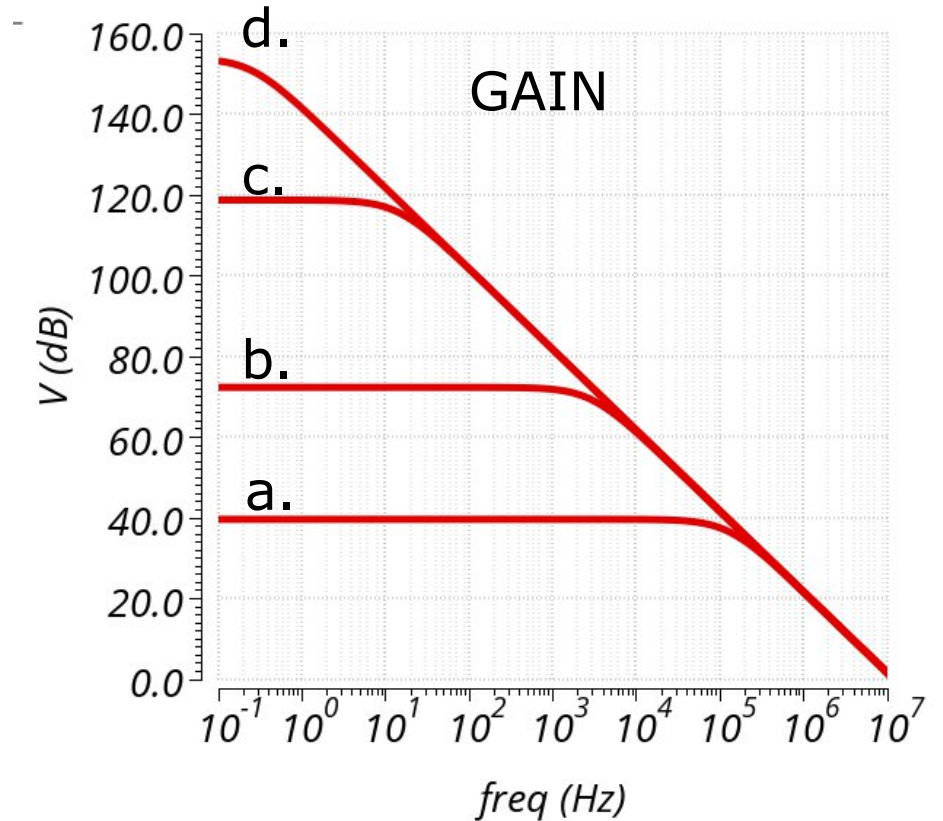
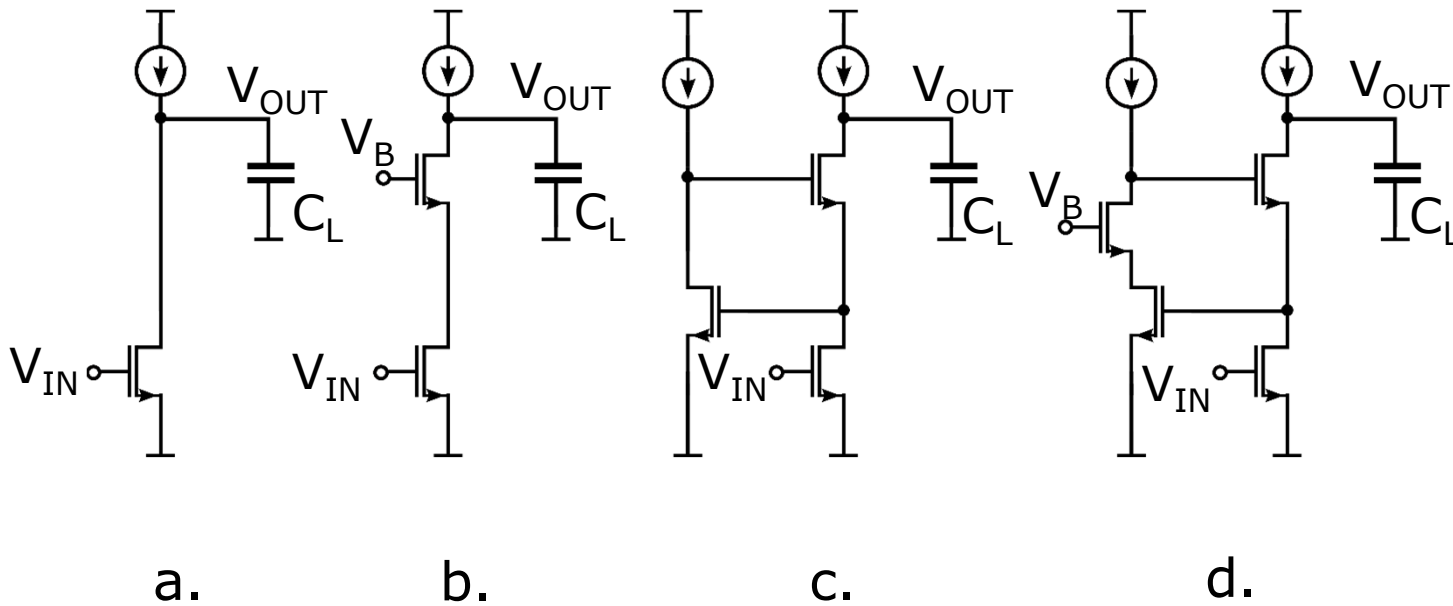


[*] M. Degranuwe, J.Rijmenants, E.A.Vittoz, and D.Man, "Adaptive biasing CMOS amplifiers," IEEE JSSC, vol.SC-17, pp522-528, Jun 1982

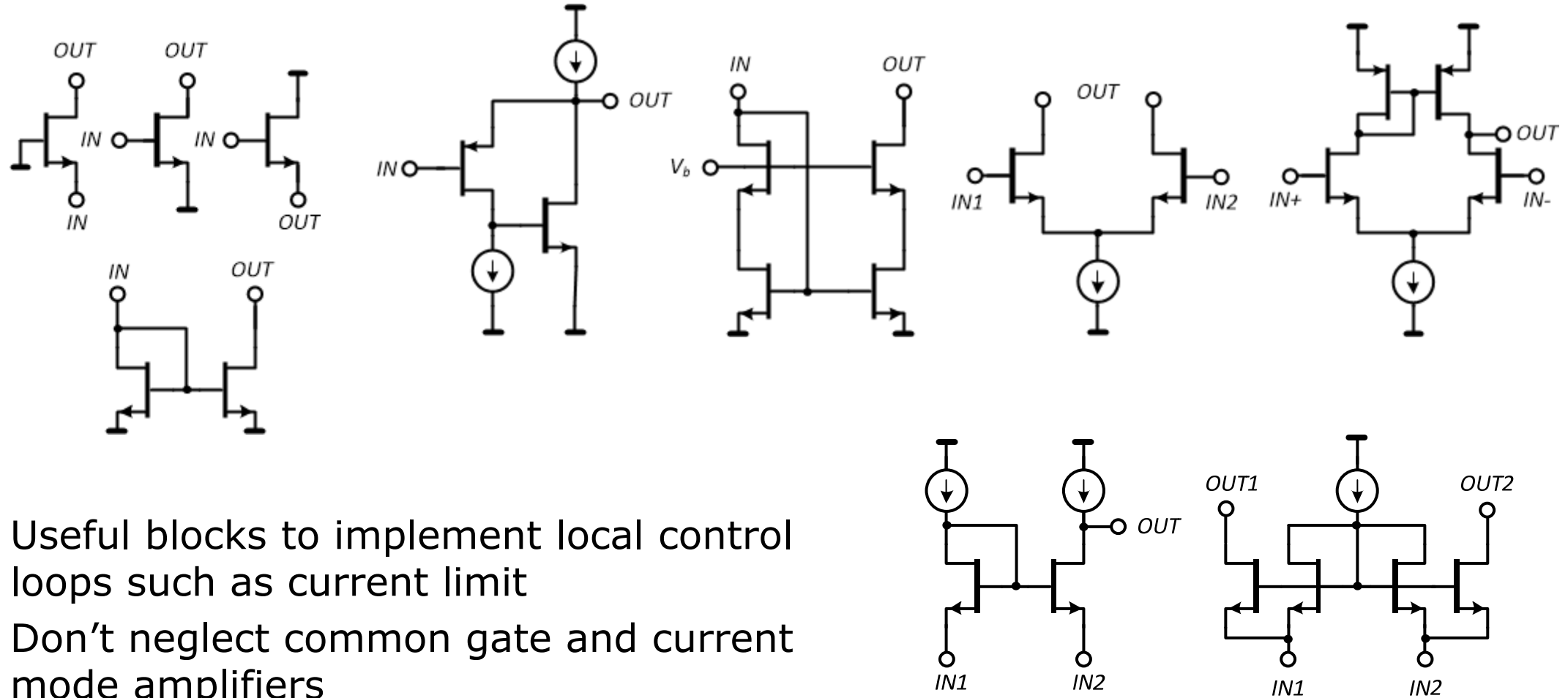
Tips to design for stability

- ❑ Limit the number of gain stages
 - Each gain stage decreases achievable bandwidth
 - Consider gain (impedance) boost
 - Consider nested gain structure
- ❑ Three or more stages can lead to conditional stability
 - Use clamps
 - Control gain
- ❑ Assure stability of each FB loop for overall system stability
 - Keep local loops single stage if possible
- ❑ Use compensation capacitors sparingly
- ❑ Consider gain reduction
- ❑ Watch for transient states (i.e. start-up; saturation in bipolar; current limit etc.)

Gain (impedance boost)



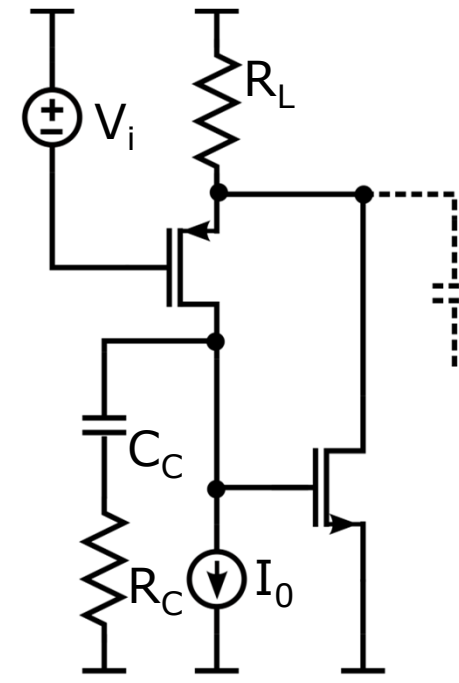
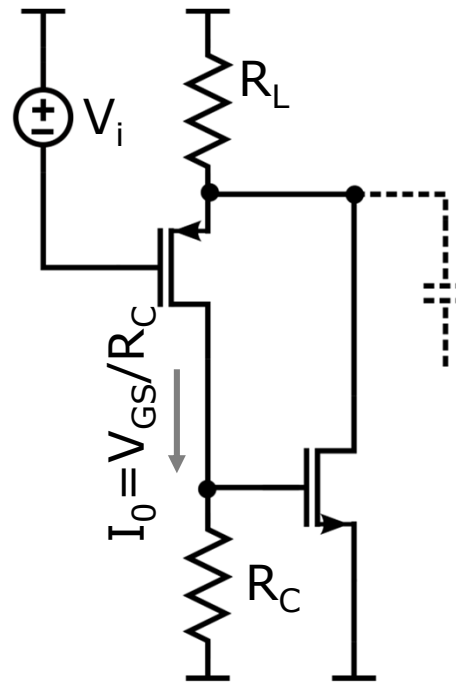
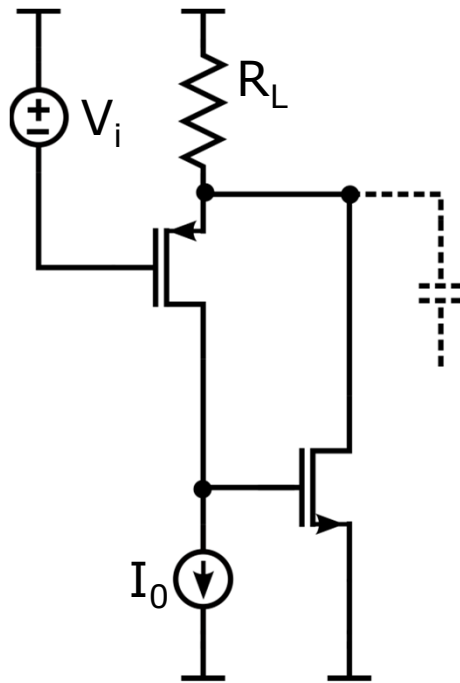
Building blocks for fast local loops^[5,6]



- Useful blocks to implement local control loops such as current limit
- Don't neglect common gate and current mode amplifiers

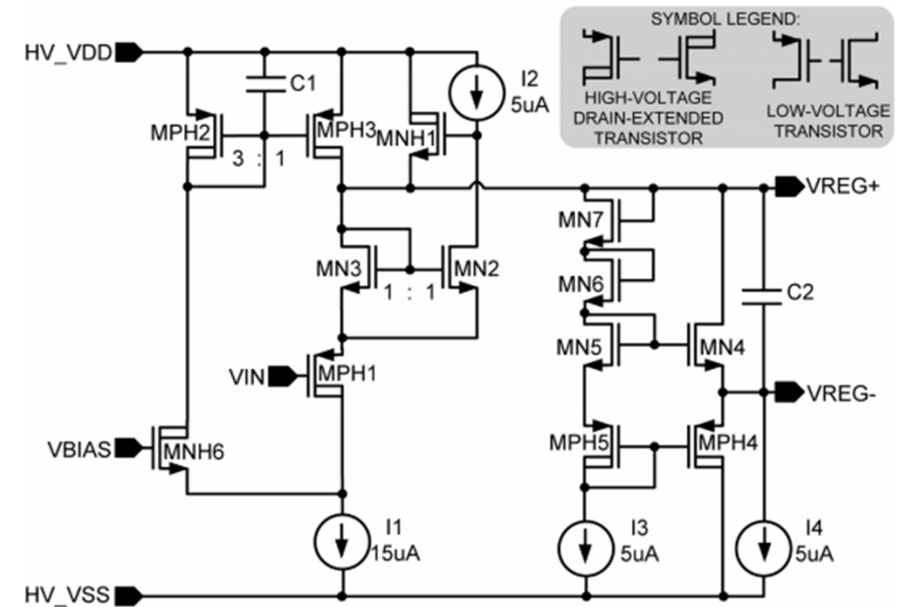
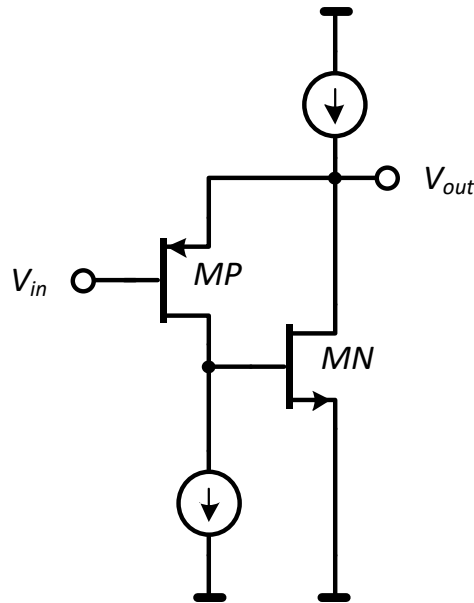
Decreasing gain

- A way to reduce impact of secondary poles
 - Decrease gain by resistor
 - Add capacitance / impedance to stabilize



Nested gain structure

- Parallel low gain path for speed and stability and high gain path for accuracy



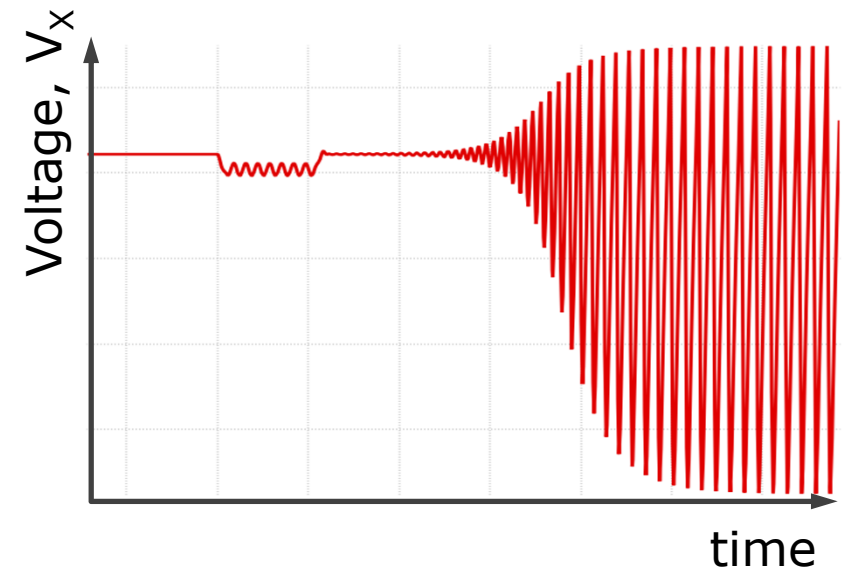
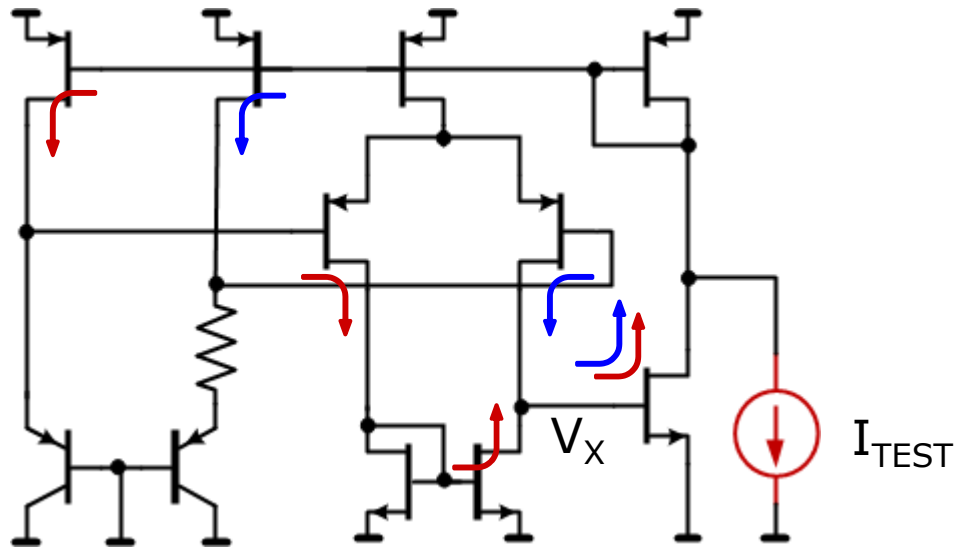
[*] V.Schaffer, M.Snoeiij, M.Ivanov, D.Trifonov, "A 36V Programmable Instrumentation Amplifier with Sub-20uV Offset and CMRR in Excess of 120dB at All Gain Settings" IEEE JSSC, vol.44, July 2009.

Common conditions to watch for

- ❑ Output stage operating conditions
 - Voltage levels
 - Sourcing or sinking current (change of transconductance)
 - Close to triode
 - C_{GS} changes
 - Class AB circuit can introduce delays and non-symmetry
- ❑ Start-up conditions
- ❑ Loading effects (voltage and current dependency)
- ❑ Parasitic capacitances (C_{GS} C_{CB} are voltage dependent)
- ❑ Parasitic inductances (i.e. wirebonds)

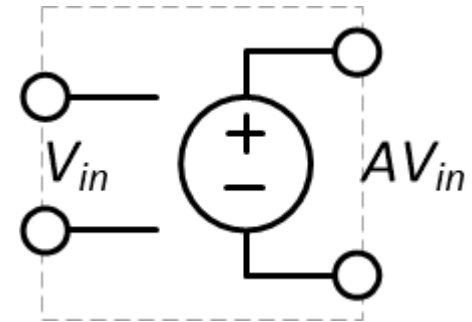
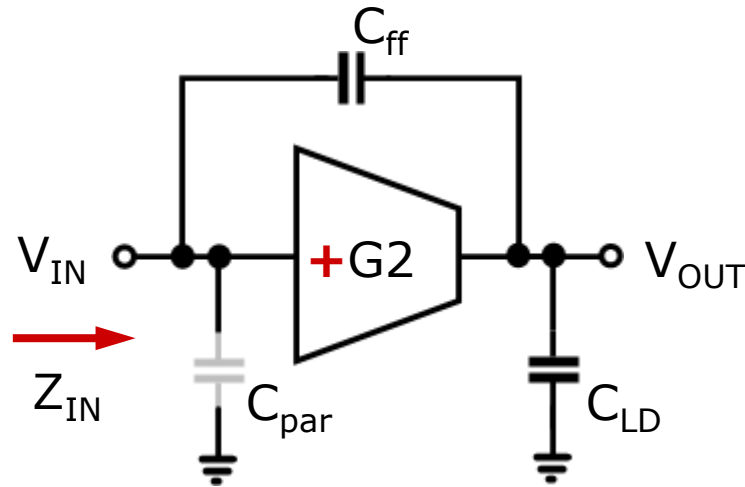
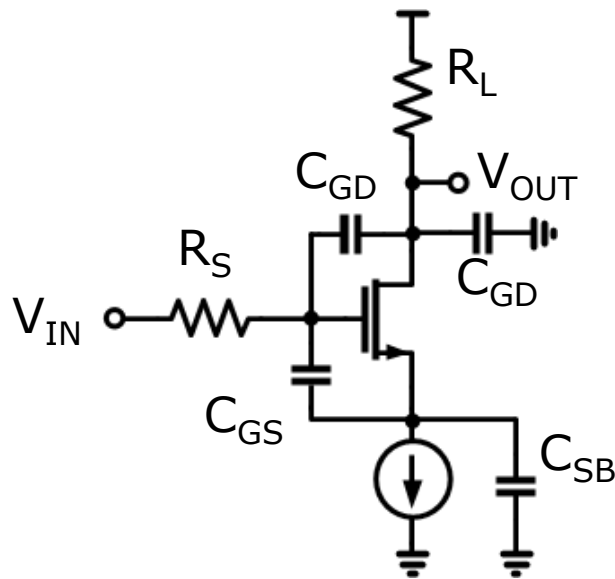
Pulsing nodes

- Apply current pulse to high impedance nodes and observe settling behavior
- Use $\sim 10\%$ bias current
- Cadence – ‘Loop finder’ or ‘STB’ analysis

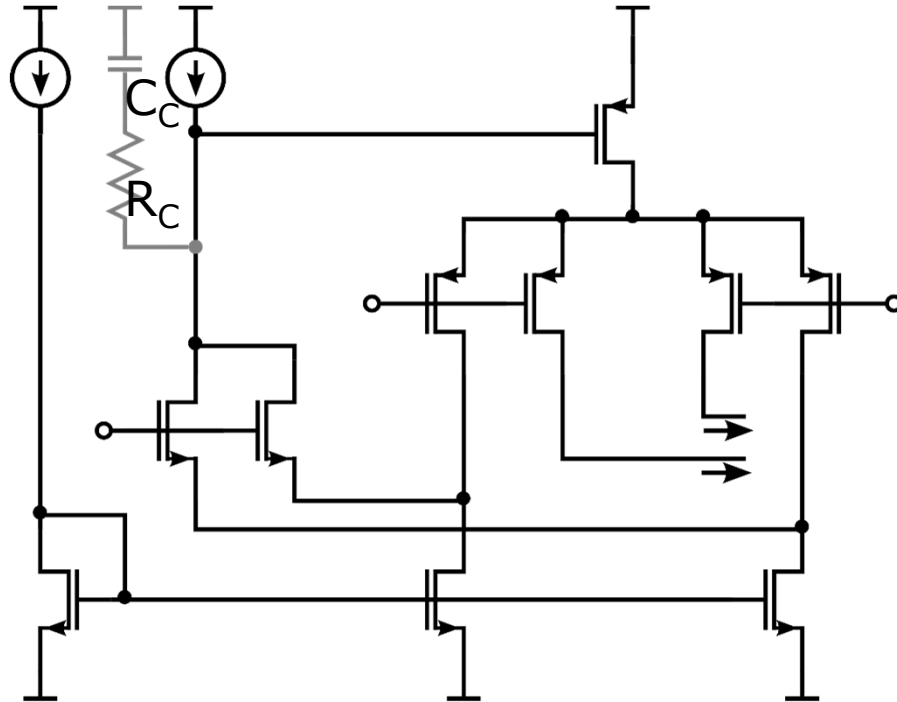


Loading effects

- Even simplest stage will introduce loading effects, parasitics and can transfer effects of subsequent stages
- An ideal voltage controlled voltage source is very helpful for quick investigation (simulation only)



Taming High Impedance Nodes



□ A practical method for taming high impedance nodes:

- Set C_C very large (1F) - BFC
- Choose largest R_C to get desired step response
- Decrease C_C until step response starts changing

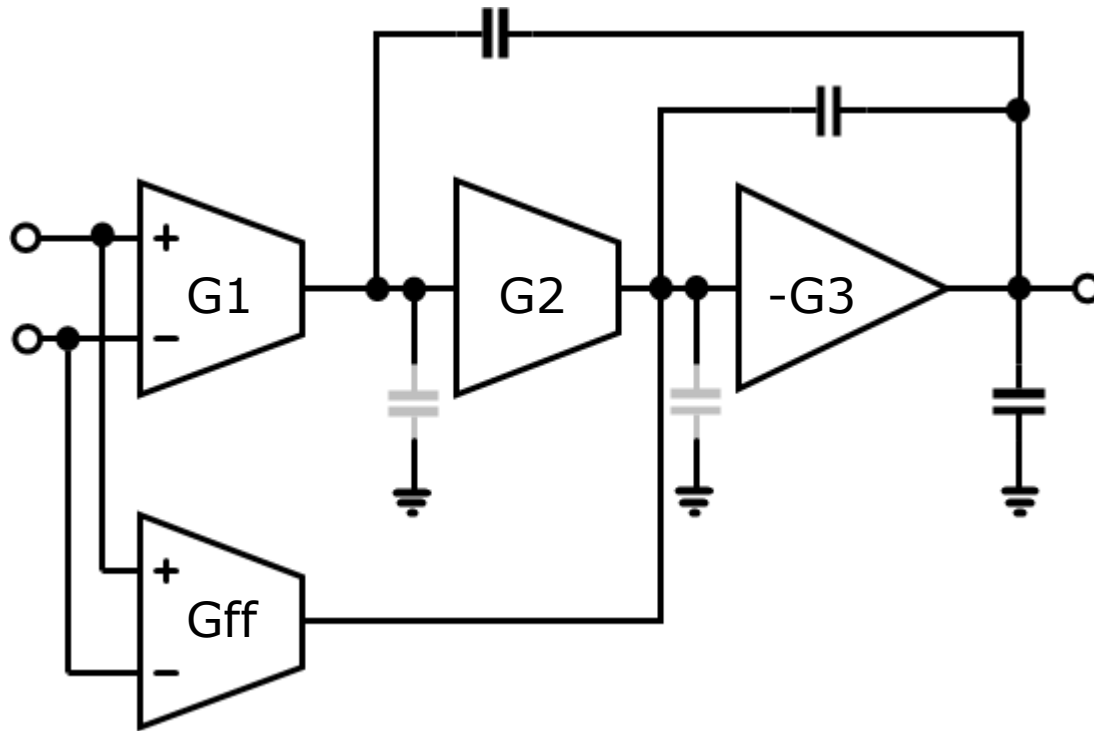
[*] M.Ivanov, M.Ismail, V.Ivanov, "High Slew Rate Micro-Power CMOS OTA with Class AB Input Stage" IEEE Xplore

Miller-Loop damping

- ❑ Multiple gain stages in the Miller Loop require special attention
- ❑ Many techniques exist to control impedance of high impedance nodes [10,12,13,14 ..]
- ❑ Worst case may not be at max or min load current and capacitance!
- ❑ Class AB biasing loop can degrade damping. This is load current dependent!
- ❑ A suggested design approach:
 - Find worse case I_{load} and C_{load} using parametric analysis and step responses
 - Look for peaking in unity gain response
 - Use STB analysis in Miller Loop (with all other outer loops opened) at worst case point. Keep phase margin > 30 degrees (over corners).

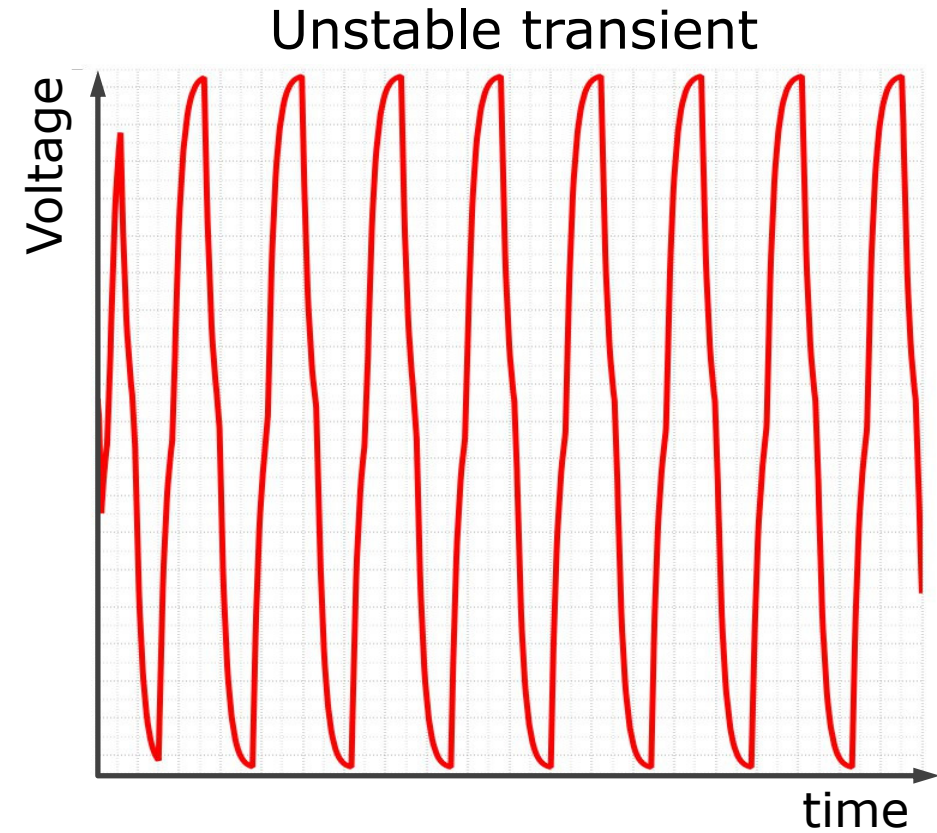
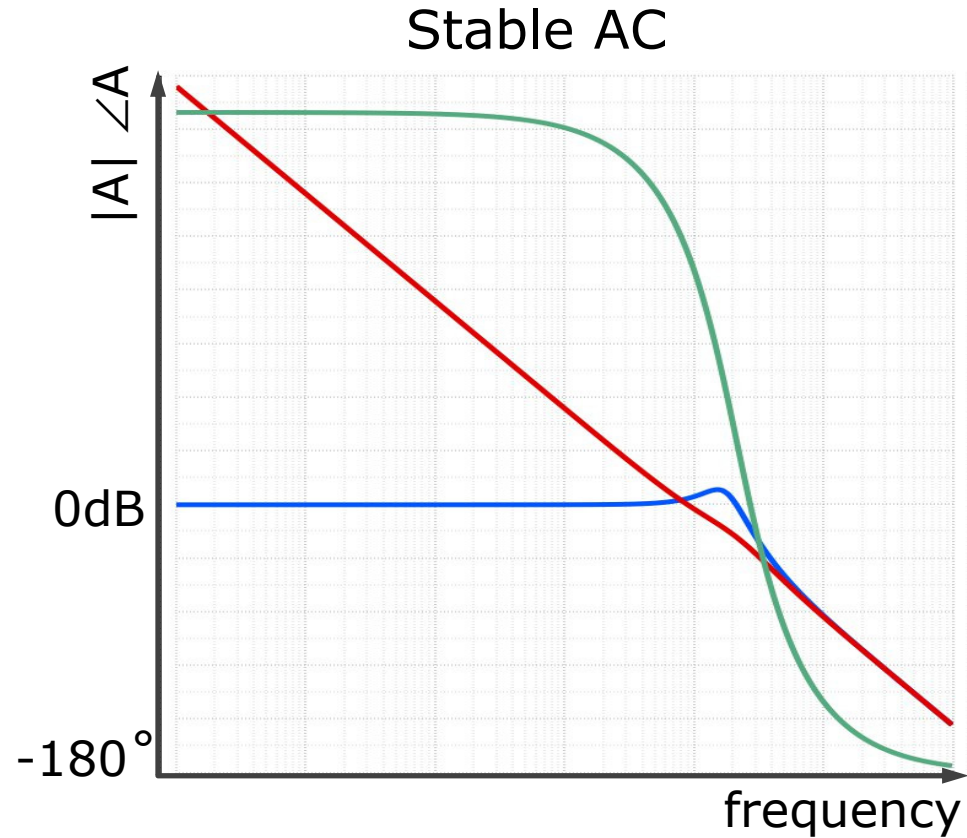
Large signal instability

- ❑ Stable for small input signal / unstable for large step
- ❑ Caused by linear range mismatch in 3- and more stage structures
- ❑ Common cause of silicon revision



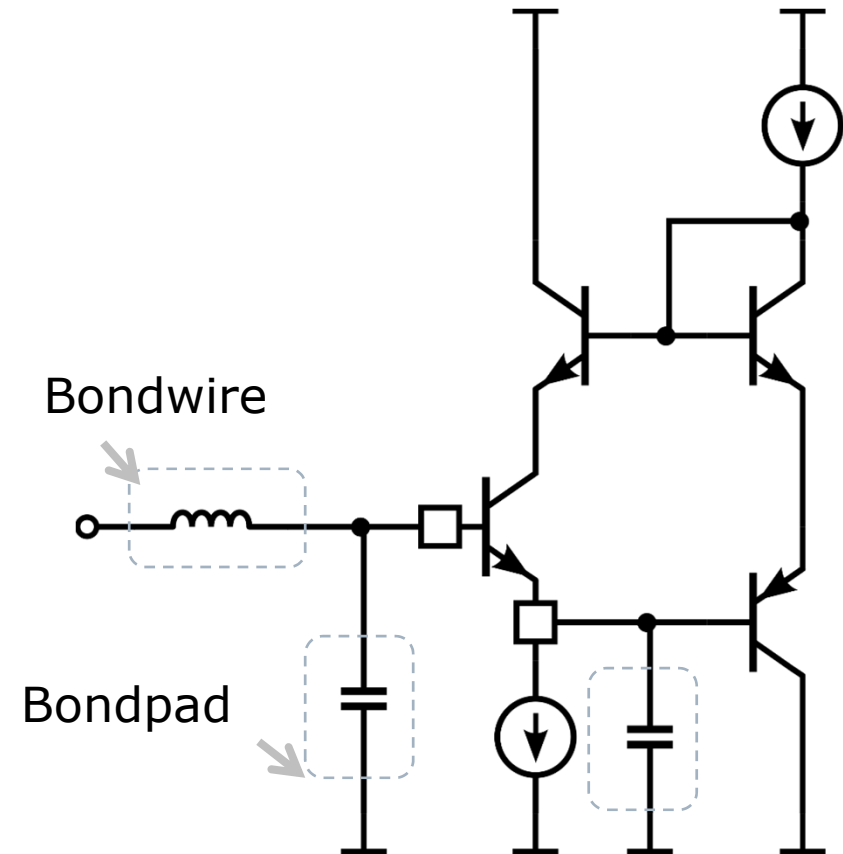
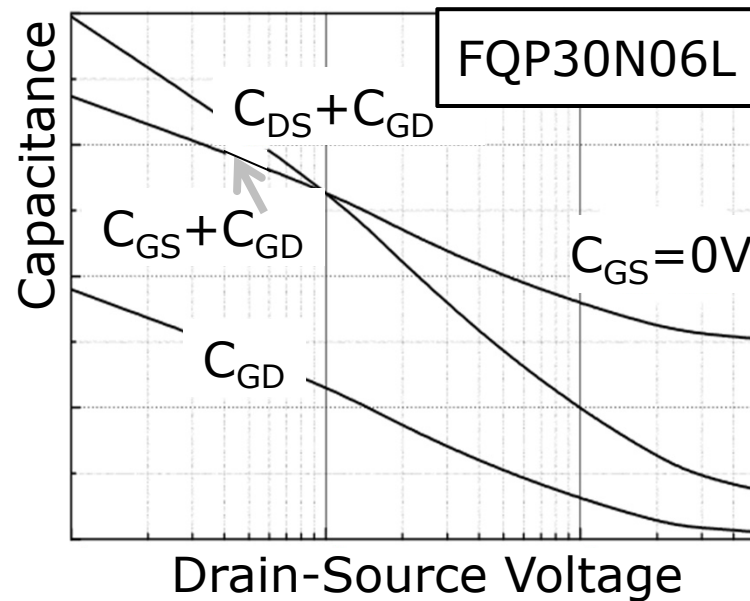
- ❑ To fix:
 - Use passive or active clamps
 - Transconductors with wide input stage

Large signal stability



Beware of Parasitic Effects

- Bondwire inductance
- Pad capacitance
- C_{DG} , C_{CB} vs voltage



Still Oscillates – where to start

- ☐ Disconnect “unnecessary” loops if possible
- ☐ Oscillation frequency
 - Close to unity gain bandwidth
 - ☐ Parasitics
 - ☐ Feedforward RHP zero
 - ☐ Loss of Miller loop gain
 - Lower frequency
 - ☐ Conditional stability
 - ☐ Overdriven gain stage
 - ☐ Saturated bipolar device
 - ☐ Significant loss of Miller loop gain (i.e. cascode transistor)
 - Higher frequency
 - ☐ Parasitic inductances, negative impedance
 - ☐ Unintended positive feedback
 - ☐ Local feedback loops

Papers to see at ISSCC 2021

- 31.4 'A Chopper-Stabilized Amplifier with -107dB IMD and 28dB Suppression of Chopper-Induced IMD' T. Rooijers, S. Karmakar, Y. Kusuda, J. H. Huijsing, K. A. A. Makinwa
- 5.6 'A 25A Hybrid Magnetic Current Sensor with 64mA Resolution, 1.8MHz Bandwidth, and a Gain Drift Compensation Scheme' A. Jouyaeian, Q. Fan, M. Motz, U. Ausserlechner, K. A. A. Makinwa
- 5.5 'A 770 kS/s Duty-Cycled Integrated-Fluxgate Magnetometer for Contactless Current Sensing' P. Garcha, V. Schaffer, B. Haroun, S. Ramaswamy, J. Wieser, J. Lang, A. Chandrakasan
- 5.8 'A 5V Dynamic Class-C Paralleled Single-Stage Amplifier with Near-Zero Dead-Zone Control and Current-Redistributive Rail-to-Rail Gm-Boosting Technique' S-T. Koh, J-H. Lee, G-G. Kang, H. Han, H-S. Kim

References

1. H. Nyquist, "Regeneration Theory," Bell System Technical J., vol. 11, no.1, pp. 126-147, 1932.
2. John M. Miller, "Dependence of the input impedance of a three-electrode vacuum tube upon the load in the plate circuit," Scientific Papers of the Bureau of Standards, vol.15, no. 351, pages 367-385 (1920)
3. R.Eschauzier "Wide Bandwidth Low Power Operational Amplifiers", Delft University Press, 1994.
4. R.Eschauzier, L.Kerklaan, J.H.Huising, "A 100-MHz 100dB Operational Amplifier with Multipath Nested Miller Compensation Structure", IEEE JSSC, Dec. 1992, pp. 1709-1717.
5. V.Ivanov, I.Filanovsky , "Operational Amplifier Speed and Accuracy Improvement", Kluwer, 2004.
6. V.Ivanov, "Analog Design Methodology – Practical Techniques for Frequency Compensation", Mead 2020
7. N. Mehta, J.H.Huijsing, V. Stojanovic, "1-mW Class-AB Amplifier with -101 dB THD+N", IEEE JSSC, Apr. 2019, pp. 948-958.
8. J.Wieser, R. Reed, "Current Source Frequency Compensation for a CMOS Amplifier" US Patent 4,484,148.
9. B.K. Ahuja, "An Improved Frequency Compensation Technique for CMOS Operational Amplifiers", IEEE JSSC, Dec. 1983, pp.629-633.
10. R.J. Reay, G.T. Kovacs, "An Unconditionally Stable Two-Stage CMOS Amplifier", JSSC, May 1995, pp. 591.
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Design of Three-Stage Class-AB 16 Ω Headphone Driver Capable of Handling Wide Range of Load Capacitance

Vijay Dhanasekaran, *Member, IEEE*, Jose Silva-Martinez, *Senior Member, IEEE*, and Edgar Sánchez-Sinenco, *Fellow, IEEE*

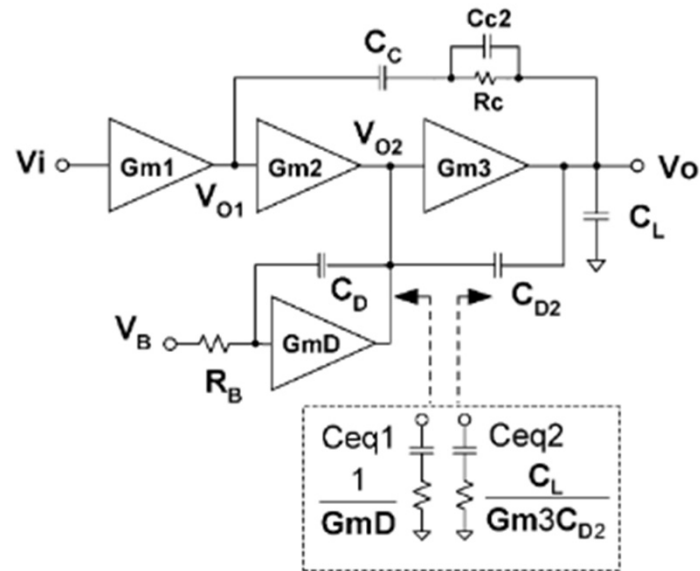
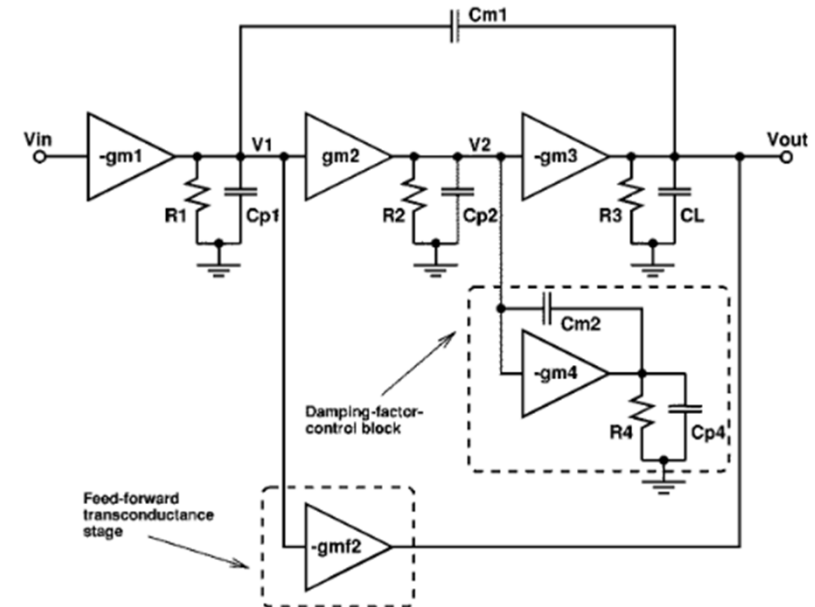


Fig. 6. Architecture of the proposed compensation scheme.

ISSCC99 / SESSION 8 / PAPER TA 8.7

TA 8.7 Damping-Factor-Control Frequency Compensation Technique for Low-Voltage Low-Power Large Capacitive Load Applications

Alex Ka Nang Leung, Philip K.T. Mok, Wing Hung Ki, Johnny K.O. Sin



Examples to improve load capacitance drive

IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 39, NO. 11, NOVEMBER 2004

AC Boosting Compensation Scheme for Low-Power Multistage Amplifiers

Xiaohong Peng and Willy Sansen, *Fellow, IEEE*

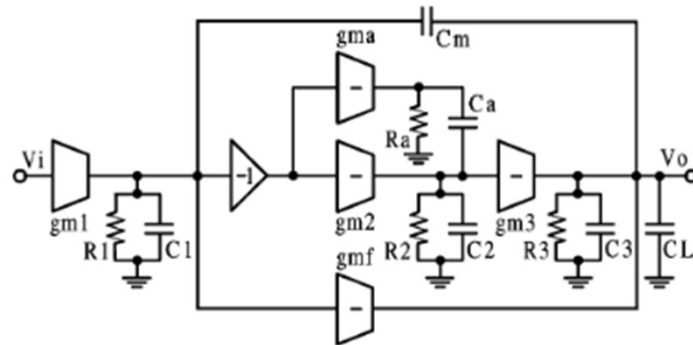


Fig. 1. Three-stage ac boosting compensation topology.

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: FUNDAMENTAL THEORY AND APPLICATIONS, VOL. 49, NO. 7, JULY 2002

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Design Methodology and Advances in Nested-Miller Compensation

Gaetano Palumbo, *Senior Member, IEEE*, and Salvatore Pennisi, *Member, IEEE*

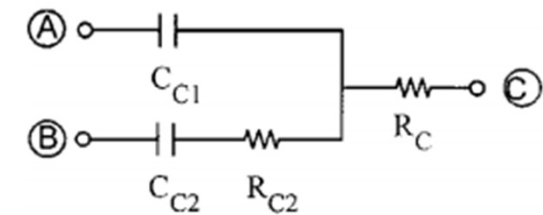


Fig. 5. Proposed two-resistor compensation network.

$$R_C = \frac{1}{g_{m3}}$$

$$C_{C1} = \frac{g_{m3} - g_{m2}}{g_{m2}} C_{C2}$$

$$R_{C2} = \frac{1}{g_{m3}} \frac{C_L}{C_{C2}}$$

2-Stage Amplifier: Stable with any Cld

IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 30, NO. 5, MAY 1995

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An Unconditionally Stable Two-Stage CMOS Amplifier

Richard J. Reay and Gregory T. A. Kovacs

